Nine stacks. Three user loads.

A measured comparison of responsiveness and resource usage, with the original historical appendix retained.

24 / 27

Fixed-load trials passed the recorded response and authentication gates.

Octane joins Python, Java, Go, Rust, Node and Bun with passes at all three loads. .NET passed 100 and 500 users, but failed 1,000. Laravel/FPM passed 100 and failed 500 and 1,000. Failed observations remain included in every comparison.

The useful conclusion

Healthy stacks delivered almost the same successful throughput at 1,000 users because of think time. Whole-backend EC2 VM CPU/RAM is the primary resource view, including OS, SSH, Docker and exporter activity; application-container readings are secondary. This tests the configured VM with a3 GiB app cap, not unrestricted 4 GiB application capacity.

The boundaries of the evidence

One trial per stack/load: no confidence intervals, statistical winner or universal user limit is justified. Historical capacity brackets include unstable repeats and later open-file validity flags. Older arrival-rate trials are operations/sec, not users.

Control	Recorded setting
Backend VM / app cap	c7i.large:2 vCPUs,advertised 4 GiB; app capped 3 GiB/no swap; DB pool 20
Workload	Unique accounts, real Argon2id login, following feeds, likes, replies, random 3-10 s waits
Fixed profile	180 s ramp + 0 s settle + 240 s steady; 5 s request deadline
Pass gates	90% valid steady responses; 90% authenticated accounts
Population	100 / 500 / 1,000 users; canonical fixture reset each trial

Dataset contains 234 inventoried trials. Main focus: 27 fixed-load runs. The historical appendix retains the original 06-Oct snapshot; later capacity searches are outside this fixed-load update. Capture/source details and metric limits appear throughout the report. Prepared 07 October 2026, Asia/Kuala_Lumpur.

100 users: all nine configurations

Summary first | steady 240 s; VM CPU/RAM primary, container detail later | gates 90% response and 90% authentication.

Stack	Result	Response / auth %	Latency mean/p99/p99.9 ms	Useful ops/s mean / peak 1 s / peak 10 s	VM CPU mean / peak %	VM used-RAM est. mean / peak MiB	DB CPU avg / max %	DB freeable avg GiB
Python 3.14	PASS	100.0000% / 100.00%	7.02 / 44.00 / 46.24	16.43 / 28.0 / 18.4	6.86 / 10.27	817.9 / 834.5	1.57 / 1.63	10.252
Java	PASS	100.0000% / 100.00%	3.46 / 8.00 / 9.00	16.32 / 28.0 / 18.2	4.52 / 10.15	1,680.0 / 1,691.5	1.69 / 1.99	10.258
PHP	PASS	100.0000% / 100.00%	55.53 / 155.04 / 224.21	16.24 / 28.0 / 17.7	33.40 / 41.35	763.8 / 775.9	4.77 / 4.95	10.246
Go	PASS	100.0000% / 100.00%	3.08 / 8.00 / 9.15	16.35 / 26.0 / 18.2	3.33 / 4.64	785.8 / 863.5	1.82 / 1.99	10.240
Rust	PASS	100.0000% / 100.00%	3.43 / 8.00 / 9.00	16.37 / 26.0 / 17.8	3.29 / 4.32	688.4 / 700.0	1.57 / 1.62	10.238
Node	PASS	100.0000% / 100.00%	4.71 / 11.00 / 13.07	16.38 / 30.0 / 18.1	4.27 / 8.32	791.1 / 815.6	1.94 / 2.04	10.236
Bun	PASS	100.0000% / 100.00%	4.43 / 11.00 / 13.07	16.36 / 29.0 / 18.3	4.13 / 5.64	757.9 / 783.0	1.95 / 2.03	10.235
.NET	PASS	100.0000% / 100.00%	3.98 / 9.00 / 31.26	16.40 / 27.0 / 18.4	4.02 / 10.45	916.1 / 930.9	2.08 / 2.20	10.261
Octane	PASS	100.0000% / 100.00%	6.44 / 12.00 / 13.09	16.29 / 30.0 / 18.0	5.65 / 7.00	1,563.0 / 1,580.2	2.11 / 2.14	10.209

Primary CPU/RAM covers the whole backend VM, including OS, Docker, SSH and monitoring. RAM is estimated total minus available; Linux reports about 3.47 GiB total. The app remains capped at 3 GiB/no swap; this is not an unrestricted 4 GiB app test.

Response/authentication gates:90%. Useful work requires a valid result within 5 s; peak 1 s/10 s figures describe bursts. RDS is separate 4vCPU/16 GiB; its steady 240 s anchor uses overlapping minute buckets. Failed PHP rows are not efficient useful work.

One run per stack/load. Mean/peak and tail observations are descriptive, not confidence estimates or a capacity ceiling. Detailed VM/container phase measurements and definitions follow.

500 users: all nine configurations

Summary first | steady 240 s; VM CPU/RAM primary, container detail later | gates 90% response and 90% authentication.

Stack	Result	Response / auth %	Latency mean/p99/p99.9 ms	Useful ops/s mean / peak 1 s / peak 10 s	VM CPU mean / peak %	VM used-RAM est. mean / peak MiB	DB CPU avg / max %	DB freeable avg GiB
Python 3.14	PASS	100.0000% / 100.00%	6.45 / 44.00 / 45.00	81.72 / 113.0 / 85.0	12.57 / 28.94	815.7 / 862.8	3.11 / 3.24	10.246
Java	PASS	100.0000% / 100.00%	2.90 / 7.00 / 8.00	81.55 / 102.0 / 85.9	5.81 / 32.39	1,371.4 / 1,388.3	3.60 / 4.00	10.252
PHP (FAIL)	FAIL	0.0000% / 42.20%	5,000.66 / 5,002.00 / 5,002.00	0.00 / 0.0 / 0.0	99.99 / 100.00	900.6 / 947.0	3.88 / 4.81	10.242
Go	PASS	100.0000% / 100.00%	2.53 / 5.00 / 7.00	81.60 / 114.0 / 84.9	4.78 / 8.72	811.3 / 952.3	3.07 / 3.63	10.233
Rust	PASS	100.0000% / 100.00%	2.91 / 7.00 / 8.00	81.71 / 110.0 / 85.1	4.80 / 16.70	702.4 / 715.6	3.49 / 3.51	10.236
Node	PASS	100.0000% / 100.00%	4.00 / 10.00 / 12.00	81.60 / 117.0 / 86.6	7.50 / 11.85	895.6 / 927.7	4.80 / 4.88	10.230
Bun	PASS	100.0000% / 100.00%	3.71 / 9.00 / 11.00	81.67 / 104.0 / 85.9	6.73 / 14.87	779.9 / 793.4	4.65 / 4.92	10.227
.NET	PASS	100.0000% / 100.00%	3.62 / 9.00 / 11.00	81.70 / 108.0 / 85.2	7.45 / 17.59	922.5 / 935.2	5.34 / 5.41	10.253
Octane	PASS	100.0000% / 100.00%	6.18 / 12.00 / 14.00	81.66 / 107.0 / 86.1	15.25 / 21.94	1,583.5 / 1,600.9	5.45 / 5.72	10.210

Primary CPU/RAM covers the whole backend VM, including OS, Docker, SSH and monitoring. RAM is estimated total minus available; Linux reports about 3.47 GiB total. The app remains capped at 3 GiB/no swap; this is not an unrestricted 4 GiB app test.

Response/authentication gates:90%. Useful work requires a valid result within 5 s; peak 1 s/10 s figures describe bursts. RDS is separate 4vCPU/16 GiB; its steady 240 s anchor uses overlapping minute buckets. Failed PHP rows are not efficient useful work.

One run per stack/load. Mean/peak and tail observations are descriptive, not confidence estimates or a capacity ceiling. Detailed VM/container phase measurements and definitions follow.

1,000 users: all nine configurations

Summary first | steady 240 s; VM CPU/RAM primary, container detail later | gates 90% response and 90% authentication.

Stack	Result	Response / auth %	Latency mean/p99/p99.9 ms	Useful ops/s mean / peak 1 s / peak 10 s	VM CPU mean / peak %	VM used-RAM est. mean / peak MiB	DB CPU avg / max %	DB freeable avg GiB
Python 3.14	PASS	100.0000% / 100.00%	6.39 / 44.00 / 46.00	163.65 / 196.0 / 169.0	19.22 / 47.63	833.0 / 879.2	5.04 / 5.25	10.236
Java	PASS	100.0000% / 100.00%	2.93 / 8.00 / 9.00	163.83 / 205.0 / 169.3	6.63 / 11.58	1,798.6 / 1,806.2	6.72 / 7.50	10.213
PHP (FAIL)	FAIL	0.0000% / 14.10%	5,000.65 / 5,002.00 / 5,002.00	0.00 / 0.0 / 0.0	99.99 / 100.00	936.1 / 972.8	2.97 / 3.19	10.246
Go	PASS	100.0000% / 100.00%	2.42 / 5.00 / 6.00	163.80 / 198.0 / 168.1	6.39 / 27.77	761.9 / 919.4	4.86 / 4.90	10.229
Rust	PASS	100.0000% / 100.00%	2.71 / 6.00 / 7.00	163.85 / 201.0 / 169.3	5.91 / 25.49	723.6 / 732.7	5.69 / 5.84	10.229
Node	PASS	99.9949% / 100.00%	3.82 / 9.00 / 13.00	163.74 / 196.0 / 169.2	11.17 / 36.15	928.0 / 979.9	8.30 / 8.54	10.223
Bun	PASS	99.9975% / 100.00%	3.56 / 9.00 / 11.00	163.88 / 203.0 / 168.4	9.64 / 44.37	790.3 / 814.2	8.10 / 8.66	10.222
.NET	FAIL	9.9552% / 5.60%	4,505.57 / 5,002.00 / 5,033.14	9.07 / 19.0 / 10.5	99.65 / 99.90	1,059.9 / 1,188.5	2.49 / 2.60	10.211
Octane	PASS	100.0000% / 100.00%	6.47 / 13.00 / 18.00	163.67 / 214.0 / 169.9	26.50 / 60.17	1,606.3 / 1,631.9	9.55 / 9.85	10.210

Primary CPU/RAM covers the whole backend VM, including OS, Docker, SSH and monitoring. RAM is estimated total minus available; Linux reports about 3.47 GiB total. The app remains capped at 3 GiB/no swap; this is not an unrestricted 4 GiB app test.

Response/authentication gates:90%. Useful work requires a valid result within 5 s; peak 1 s/10 s figures describe bursts. RDS is separate 4vCPU/16 GiB; its steady 240 s anchor uses overlapping minute buckets. Failed PHP rows are not efficient useful work.

One run per stack/load. Mean/peak and tail observations are descriptive, not confidence estimates or a capacity ceiling. Detailed VM/container phase measurements and definitions follow.

How to read this report

Common definitions prevent success counts, latency tails and resource budgets from being conflated.

Successful work

An operation is successful only when the expected valid HTTP result arrives within the 5-second deadline. Failed statuses, invalid bodies and timeouts remain failures. Successful operations/sec is the throughput measure used for comparison.

Latency and slow-tail cutoffs

p99 is the boundary for the slowest 1% of recorded latencies; p99.9 is the boundary for the slowest 0.1%. Neither is the mean of that tail. Response latency includes failed/time-out operations. All numbers are per trial, not pooled statistical estimates.

Scheduled steady throughput

Mean attempted/successful rates use the full configured 240-second steady duration. One-second and rolling-ten-second peaks use only actual completions in the scheduled steady window. Phase-tagged late completions are reported as spillover, never clamped into the last second.

VM CPU and RAM primary

Whole-backend VM CPU uses first 8/proc/stat counter deltas:100 x (1 - (idle+iowait)/total). It includes steal, tracked separately, plus OS/SSH/Docker/exporter work. VM used-RAM estimate is MemTotal - MemAvailable, not RSS/all cache; observed MemTotal≈3553.54 MiB,advertised 4 GiB. App container is still capped 3 GiB/no swap; this is tested VM configuration, not unrestricted RAM capacity.

RDS

RDS is a separate allocation. Freeable memory is available/headroom, not exact used memory. RDS queries are anchored to the 240-second steady window and rounded to overlapping minute buckets (4-5 per trial). They are not exact second-by-second backend samples. DB tail estimates are especially sparse.

Validity

Missing/incomplete telemetry, constrained generators, open-file exhaustion, bad configuration or failed fixture restoration cannot define capacity. Normal backend overload is failure evidence when infrastructure is qualified.

Locked implementations

Versions and production topologies, not interchangeable language microbenchmarks.

Stack	Runtime	Framework	Request / pool topology
Python 3.14	3.14.8	FastAPI 0.142.2	2 async / pool 10 each
Java	Temurin 27+35	Spring Boot 4.1.1	Virtual threads / Hikari 20
PHP	8.5.11	Laravel 13.34.0	nginx + FPM 20 / one connection each
Go	1.27.1	net/http + pgx	One process / pgxpool 20
Rust	1.99.0	Axum 0.8.9	Tokio 2 / SQLx 20
Node	26.10.0	Express 5.2.1	Cluster 2 / pool 10 each
Bun	1.4.2	Express 5.2.1	Same Express code / cluster 2
.NET	10.0.12	ASP.NET Core 10.0.12	Kestrel async / Npgsql 20
Octane	PHP 8.5.11	Laravel 13.35 / Octane 2.21	RoadRunner 2025.1.15 /20 cached PDO

Python reference caveat

Python 3.14.8 is the stable interpreter reference. Its modern dependency lock retains Pydantic 2.14.0b2 / Core 2.49.0 from the experimental stack. It is not an all-stable/latest dependency claim. Historical Python 3.15.0rc3 JIT/free-threaded cells remain prerelease evidence.

Shared controls

Octane uses Laravel 13.35.0, one patch newer than FPM 13.34.0, plus persistent workers/connections. This is a configuration comparison, not a pure single-variable PHP speed test. Identical API/schema and UTC microsecond keyset pagination; password Argon2id memory 65,536 KiB, time 3, parallelism 1; real salted fixture PHCs; JWT validation; no application caches or denormalized counters. Fixture: 50,000 users, 500,000 posts, 2,500,000 follows, 1,000,000 likes and 250,000 comments.

Infrastructure

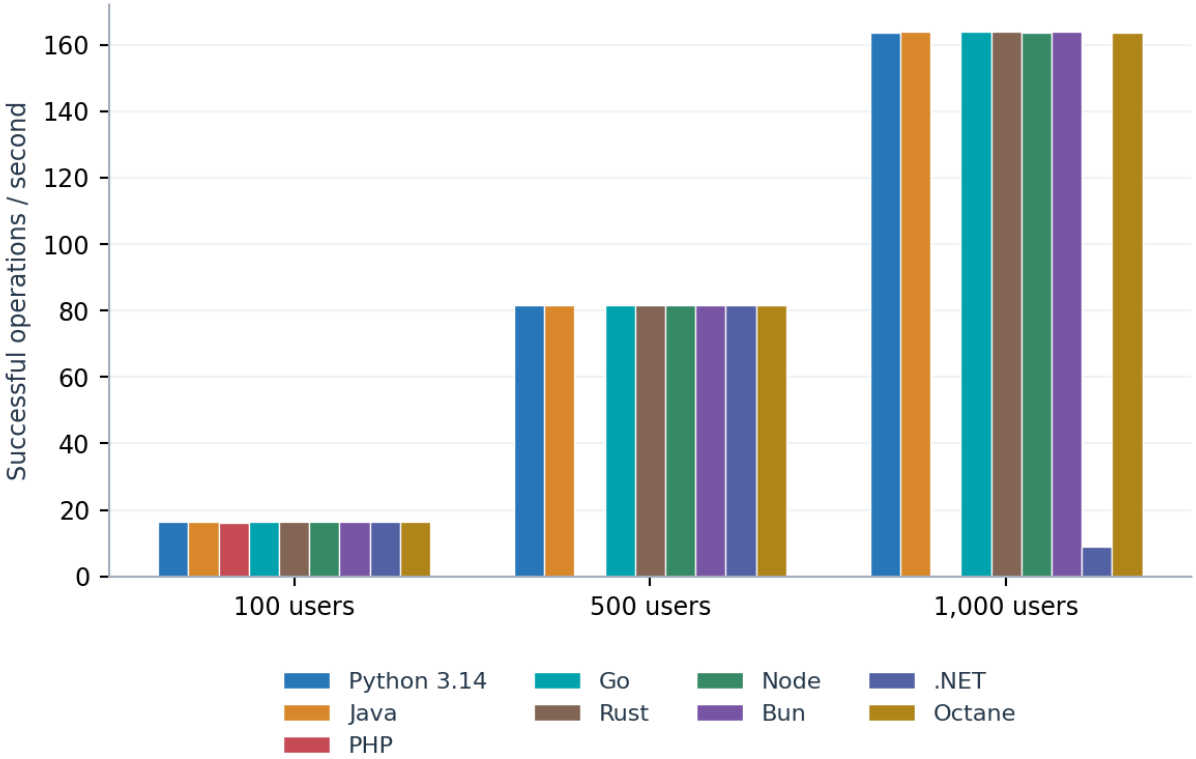
AWS ap-southeast-1. Backend c7i.large, 2 vCPUs with Docker 3 GiB/no-swap; Rocky Linux 10.2, Docker CE 29.8.2 and Compose 5.6.0. k6 uses a separate c7i.8xlarge generator (32 vCPUs/64 GiB) with four source IPs. RDS PostgreSQL 17.6, db.m7i.xlarge (4 vCPUs/16 GiB), encrypted gp 3. Native and fixed-load runs require a 250,000 open-file limit.

Historical Python two-process hashing concurrency and native implementation details differ; comparisons concern the deployed stack and its workload, not a single instruction-set or isolated language-speed benchmark.

Pass / fail at fixed population

Both operation and authentication gates must pass. Failed PHP-FPM and .NET trials are retained.

Stack	100 users	500 users	1,000 users
Python 3.14	PASS	PASS	PASS
Java	PASS	PASS	PASS
PHP	PASS	FAIL	FAIL
Go	PASS	PASS	PASS
Rust	PASS	PASS	PASS
Node	PASS	PASS	PASS
Bun	PASS	PASS	PASS
.NET	PASS	PASS	FAIL
Octane	PASS	PASS	PASS



Think-time pacing produces approximately 16, 82 and 164 useful operations/sec in healthy trials. PHP at 500 users/1000 did not return successful steady operations; zero useful throughput cannot be read as a throughput win. All trial rows are descriptive single observations.

Steady response latency

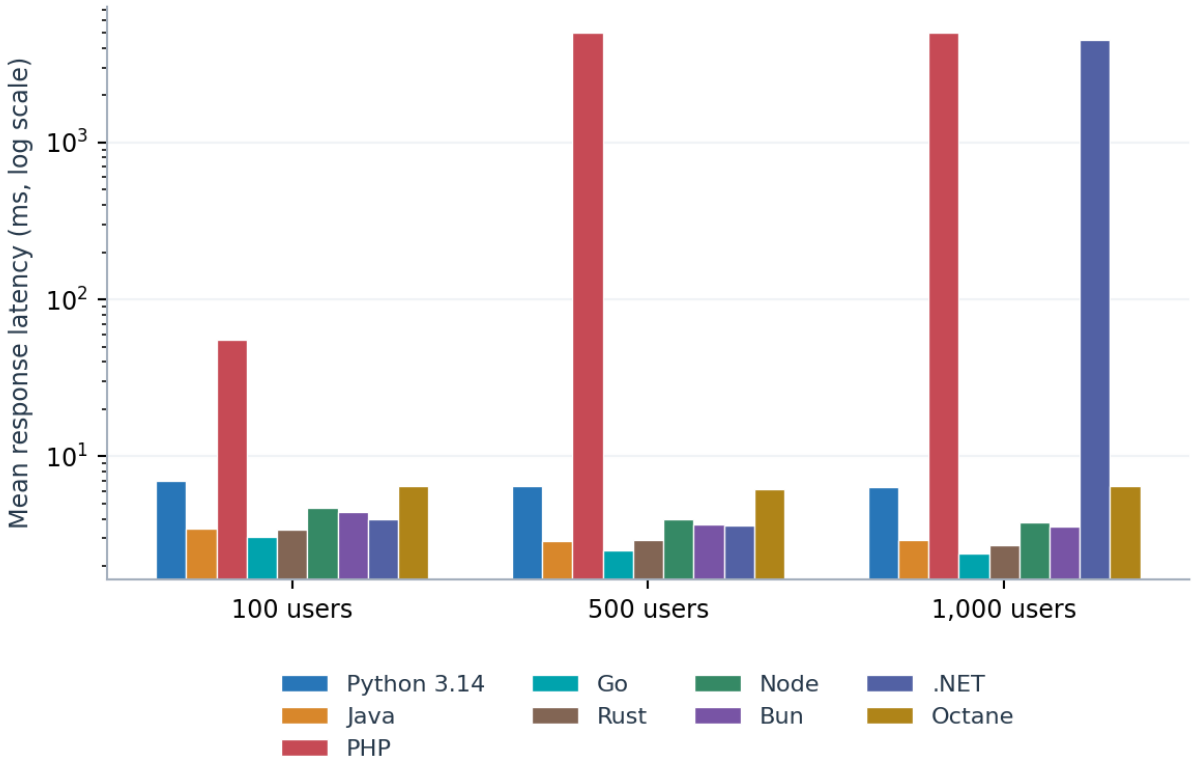
Milliseconds, including failures/timeouts. p99 / p99.9 mark slow-tail cutoffs.

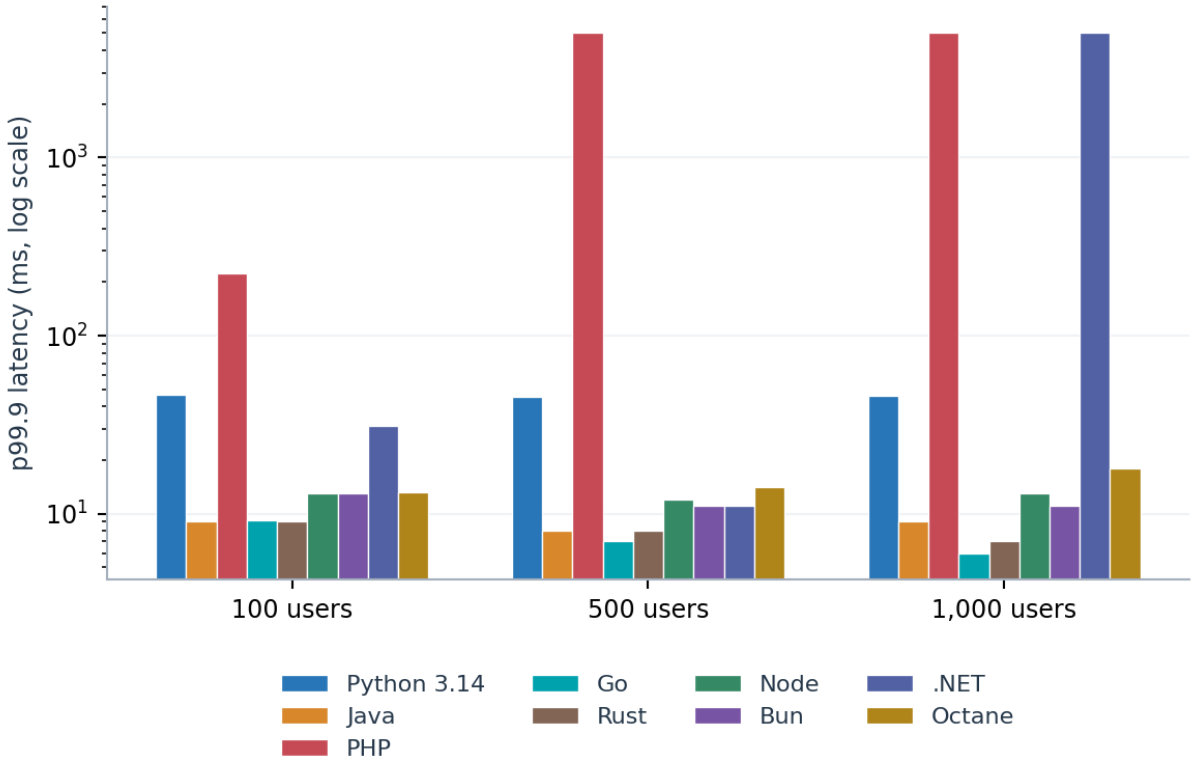
Stack	Users	Mean	p50	p95	p99 (1%)	p99.9 (0.1%)
Python 3.14	100	7.02	4.00	43.00	44.00	46.24
Java	100	3.46	3.00	6.00	8.00	9.00
PHP	100	55.53	48.00	102.00	155.04	224.21
Go	100	3.08	2.00	7.00	8.00	9.15
Rust	100	3.43	3.00	7.00	8.00	9.00
Node	100	4.71	4.00	9.00	11.00	13.07
Bun	100	4.43	3.00	9.00	11.00	13.07
.NET	100	3.98	3.00	8.00	9.00	31.26
Octane	100	6.44	6.00	11.00	12.00	13.09
Python 3.14	500	6.45	4.00	43.00	44.00	45.00
Java	500	2.90	2.00	6.00	7.00	8.00
PHP (FAIL)	500	5,000.66	5,001.00	5,001.00	5,002.00	5,002.00
Go	500	2.53	2.00	5.00	5.00	7.00
Rust	500	2.91	3.00	5.00	7.00	8.00
Node	500	4.00	3.00	8.00	10.00	12.00
Bun	500	3.71	3.00	8.00	9.00	11.00
.NET	500	3.62	3.00	8.00	9.00	11.00
Octane	500	6.18	5.00	10.00	12.00	14.00
Python 3.14	1,000	6.39	4.00	43.00	44.00	46.00
Java	1,000	2.93	2.00	6.00	8.00	9.00
PHP (FAIL)	1,000	5,000.65	5,001.00	5,001.00	5,002.00	5,002.00
Go	1,000	2.42	2.00	4.00	5.00	6.00
Rust	1,000	2.71	2.00	5.00	6.00	7.00
Node	1,000	3.82	3.00	8.00	9.00	13.00
Bun	1,000	3.56	3.00	8.00	9.00	11.00
.NET	1,000	4,505.57	5,001.00	5,001.00	5,002.00	5,033.14
Octane	1,000	6.47	6.00	11.00	13.00	18.00

Each row is a separate 240-second steady population. Python at 1,000 users comment endpoint drives its 44-47 ms tail: comment mean 11.66 ms / p99 45 ms / p99.9 47.08 ms versus feed mean 5.26 ms / p99.9 12.10 ms. This identifies an observed endpoint, not a causal GIL/network/DB diagnosis. Failed PHP and .NET rows include deadline-length requests.

Latency profiles across loads

The logarithmic scale keeps healthy millisecond response times and timeout-heavy failures visible together.





Useful throughput and outcomes

Phase-qualified counts divided by 240 seconds, with exact authentication counts.

Stack	Users	Attempts	Successes	Success %	Auth / users	Useful ops/s
Python 3.14	100	3,942	3,942	100.0000%	100 / 100	16.43
Java	100	3,918	3,918	100.0000%	100 / 100	16.32
PHP	100	3,897	3,897	100.0000%	100 / 100	16.24
Go	100	3,925	3,925	100.0000%	100 / 100	16.35
Rust	100	3,929	3,929	100.0000%	100 / 100	16.37
Node	100	3,931	3,931	100.0000%	100 / 100	16.38
Bun	100	3,926	3,926	100.0000%	100 / 100	16.36
.NET	100	3,935	3,935	100.0000%	100 / 100	16.40
Octane	100	3,910	3,910	100.0000%	100 / 100	16.29
Python 3.14	500	19,612	19,612	100.0000%	500 / 500	81.72
Java	500	19,571	19,571	100.0000%	500 / 500	81.55
PHP (FAIL)	500	10,411	0	0.0000%	211 / 500	0.00
Go	500	19,584	19,584	100.0000%	500 / 500	81.60
Rust	500	19,611	19,611	100.0000%	500 / 500	81.71
Node	500	19,583	19,583	100.0000%	500 / 500	81.60
Bun	500	19,601	19,601	100.0000%	500 / 500	81.67
.NET	500	19,609	19,609	100.0000%	500 / 500	81.70
Octane	500	19,599	19,599	100.0000%	500 / 500	81.66
Python 3.14	1,000	39,275	39,275	100.0000%	1,000 / 1,000	163.65
Java	1,000	39,319	39,319	100.0000%	1,000 / 1,000	163.83
PHP (FAIL)	1,000	20,827	0	0.0000%	141 / 1,000	0.00
Go	1,000	39,313	39,313	100.0000%	1,000 / 1,000	163.80
Rust	1,000	39,324	39,324	100.0000%	1,000 / 1,000	163.85
Node	1,000	39,300	39,298	99.9949%	1,000 / 1,000	163.74
Bun	1,000	39,331	39,330	99.9975%	1,000 / 1,000	163.88
.NET	1,000	21,858	2,176	9.9552%	56 / 1,000	9.07
Octane	1,000	39,281	39,281	100.0000%	1,000 / 1,000	163.67

Node at 1,000 users: 39,298 / 39,300 successful operations (2 like network errors). Bun at 1,000 users: 39,330 / 39,331 (1 like network error). Neither had steady timeouts. All healthy stacks qualify against 90% while remaining near 100%; higher threshold reclassification is not a new saturation experiment.

Requests, bursts and peak windows

Rates in operations/sec. Peaks are windowed observations, not sustainable capacity.

Stack	Users	Attempt mean	Success mean	Success peak 1 s	Success peak 10 s	Attempt peak 1 s
Python 3.14	100	16.43	16.43	28.00	18.40	28.00
Java	100	16.32	16.32	28.00	18.20	28.00
PHP	100	16.24	16.24	28.00	17.70	29.00
Go	100	16.35	16.35	26.00	18.20	26.00
Rust	100	16.37	16.37	26.00	17.80	26.00
Node	100	16.38	16.38	30.00	18.10	30.00
Bun	100	16.36	16.36	29.00	18.30	29.00
.NET	100	16.40	16.40	27.00	18.40	27.00
Octane	100	16.29	16.29	30.00	18.00	30.00
Python 3.14	500	81.72	81.72	113.00	85.00	110.00
Java	500	81.55	81.55	102.00	85.90	102.00
PHP (FAIL)	500	43.38	0.00	0.00	0.00	60.00
Go	500	81.60	81.60	114.00	84.90	115.00
Rust	500	81.71	81.71	110.00	85.10	110.00
Node	500	81.60	81.60	117.00	86.60	114.00
Bun	500	81.67	81.67	104.00	85.90	104.00
.NET	500	81.70	81.70	108.00	85.20	108.00
Octane	500	81.66	81.66	107.00	86.10	105.00
Python 3.14	1,000	163.65	163.65	196.00	169.00	195.00
Java	1,000	163.83	163.83	205.00	169.30	205.00
PHP (FAIL)	1,000	86.78	0.00	0.00	0.00	113.00
Go	1,000	163.80	163.80	198.00	168.10	198.00
Rust	1,000	163.85	163.85	201.00	169.30	201.00
Node	1,000	163.75	163.74	196.00	169.20	196.00
Bun	1,000	163.88	163.88	203.00	168.40	203.00
.NET	1,000	91.08	9.07	19.00	10.50	115.00
Octane	1,000	163.67	163.67	214.00	169.90	215.00

One-second peaks are noisy. The rolling 10-second average is a smoother burst statistic; neither substitutes for scheduled-window mean throughput. End-of-window spillover appears on individual trial sheets. Requests with errors do not become successful work.

Full backend VM CPU distribution

PRIMARY | steady 240 s; all host activity; mean/peak and sparse descriptive sample tails.

Stack	Users	Samples n	Mean %	Peak %	p99 %	p99.9 %
Python 3.14	100	51	6.86	10.27	9.17	10.16
Java	100	52	4.52	10.15	8.06	9.94
PHP	100	52	33.40	41.35	39.77	41.19
Go	100	53	3.33	4.64	4.38	4.61
Rust	100	52	3.29	4.32	4.22	4.31
Node	100	54	4.27	8.32	6.54	8.14
Bun	100	52	4.13	5.64	5.43	5.62
.NET	100	53	4.02	10.45	8.16	10.22
Octane	100	52	5.65	7.00	6.65	6.97
Python 3.14	500	51	12.57	28.94	21.33	28.18
Java	500	53	5.81	32.39	19.27	31.08
PHP (FAIL)	500	30	99.99	100.00	100.00	100.00
Go	500	53	4.78	8.72	7.13	8.56
Rust	500	53	4.80	16.70	10.82	16.11
Node	500	53	7.50	11.85	10.14	11.68
Bun	500	52	6.73	14.87	11.44	14.53
.NET	500	52	7.45	17.59	13.24	17.16
Octane	500	52	15.25	21.94	19.00	21.64
Python 3.14	1,000	50	19.22	47.63	34.22	46.29
Java	1,000	52	6.63	11.58	10.14	11.43
PHP (FAIL)	1,000	31	99.99	100.00	100.00	100.00
Go	1,000	53	6.39	27.77	16.86	26.68
Rust	1,000	53	5.91	25.49	15.74	24.51
Node	1,000	52	11.17	36.15	23.99	34.93
Bun	1,000	53	9.64	44.37	26.81	42.62
.NET	1,000	48	99.65	99.90	99.90	99.90
Octane	1,000	52	26.50	60.17	43.58	58.51

Includes all VM processes, SSH collection, OS, Docker, exporters and steal. Idle+iowait are excluded from busy in the recorded first 8-counter formula. CPU 100% represents non-idle/non-I/O-wait across both VM vCPUs, including steal. Sparsep 99/p99.9 are not reliable rare-event guarantees. No subtraction of container/VM peaks is meaningful.

Full backend VM RAM distribution

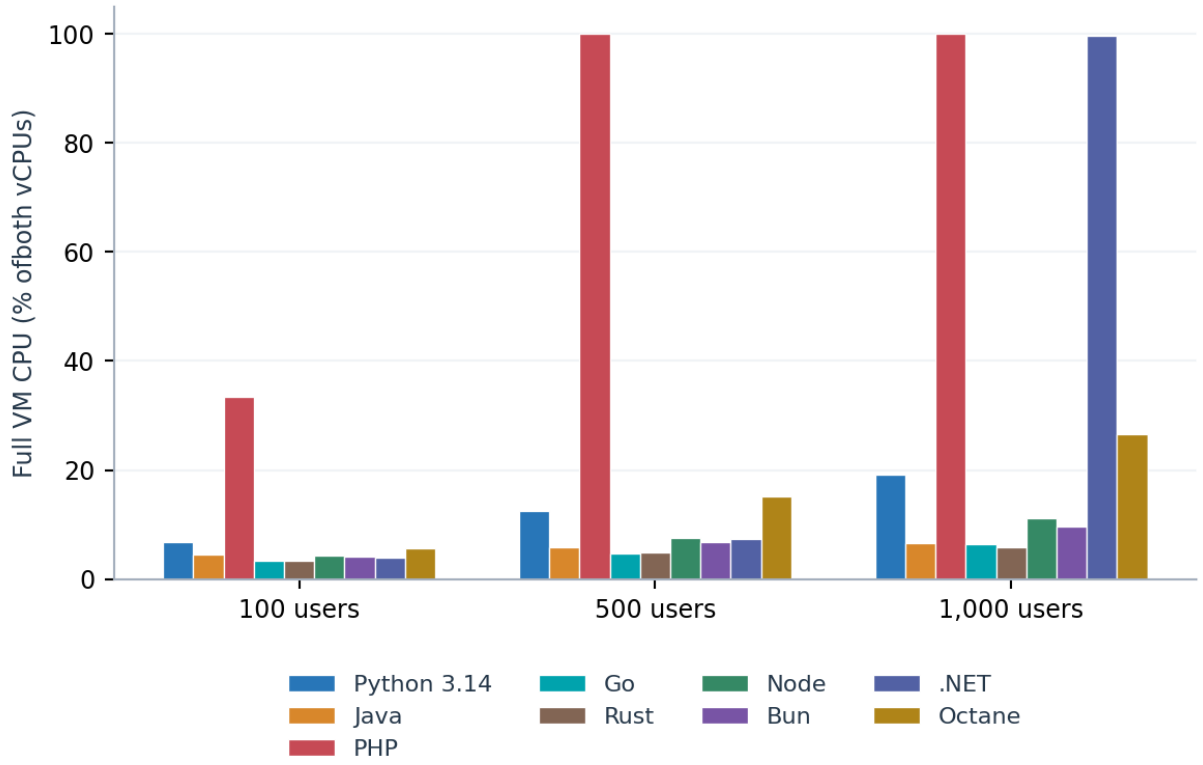
PRIMARY | steady 240 s; used-RAM estimate=MemTotal - MemAvailable; sampled MiB.

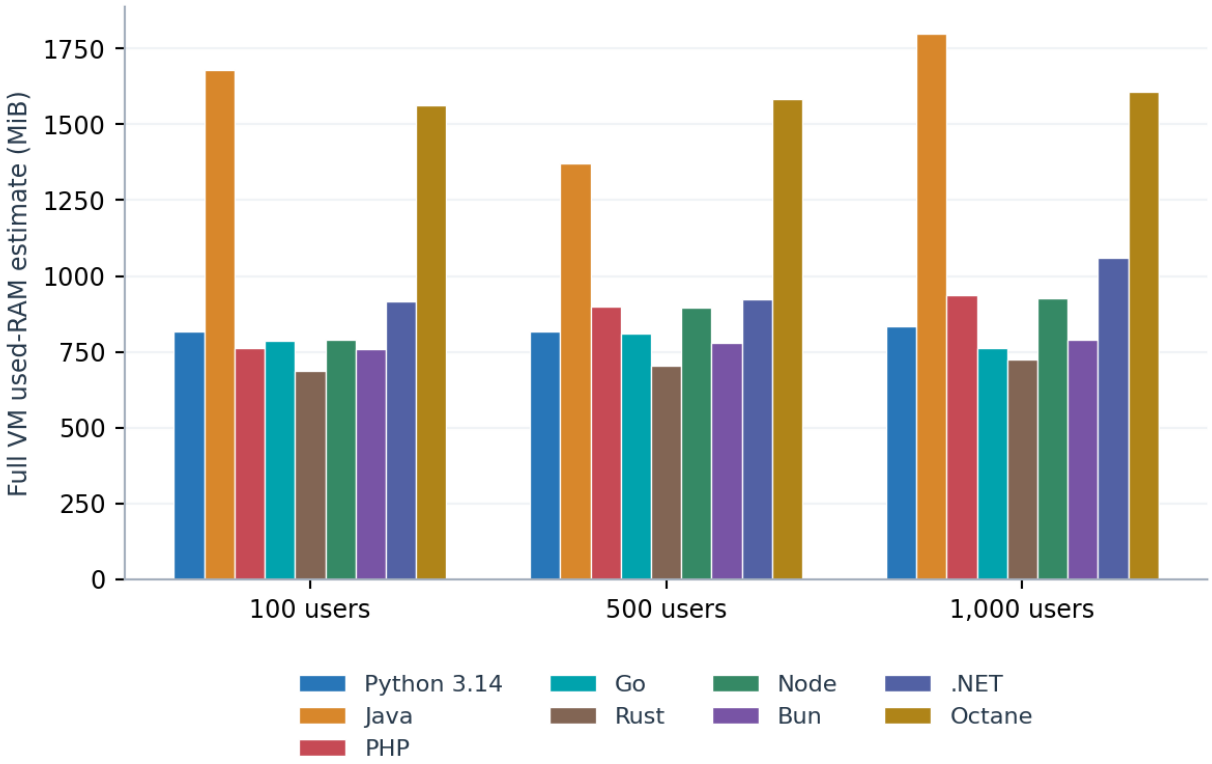
Stack	Users	Samples n	Mean MiB	Peak MiB	p99 MiB	p99.9 MiB
Python 3.14	100	51	817.86	834.54	834.16	834.50
Java	100	52	1,679.98	1,691.50	1,690.79	1,691.43
PHP	100	52	763.78	775.94	775.92	775.94
Go	100	53	785.76	863.45	862.00	863.31
Rust	100	52	688.39	700.03	699.63	699.99
Node	100	54	791.10	815.58	812.64	815.28
Bun	100	52	757.91	782.99	780.87	782.78
.NET	100	53	916.07	930.86	928.05	930.58
Octane	100	52	1,563.03	1,580.23	1,576.24	1,579.83
Python 3.14	500	51	815.65	862.82	854.23	861.96
Java	500	53	1,371.38	1,388.31	1,388.12	1,388.29
PHP (FAIL)	500	30	900.57	947.02	942.51	946.57
Go	500	53	811.27	952.33	952.26	952.32
Rust	500	53	702.37	715.61	712.52	715.30
Node	500	53	895.60	927.70	918.51	926.78
Bun	500	52	779.89	793.42	792.50	793.33
.NET	500	52	922.49	935.25	933.29	935.05
Octane	500	52	1,583.55	1,600.93	1,598.26	1,600.66
Python 3.14	1,000	50	833.02	879.20	868.74	878.15
Java	1,000	52	1,798.60	1,806.21	1,805.11	1,806.10
PHP (FAIL)	1,000	31	936.12	972.77	972.22	972.71
Go	1,000	53	761.87	919.43	904.58	917.94
Rust	1,000	53	723.56	732.75	730.38	732.51
Node	1,000	52	928.03	979.94	973.73	979.32
Bun	1,000	53	790.29	814.23	809.43	813.75
.NET	1,000	48	1,059.86	1,188.52	1,175.28	1,187.19
Octane	1,000	52	1,606.25	1,631.93	1,629.21	1,631.66

Linux observed total≈3553.54 MiB; EC2 advertises 4 GiB. This estimate incorporates whole VM pressure and available reclaimable memory, not just app RSS or all cache. Docker app cap remains 3072 MiB; full VM readings do not mean the app was granted unrestricted 4 GiB.

Full VM resource profiles by load

PRIMARY | sampled steady means; failed runs describe overload rather than efficiency.





Application-container CPU distribution

Steady 240 s only. Percent of the two-vCPU allocation; PHP failed rows describe overload. Ramp/login peaks are in trial sheets/time profiles.

Stack	Users	Samples	Mean %	Peak %	p99 %	p99.9 %
Python 3.14	100	51	2.89	6.62	6.42	6.61
Java	100	52	1.77	7.48	7.39	7.47
PHP	100	52	28.22	49.17	48.32	49.08
Go	100	53	0.32	0.56	0.52	0.56
Rust	100	52	0.30	0.47	0.45	0.47
Node	100	54	1.14	2.02	1.98	2.02
Bun	100	52	1.10	1.99	1.88	1.97
.NET	100	53	0.78	1.15	1.13	1.14
Octane	100	52	2.58	3.96	3.61	3.93
Python 3.14	500	51	6.74	10.60	10.43	10.58
Java	500	53	2.07	7.75	6.90	7.67
PHP (FAIL)	500	30	96.88	97.53	97.52	97.52
Go	500	53	1.27	1.61	1.61	1.61
Rust	500	53	1.06	1.37	1.32	1.36
Node	500	53	3.69	5.84	5.57	5.81
Bun	500	52	2.92	4.91	4.48	4.87
.NET	500	52	3.38	4.34	4.21	4.33
Octane	500	52	10.87	13.64	13.12	13.59
Python 3.14	1,000	50	12.38	18.79	18.40	18.75
Java	1,000	52	2.94	7.88	7.40	7.84
PHP (FAIL)	1,000	31	96.08	97.24	97.19	97.24
Go	1,000	53	2.18	2.68	2.66	2.68
Rust	1,000	53	1.73	1.97	1.96	1.97
Node	1,000	52	6.48	9.62	9.31	9.59
Bun	1,000	53	4.70	6.34	6.32	6.34
.NET	1,000	48	98.06	99.50	99.45	99.50
Octane	1,000	52	21.03	26.65	26.45	26.63

Docker 100% per one logical CPU/vCPU becomes normalized 50% for this 2-CPU allocation. p99.9 with a few dozen samples sits close to the sampled maximum; it is a descriptive interpolated quantile, not a high-confidence rare-event guarantee.

Application-container memory

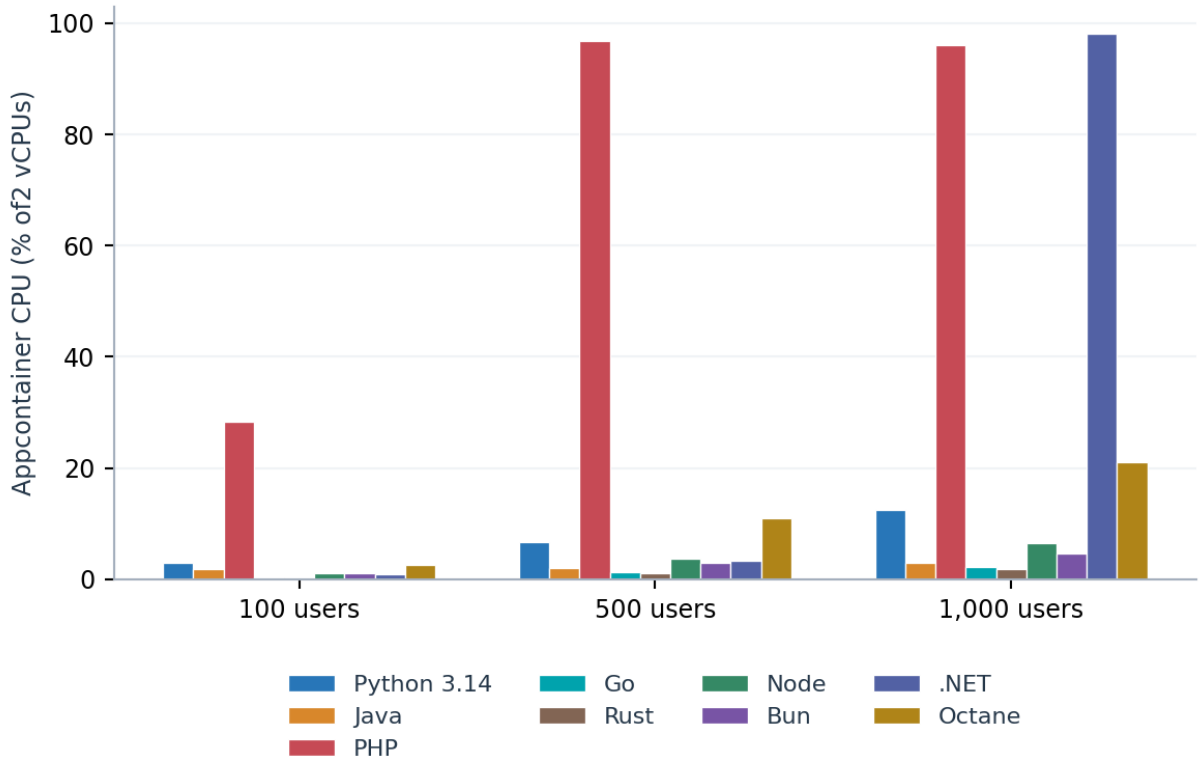
Steady 240 s only; sampled working set MiB, ceiling 3,072. PHP failed rows are overload. Ramp/login peaks are in trial sheets/time profiles.

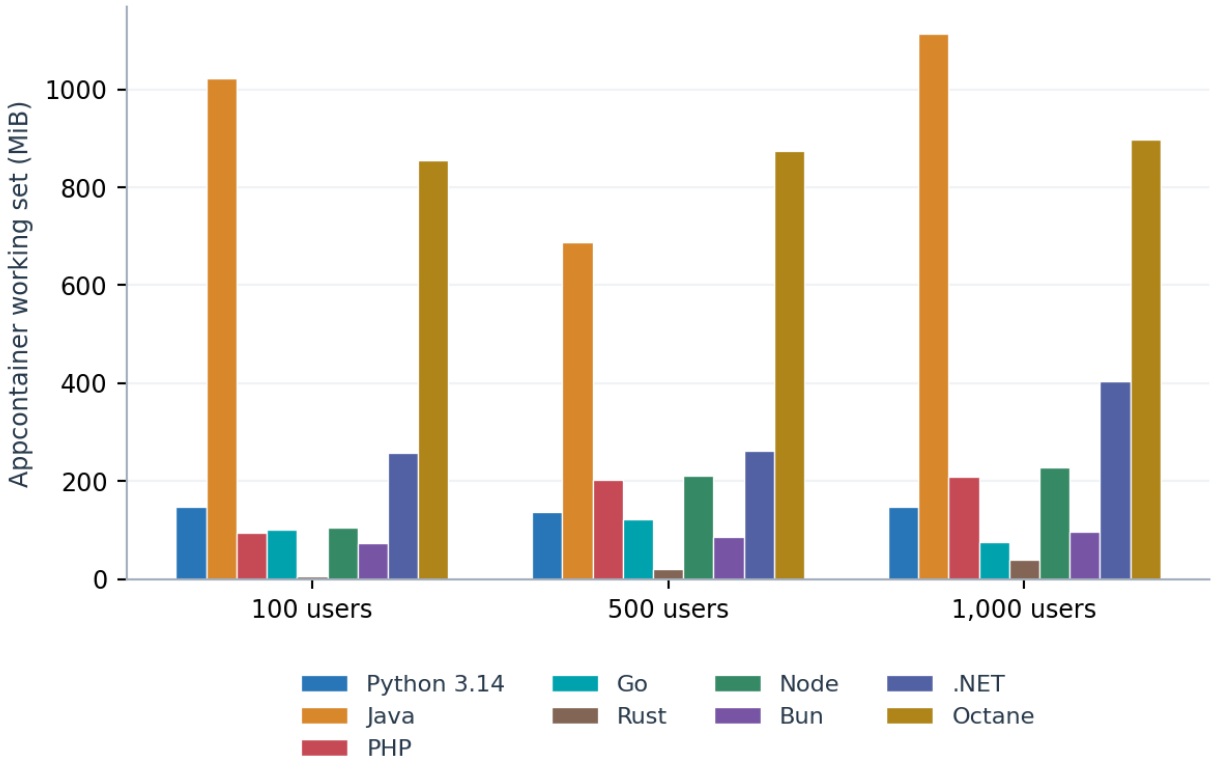
Stack	Users	Samples	Mean MiB	Peak MiB	p99 MiB	p99.9 MiB
Python 3.14	100	51	146.74	155.80	151.30	155.35
Java	100	52	1,021.50	1,022.00	1,022.00	1,022.00
PHP	100	52	94.10	94.48	94.47	94.48
Go	100	53	101.93	175.00	175.00	175.00
Rust	100	52	5.99	6.11	6.11	6.11
Node	100	54	105.09	108.30	107.82	108.25
Bun	100	52	73.27	75.44	75.02	75.40
.NET	100	53	257.05	259.10	259.10	259.10
Octane	100	52	855.48	856.40	856.40	856.40
Python 3.14	500	51	137.89	141.70	141.65	141.69
Java	500	53	688.41	689.60	689.55	689.59
PHP (FAIL)	500	30	203.01	234.20	234.14	234.19
Go	500	53	122.05	271.50	271.45	271.49
Rust	500	53	21.07	21.51	21.45	21.50
Node	500	53	210.23	217.00	216.58	216.96
Bun	500	52	86.56	89.13	88.93	89.11
.NET	500	52	262.32	266.00	265.69	265.97
Octane	500	52	874.84	876.40	876.40	876.40
Python 3.14	1,000	50	147.37	149.50	148.67	149.42
Java	1,000	52	1,113.01	1,113.09	1,113.09	1,113.09
PHP (FAIL)	1,000	31	210.05	241.20	241.17	241.20
Go	1,000	53	75.66	249.80	217.56	246.58
Rust	1,000	53	39.85	40.11	40.09	40.11
Node	1,000	52	228.60	233.00	232.85	232.98
Bun	1,000	53	96.56	99.35	98.99	99.31
.NET	1,000	48	403.22	539.10	538.54	539.04
Octane	1,000	52	896.80	897.40	897.40	897.40

Docker CLI memory values are sampled/rounded working-set readings. They are not exact heap, RSS or guaranteed transient-peak measurements. This is the app container secondary view, not whole VM usage. Cross-runtime footprints depend on topology/GC. Do notsubtract VM/container peaks as overhead.

Container resources: secondary view

Failed runs remain visible with their status; resource efficiency requires useful responses.





Authentication and phase contrast

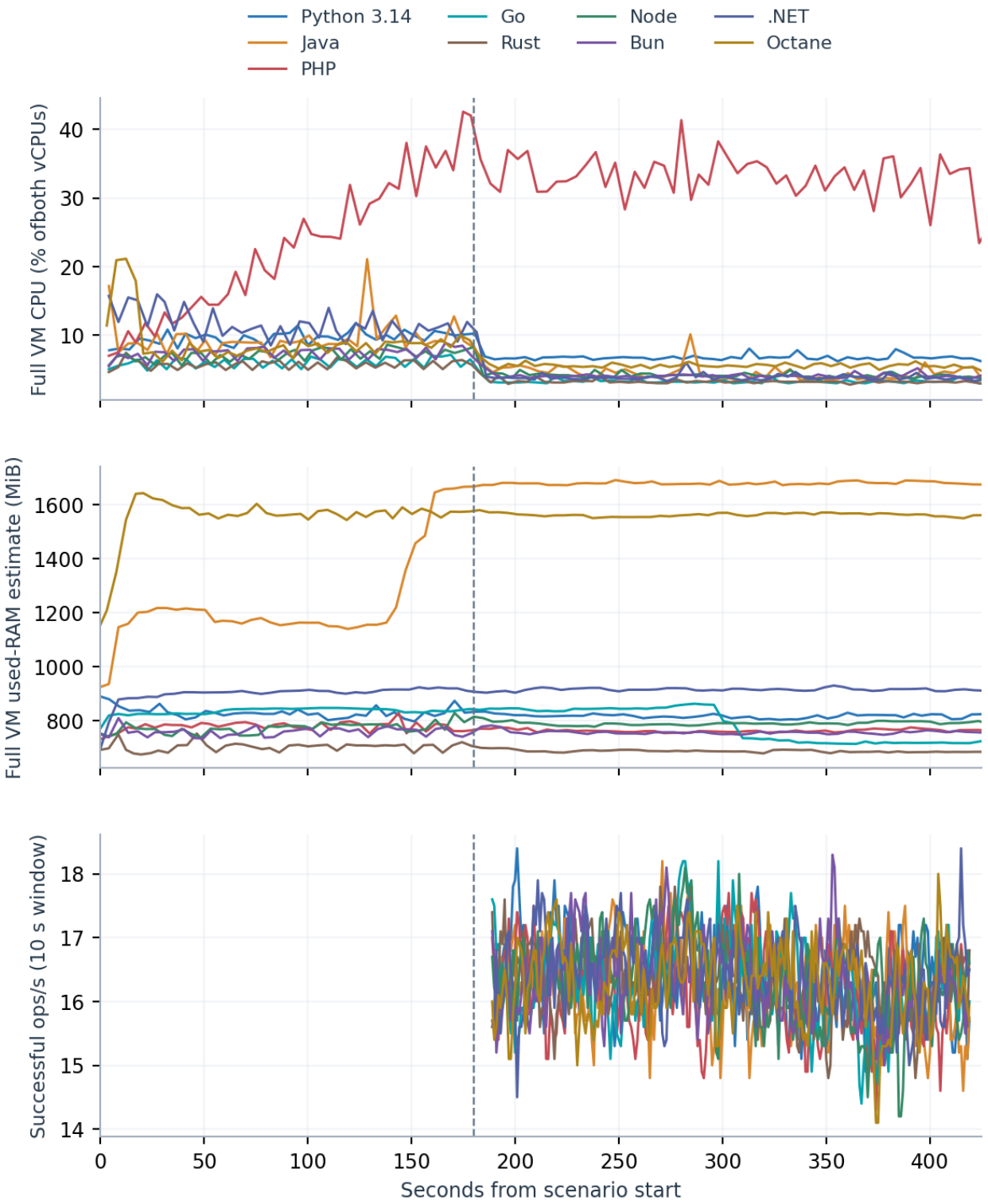
Login work happens during the ramp and can destabilize subsequent activity.

Stack	Users	Auth count	Auth %	RampVM CPU %	SteadyVM CPU %	Timeouts
Python 3.14	100	100	100.00%	9.66	6.86	0
Java	100	100	100.00%	9.61	4.52	0
PHP	100	100	100.00%	22.79	33.40	0
Go	100	100	100.00%	6.04	3.33	0
Rust	100	100	100.00%	5.77	3.29	0
Node	100	100	100.00%	7.14	4.27	0
Bun	100	100	100.00%	7.40	4.13	0
.NET	100	100	100.00%	11.64	4.02	0
Octane	100	100	100.00%	9.12	5.65	0
Python 3.14	500	500	100.00%	27.02	12.57	0
Java	500	500	100.00%	30.36	5.81	0
PHP (FAIL)	500	211	42.20%	81.82	99.99	10,411
Go	500	500	100.00%	19.34	4.78	0
Rust	500	500	100.00%	17.55	4.80	0
Node	500	500	100.00%	22.87	7.50	0
Bun	500	500	100.00%	23.13	6.73	0
.NET	500	500	100.00%	42.41	7.45	0
Octane	500	500	100.00%	29.81	15.25	0
Python 3.14	1,000	1,000	100.00%	49.18	19.22	0
Java	1,000	1,000	100.00%	59.61	6.63	0
PHP (FAIL)	1,000	141	14.10%	96.15	99.99	20,827
Go	1,000	1,000	100.00%	35.53	6.39	0
Rust	1,000	1,000	100.00%	31.05	5.91	0
Node	1,000	1,000	100.00%	41.50	11.17	0
Bun	1,000	1,000	100.00%	42.31	9.64	0
.NET	1,000	56	5.60%	99.32	99.65	19,682
Octane	1,000	1,000	100.00%	57.25	26.50	0

PHP at 500 users authenticated 211/500 (42.2%); PHP at 1,000 users authenticated 141/1000 (14.1%). Its failed startup/activity behavior does not represent already-authenticated capacity. Login latency and phase-specific endpoint detail appear on each trial sheet.

100 users: measured time profiles

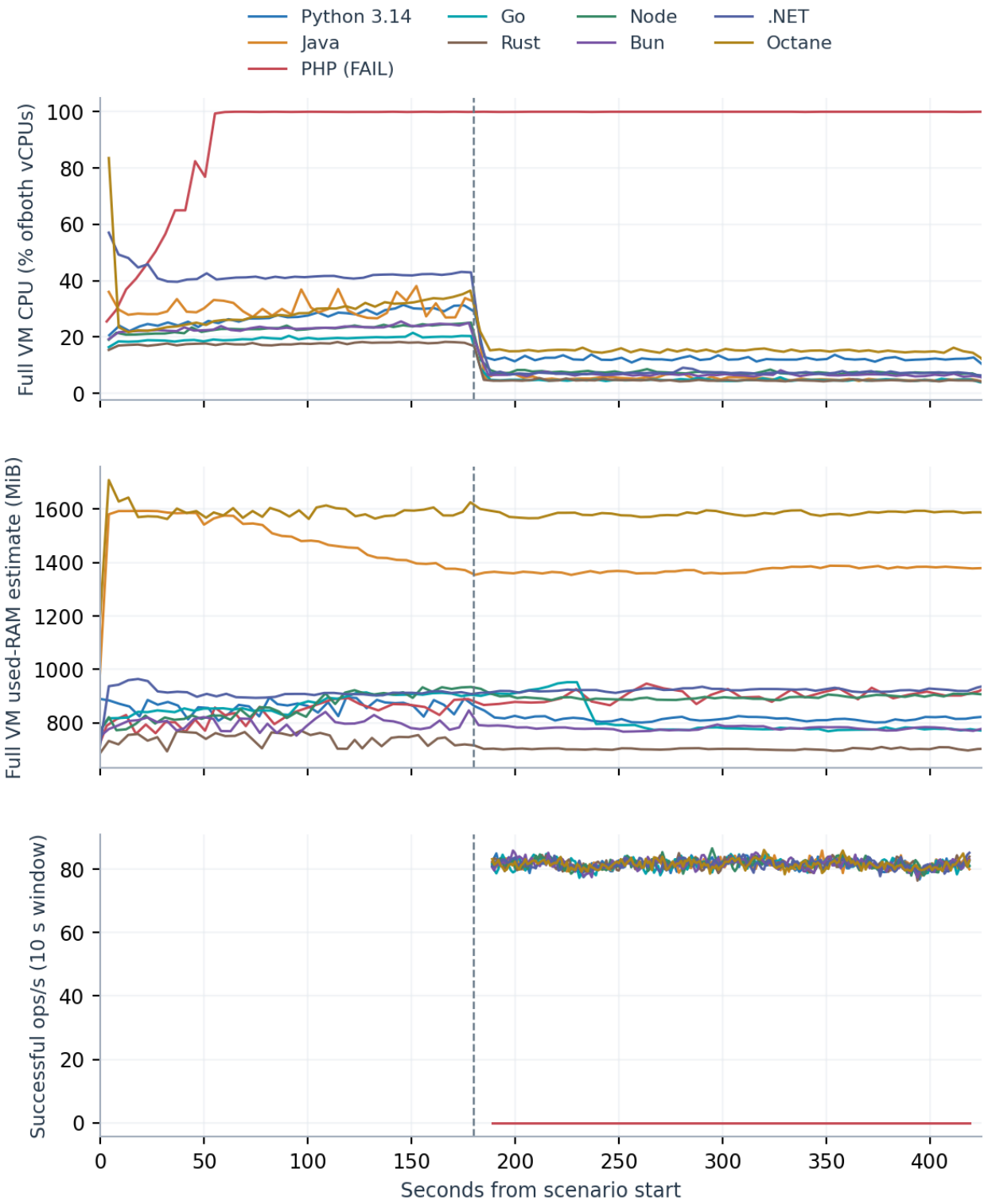
PRIMARYfull VM CPU/RAM; all nine configurations. Time zero=scenario start; dashed line=ramp end 180 s.



Full VM samples include OS/SSH/Docker/exporter activity and may have irregular cadence; RAM is MemTotal - MemAvailable. RPS starts at the steady window; missing/ramp RPS is not rendered as fabricated zero traffic. Failed PHP activity is part of the comparison, not removed.

500 users: measured time profiles

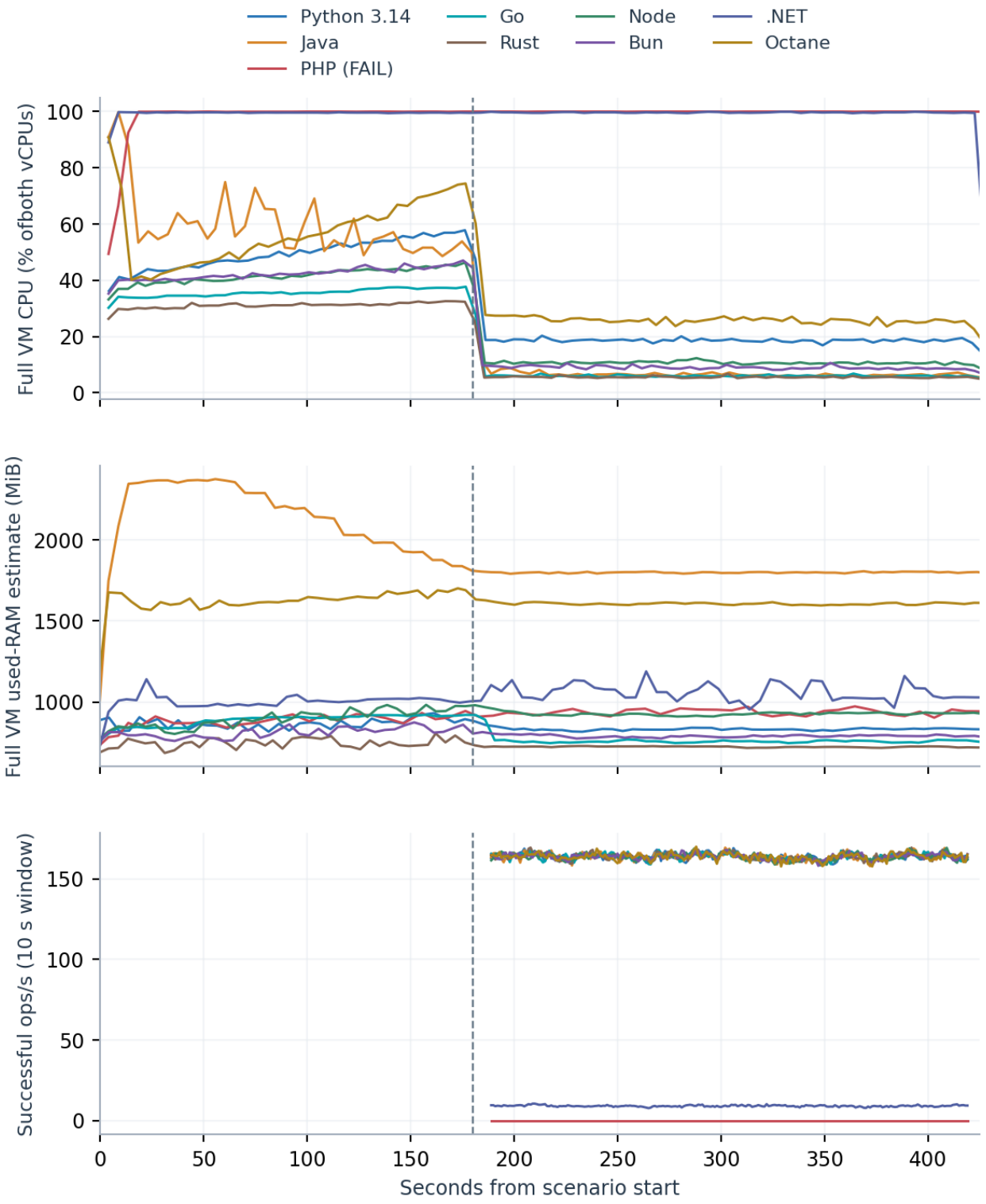
PRIMARYfull VM CPU/RAM; all nine configurations. Time zero=scenario start; dashed line=ramp end 180 s.



Full VM samples include OS/SSH/Docker/exporter activity and may have irregular cadence; RAM is MemTotal - MemAvailable. RPS starts at the steady window; missing/ramp RPS is not rendered as fabricated zero traffic. Failed PHP activity is part of the comparison, not removed.

1,000 users: measured time profiles

PRIMARYfull VM CPU/RAM; all nine configurations. Time zero=scenario start; dashed line=ramp end 180 s.



Full VM samples include OS/SSH/Docker/exporter activity and may have irregular cadence; RAM is MemTotal - MemAvailable. RPS starts at the steady window; missing/ramp RPS is not rendered as fabricated zero traffic. Failed PHP activity is part of the comparison, not removed.

RDS capacity headroom

Independent database allocation; CloudWatch minute-resolution observations.

Stack	Users	CPU avg %	CPU max %	Conn avg	Conn max	Freeable avg GiB
Python 3.14	100	1.57	1.63	6.60	7.00	10.25
Java	100	1.69	1.99	5.00	5.00	10.26
PHP	100	4.77	4.95	1.60	4.00	10.25
Go	100	1.82	1.99	3.20	4.00	10.24
Rust	100	1.57	1.62	4.00	4.00	10.24
Node	100	1.94	2.04	5.20	6.00	10.24
Bun	100	1.95	2.03	5.40	6.00	10.24
.NET	100	2.08	2.20	3.20	4.00	10.26
Octane	100	2.11	2.14	21.00	21.00	10.21
Python 3.14	500	3.11	3.24	9.00	9.00	10.25
Java	500	3.60	4.00	7.00	7.00	10.25
PHP (FAIL)	500	3.88	4.81	5.40	14.00	10.24
Go	500	3.07	3.63	5.80	6.00	10.23
Rust	500	3.49	3.51	5.00	5.00	10.24
Node	500	4.80	4.88	7.00	8.00	10.23
Bun	500	4.65	4.92	8.20	10.00	10.23
.NET	500	5.34	5.41	5.60	6.00	10.25
Octane	500	5.45	5.72	21.00	21.00	10.21
Python 3.14	1,000	5.04	5.25	12.00	12.00	10.24
Java	1,000	6.72	7.50	21.00	21.00	10.21
PHP (FAIL)	1,000	2.97	3.19	2.00	3.00	10.25
Go	1,000	4.86	4.90	7.40	8.00	10.23
Rust	1,000	5.69	5.84	6.40	7.00	10.23
Node	1,000	8.30	8.54	9.60	12.00	10.22
Bun	1,000	8.10	8.66	9.80	12.00	10.22
.NET	1,000	2.49	2.60	21.00	21.00	10.21
Octane	1,000	9.55	9.85	21.00	21.00	10.21

FreeableMemory is memory headroom, not an exact amount used by the application. All 27 RDS queries use the 240-second steady interval, rounded to minute boundaries;4-5 reported buckets can overlap ramp or drain edges. Do not interpret them as exact second-by-second backend samples. Low average DB CPU does not exclude transient lock/pool/connection waits.

RDS I/O and response times

Averages from the 240-second steady anchor, rounded to CloudWatch minute buckets; read/write latency in ms.

Stack	Users	Read IOPS	Write IOPS	Read ms	Write ms	Queue depth
Python 3.14	100	0.75	20.64	0.200	1.975	0.0437
Java	100	0.52	19.40	0.240	1.199	0.0250
PHP	100	0.26	19.52	0.080	1.224	0.0245
Go	100	0.26	14.89	0.080	0.930	0.0174
Rust	100	0.26	18.51	0.080	1.106	0.0231
Node	100	0.26	18.74	0.160	1.217	0.0239
Bun	100	0.00	19.19	0.000	1.174	0.0243
.NET	100	0.26	19.10	0.160	1.233	0.0244
Octane	100	0.26	20.17	0.160	1.160	0.0255
Python 3.14	500	0.26	58.21	0.160	1.020	0.0611
Java	500	0.26	59.10	0.080	1.044	0.0625
PHP (FAIL)	500	0.26	9.82	0.160	1.342	0.0148
Go	500	0.26	58.45	0.080	1.053	0.0620
Rust	500	0.26	60.28	0.160	1.060	0.0644
Node	500	0.26	61.28	0.160	1.064	0.0661
Bun	500	0.26	57.35	0.160	1.006	0.0599
.NET	500	0.26	60.87	0.160	1.083	0.0662
Octane	500	0.26	58.95	0.000	1.018	0.0629
Python 3.14	1,000	0.26	101.03	0.160	0.935	0.0974
Java	1,000	0.26	106.85	0.160	0.993	0.1048
PHP (FAIL)	1,000	0.26	6.53	0.080	0.893	0.0108
Go	1,000	0.26	109.27	0.160	0.999	0.1086
Rust	1,000	0.26	104.55	0.080	0.989	0.1025
Node	1,000	0.26	107.09	0.080	0.966	0.1052
Bun	1,000	0.26	99.87	0.080	0.966	0.0993
.NET	1,000	0.26	14.30	0.160	1.129	0.0190
Octane	1,000	0.26	110.70	0.080	0.971	0.1076

These are database-service metrics and do not replace end-to-end response latency. No statistical p99.9 claim is meaningful with a few minute buckets; detailed exports retain bucket counts and descriptive quantiles.

RDS throughput and generator checks

Data traffic in KiB/s; generator CPU is percent of its own host, not the backend allocation.

Stack	Users	Read KiB/s	Write KiB/s	GenCPU avg %	GenCPU peak %	Gen avail MiB min
Python 3.14	100	0.40	1,196.85	0.18	0.22	61,141
Java	100	0.27	1,021.60	0.16	0.22	61,498
PHP	100	0.13	1,176.44	0.16	0.24	61,509
Go	100	0.13	567.71	0.16	0.21	61,499
Rust	100	0.13	814.03	0.15	0.22	61,507
Node	100	0.13	1,026.02	0.17	0.24	61,494
Bun	100	0.00	1,029.99	0.17	0.24	61,501
.NET	100	0.13	1,026.43	0.16	0.22	61,493
Octane	100	0.13	1,208.02	0.16	0.26	61,426
Python 3.14	500	0.13	1,362.48	0.37	0.50	61,223
Java	500	0.13	1,369.60	0.29	0.43	61,339
PHP (FAIL)	500	0.13	932.92	0.22	0.30	61,352
Go	500	0.13	1,523.06	0.29	0.40	61,332
Rust	500	0.13	1,539.88	0.29	0.42	61,342
Node	500	0.13	1,546.10	0.34	0.51	61,337
Bun	500	0.13	1,345.31	0.34	0.48	61,324
.NET	500	0.13	1,552.81	0.29	0.44	61,314
Octane	500	0.13	1,356.79	0.28	0.43	61,262
Python 3.14	1,000	0.13	1,278.59	0.59	0.83	61,093
Java	1,000	0.13	1,556.38	0.46	0.76	61,124
PHP (FAIL)	1,000	0.13	709.92	0.34	0.49	61,152
Go	1,000	0.13	1,776.39	0.46	0.77	61,116
Rust	1,000	0.13	1,519.41	0.45	0.72	61,125
Node	1,000	0.13	1,543.07	0.53	0.75	61,129
Bun	1,000	0.13	1,395.16	0.54	0.76	61,123
.NET	1,000	0.13	974.73	0.48	2.08	61,133
Octane	1,000	0.13	1,395.31	0.43	0.70	61,060

Generator and database have independent budgets. Data quality flags and PostgreSQL wait summaries are included in each detailed sheet/export. Missing telemetry must never be interpreted as zero usage.

RDS CPU: sparse bucket quantiles

Percent of the independent RDS allocation; distribution of minute-AVERAGE buckets.

Stack	Users	Buckets n	Mean %	Peak Maximum %	p99 avg %	p99.9 avg %
Python 3.14	100	4	1.57	1.63	1.63	1.63
Java	100	4	1.69	1.99	1.98	1.99
PHP	100	5	4.77	4.95	4.95	4.95
Go	100	5	1.82	1.99	1.98	1.99
Rust	100	4	1.57	1.62	1.62	1.62
Node	100	5	1.94	2.04	2.04	2.04
Bun	100	4	1.95	2.03	2.03	2.03
.NET	100	4	2.08	2.20	2.20	2.20
Octane	100	4	2.11	2.14	2.14	2.14
Python 3.14	500	5	3.11	3.24	3.24	3.24
Java	500	4	3.60	4.00	4.00	4.00
PHP (FAIL)	500	5	3.88	4.81	4.79	4.81
Go	500	5	3.07	3.63	3.62	3.63
Rust	500	4	3.49	3.51	3.51	3.51
Node	500	4	4.80	4.88	4.88	4.88
Bun	500	5	4.65	4.92	4.91	4.92
.NET	500	4	5.34	5.41	5.41	5.41
Octane	500	4	5.45	5.72	5.72	5.72
Python 3.14	1,000	5	5.04	5.25	5.25	5.25
Java	1,000	5	6.72	7.50	7.49	7.50
PHP (FAIL)	1,000	5	2.97	3.19	3.19	3.19
Go	1,000	4	4.86	4.90	4.90	4.90
Rust	1,000	5	5.69	5.84	5.83	5.84
Node	1,000	4	8.30	8.54	8.54	8.54
Bun	1,000	5	8.10	8.66	8.66	8.66
.NET	1,000	4	2.49	2.60	2.60	2.60
Octane	1,000	4	9.55	9.85	9.85	9.85

Only 4-5 buckets per run. p99/p99.9 interpolate minute-AVERAGE samples and sit close to their largest observation; they do not estimate an operational rare-event tail. Peak is the largest reported Maximum bucket. All queries are anchored to steady 240 s with rounded minute-boundary overlap.

RDS freeable memory: headroom

GiB. Not used RAM; lowest/highest values are minute-AVERAGE bucket observations.

Stack	Users	Buckets n	Mean GiB	Lowest avg GiB	Highest avg GiB	p99 avg GiB	p99.9 avg GiB
Python 3.14	100	5	10.252	10.250	10.254	10.254	10.254
Java	100	5	10.258	10.257	10.260	10.259	10.260
PHP	100	5	10.246	10.245	10.249	10.249	10.249
Go	100	5	10.240	10.237	10.242	10.242	10.242
Rust	100	5	10.238	10.237	10.239	10.239	10.239
Node	100	5	10.236	10.235	10.237	10.237	10.237
Bun	100	5	10.235	10.232	10.237	10.237	10.237
.NET	100	5	10.261	10.259	10.263	10.263	10.263
Octane	100	5	10.209	10.208	10.213	10.212	10.213
Python 3.14	500	5	10.246	10.243	10.248	10.248	10.248
Java	500	5	10.252	10.250	10.255	10.255	10.255
PHP (FAIL)	500	5	10.242	10.229	10.248	10.248	10.248
Go	500	5	10.233	10.231	10.236	10.236	10.236
Rust	500	5	10.236	10.234	10.237	10.237	10.237
Node	500	5	10.230	10.228	10.232	10.232	10.232
Bun	500	5	10.227	10.219	10.230	10.230	10.230
.NET	500	5	10.253	10.251	10.256	10.256	10.256
Octane	500	5	10.210	10.208	10.212	10.212	10.212
Python 3.14	1,000	5	10.236	10.233	10.238	10.238	10.238
Java	1,000	5	10.213	10.212	10.215	10.215	10.215
PHP (FAIL)	1,000	5	10.246	10.243	10.248	10.248	10.248
Go	1,000	5	10.229	10.225	10.236	10.235	10.235
Rust	1,000	5	10.229	10.227	10.233	10.233	10.233
Node	1,000	5	10.223	10.218	10.226	10.226	10.226
Bun	1,000	5	10.222	10.216	10.225	10.225	10.225
.NET	1,000	5	10.211	10.210	10.213	10.213	10.213
Octane	1,000	5	10.210	10.208	10.213	10.213	10.213

AWS Minimum statistics were not collected: lowest recorded minute-average freeable memory is not an instantaneous minimum. Sparse 4-5-bucket percentiles are descriptive only. db.m7i.xlarge has 16 GiB nominal RAM; subtracting freeable from 16 GiB does not yield exact application-used memory.

PostgreSQL waits and transaction deltas

Steady collector samples. Wait-event counts are session gauges, not request latencies.

Stack	Users	Client mean	Active mean	Lock peak	IO peak	Commits delta
Python 3.14	100	6.61	1.00	n/a	n/a	4,043
Java	100	4.98	1.00	n/a	1.00	4,242
PHP	100	1.00	1.04	n/a	n/a	11,141
Go	100	3.36	1.02	n/a	n/a	4,034
Rust	100	3.96	1.04	n/a	n/a	3,943
Node	100	5.07	1.06	n/a	n/a	4,034
Bun	100	5.17	1.06	n/a	1.00	3,957
.NET	100	3.68	1.02	n/a	n/a	8,074
Octane	100	20.96	1.02	n/a	1.00	7,309
Python 3.14	500	8.88	1.08	n/a	1.00	19,434
Java	500	6.79	1.13	n/a	1.00	20,688
PHP (FAIL)	500	1.53	1.03	n/a	n/a	9,884
Go	500	5.83	1.04	n/a	1.00	19,471
Rust	500	4.79	1.19	n/a	1.00	19,571
Node	500	6.92	1.21	n/a	1.00	19,119
Bun	500	7.54	1.13	n/a	1.00	19,230
.NET	500	5.44	1.13	n/a	n/a	39,618
Octane	500	20.87	1.10	n/a	1.00	36,348
Python 3.14	1,000	11.64	1.22	n/a	1.00	38,651
Java	1,000	20.63	1.21	n/a	1.00	41,057
PHP (FAIL)	1,000	1.39	1.00	n/a	n/a	7,856
Go	1,000	7.04	1.19	n/a	1.00	39,061
Rust	1,000	6.15	1.19	n/a	1.00	39,053
Node	1,000	9.02	1.17	n/a	1.00	38,789
Bun	1,000	9.96	1.32	n/a	1.00	39,076
.NET	1,000	20.52	1.06	n/a	n/a	42,709
Octane	1,000	20.52	1.37	n/a	1.00	73,755

Absent event groups are unavailable/not observed in the exported summary, not proof that all waits were absent. Client includes idle network/client waits; active/no-wait does not mean a blocked query. Transaction/cache counters are database-wide deltas across sampled phase boundaries and include monitoring traffic.

Historical Python capacity

Legacy 300/120/300,99% gates. Recorded endpoints require review; they are not definitive ceilings.

Configuration	Pass endpoint	Fail endpoint	Recorded status	Later review
Baseline 3.10.22	2,031	2,109	bracket found	0 FD flags
Baseline 3.11.17	2,187	2,265	unstable or invalid boundary	0 FD flags
Baseline 3.12.13	2,148	2,206	unstable boundary	1 FD flags
Baseline 3.13.16	2,118	2,206	unstable boundary	3 FD flags
Baseline 3.14.8	2,118	2,206	bracket found	2 FD flags
Baseline 3.15.0rc3	2,118	2,206	bracket found	0 FD flags
Modern standard-jit0-gil1-async	2,206	2,306	bracket found	0 FD flags
Modern jit-jit0-gil1-async	2,206	2,306	bracket found	0 FD flags
Modern jit-jit1-gil1-async	2,206	2,306	bracket found	0 FD flags
Modern free-threaded-jit0-gil1-async	2,137	2,206	bracket found	0 FD flags
Modern free-threaded-jit0-gil0-async	2,206	2,306	bracket found	0 FD flags
Modern free-threaded-jit0-gil1-threaded	1,240	1,309	bracket found	1 FD flags
Modern free-threaded-jit0-gil0-threaded	1,309	1,409	bracket found	0 FD flags

Python 3.15.0rc3 is prerelease; modern dependencies include Pydantic 2.14.0b2. Some repeat results were unstable. Seven Python trials have later file-descriptor/open-file validity flags; authoritative ceilings cannot be inferred from a bracket containing invalid or contradictory observations. Old missing steady means/p99.9 remain unavailable, not replaced by all-phase averages.

Historical native capacity

Starting hints were remeasured; all six native implementations used 2 vCPUs.

Stack	Pass endpoint	Fail endpoint	Recorded status	Review caveat
Java	1,937	2,025	bracket found	0 FD flags
PHP	302	363	bracket found	0 FD flags
Go	3,176	3,267	bracket found	0 FD flags
Rust	992	1,091	bracket found	5 FD flags
Node	2,795	2,863	bracket found	0 FD flags
Bun	2,863	2,952	bracket found	0 FD flags

Why these are provisional

Five Rust trials were later identified with open-file exhaustion. Those observations cannot establish Rust capacity even if the original controller marked a bracket complete. The later fixed-load suite raised and verified nofile 250000. Different timing/gates mean the new fixed-load runs are not confirmations of old 99%-capacity brackets.

No universal ordering

Login ramp, Argon2 implementation, application connection/worker behavior, token reuse, think times and DB/cache state shape these results. A limit for login plus activity is not a limit for already-authenticated users; a pool/FD bottleneck is not a property of the language alone.

What to do with the evidence

Retain raw observations and flags. Repair known infrastructure constraints, then run a fresh capacity campaign with fixed methodology and adequate repeatability. Existing unstable/invalid historical bounds remain visible rather than rewritten as definitive rankings.

Arrival-rate pilot and other archives

These trials answer different questions and are kept out of user-capacity comparisons.

Experiment directory	Trial manifests	Role
cloud-smoke	1	Preflight / smoke / invalid archive
fixed-load-20261006	21	Fixed-load comparison
native-capacity	54	Historical capacity
native-capacity-preflight-1791217587	1	Preflight / smoke / invalid archive
pilot	48	Arrival-rate pilot (RPS)
user-capacity	102	Historical capacity
user-capacity-invalid-20261004	1	Preflight / smoke / invalid archive

The 48-trial pilot actually used 30-second warmup and 30-second measurement, open arrival rates, token preparation for non-login operations and separate qualification rules. The cloud-smoke run used 30-second warmup and 60-second measurement; planned 120/300 defaults are not the recorded pilot protocol. Its offered operations/sec must never be renamed concurrent users. Cloud smoke proves workflow; a failed native preflight has no measured server capacity; the archived invalid campaign contains constrained evidence, not a usable limit.

Historical metric availability

Only bounded saved summaries and telemetry/RDS summaries were used for old trials. No 13 GiB rescan of historical raw files was performed. All-phase legacy mean latency is a distinct field and is not substituted for fixed-load steady means. Missing p99.9, exact burst rates or resources stay unavailable.

Interpretation and next experiments

A practical reading of this snapshot, with conclusions scaled to its evidence.

1. Fixed-load responsiveness

At 100/500/1000, seven configurations delivered the paced workload with near-perfect success. Compare latency and backend resources at the same population rather than declaring a throughput champion.

2. PHP behavior needs engineering diagnosis

At 500/1000 the recorded authentication fractions and zero steady success are clear failures of this deployment/workload. Full VM non-idle/non-I/O-wait CPU was about 99.99%mean and 100%sampled peak at 500/1000. Estimatednon-available RAM was about 901/936 MiB mean (947/973MiBpeak) against observed MemTotal 3554 MiB. Secondaryapp containerCPU was 96.88%/96.08%; RDSCPU averaged 3.88%/2.97% with about 10.24 GiB freeable headroom. This supports CPU/application pressure, not RAM or RDS-CPU exhaustion. It does not exclude locks/IO or assign an Argon2/connection-specific cause without profiling.

3. Tail measurements need repetitions

Single trials and sparse resource samples support descriptive contrasts. Repeat randomized trials before attaching confidence intervals, p-values or operational tail promises.

4. Repair constrained capacity evidence

Exclude file-descriptor-constrained observations as ceilings. Re-run capacity with verified nofile 250000, identical profile/gates and calibrated generator; measure hints anew.

5. Separate already-authenticated demand

A separately designed preauthenticated/longer-ramp scenario can answer a different operational question. It must retain its own identity and must not be pooled with this login-and-activity experiment.

Detailed 27-trial sheets follow. The exhaustive inventory and complete metric JSON/CSV preserve additional fields, definitions and provenance without squeezing hundreds of columns into unreadable print tables.

Python 3.14 / 100 users

Detailed fixed-load observation | PASS | trial e02ce1700d3a-0000

Attempts / successful	Operation success	Authenticated	Timeout / network
3,942 / 3,942	100.0000%	100 / 100	0 / 0

Mean ms	p50 ms	p95 ms	p99 ms	p99.9 ms
7.02	4.00	43.00	44.00	46.24

VM phase PRIMARY	VM CPU mean/peak %	VM RAM est. mean/peak MiB	n
ramp	9.66 / 11.81	829.36 / 880.00	38
steady	6.86 / 10.27	817.86 / 834.54	51

Container SECONDARY	CPU mean/peak %	Working set mean/peak MiB
ramp	4.66 / 13.51	150.50 / 209.70
steady	2.89 / 6.62	146.74 / 155.80

Steady endpoint	Attempts	Success	Mean ms	p95 ms	p99 ms	p99.9 ms
login	1	1	120.00	120.00	120.00	120.00
comment	1,196	1,196	11.81	44.00	44.00	50.81
feed	1,201	1,201	6.65	9.00	10.00	11.80
comments	233	233	3.09	4.00	4.00	6.00
unlike	325	325	3.51	4.00	5.00	7.00
like	867	867	3.57	4.00	5.00	11.02
post	119	119	4.20	5.00	5.00	5.00

Ramp login: 99 / 99 successful; mean 126.66 ms; p99 164.30 ms; p99.9 177.53 ms.

Rate definition	Attempted	Successful	HTTP completed
Mean / configured 240 s	16.43	16.43	16.43
Peak 1 s inside window	28.00	28.00	28.00
Peak rolling 10 s	18.40	18.40	18.40

Window spillover counts: {'early': {'attempted': 0, 'successful': 0, 'http_completed': 0}, 'late': {'attempted': 0, 'successful': 0, 'http_completed': 0}}. Completed journeys: 1,200. Generator resource errors: 0.

RDS average / maximum: CPU 1.57 / 1.63%; connections 6.60 / 7.00; freeable 10.25 GiB; read/write 0.75 / 20.64 IOPS.

Validity: valid-recorded-evidence. Flags: none recorded. Fixture restored: True.

Java / 100 users

Detailed fixed-load observation | PASS | trial e02ce1700d3a-0003

Attempts / successful	Operation success	Authenticated	Timeout / network
3,918 / 3,918	100.0000%	100 / 100	0 / 0

Mean ms	p50 ms	p95 ms	p99 ms	p99.9 ms
3.46	3.00	6.00	8.00	9.00

VM phase PRIMARY	VM CPU mean/peak %	VM RAM est. mean/peak MiB	n
ramp	9.61 / 21.10	1,251.34 / 1,667.32	39
steady	4.52 / 10.15	1,679.98 / 1,691.50	52

Container SECONDARY	CPU mean/peak %	Working set mean/peak MiB
ramp	6.15 / 20.96	599.62 / 1,020.00
steady	1.77 / 7.48	1,021.50 / 1,022.00

Steady endpoint	Attempts	Success	Mean ms	p95 ms	p99 ms	p99.9 ms
login	1	1	169.00	169.00	169.00	169.00
feed	1,188	1,188	5.15	7.00	8.00	9.00
like	852	852	2.65	3.00	4.00	5.00
comment	1,192	1,192	2.84	4.00	4.00	8.43
post	116	116	2.92	4.00	4.00	4.00
comments	228	228	1.81	3.00	3.00	5.55
unlike	341	341	2.59	3.00	4.00	4.00

Ramp login: 99 / 99 successful; mean 145.62 ms; p99 338.70 ms; p99.9 457.77 ms.

Rate definition	Attempted	Successful	HTTP completed
Mean / configured 240 s	16.32	16.32	16.32
Peak 1 s inside window	28.00	28.00	28.00
Peak rolling 10 s	18.10	18.20	18.20

Window spillover counts: {'early': {'attempted': 0, 'successful': 0, 'http_completed': 0}, 'late': {'attempted': 0, 'successful': 0, 'http_completed': 0}}. Completed journeys: 1,187. Generator resource errors: 0.

RDS average / maximum: CPU 1.69 / 1.99%; connections 5.00 / 5.00; freeable 10.26 GiB; read/write 0.52 / 19.40 IOPS.

Validity: valid-recorded-evidence. Flags: none recorded. Fixture restored: True.

PHP / 100 users

Detailed fixed-load observation | PASS | trial e02ce1700d3a-0006

Attempts / successful	Operation success	Authenticated	Timeout / network
3,897 / 3,897	100.0000%	100 / 100	0 / 0

Mean ms	p50 ms	p95 ms	p99 ms	p99.9 ms
55.53	48.00	102.00	155.04	224.21

VM phase PRIMARY	VM CPU mean/peak %	VM RAM est. mean/peak MiB	n
ramp	22.79 / 42.57	780.06 / 829.67	40
steady	33.40 / 41.35	763.78 / 775.94	52

Container SECONDARY	CPU mean/peak %	Working set mean/peak MiB
ramp	17.80 / 33.80	98.11 / 157.50
steady	28.22 / 49.17	94.10 / 94.48

Steady endpoint	Attempts	Success	Mean ms	p95 ms	p99 ms	p99.9 ms
login	1	1	162.00	162.00	162.00	162.00
feed	1,196	1,196	58.47	106.00	171.10	230.83
comment	1,175	1,175	55.39	103.00	153.52	219.13
like	867	867	53.67	97.00	140.02	177.67
comments	221	221	51.71	93.00	152.40	175.72
post	113	113	49.81	84.40	99.16	106.22
unlike	324	324	54.44	107.40	153.00	194.09

Ramp login: 99 / 99 successful; mean 188.26 ms; p99 324.32 ms; p99.9 514.83 ms.

Rate definition	Attempted	Successful	HTTP completed
Mean / configured 240 s	16.24	16.24	16.24
Peak 1 s inside window	29.00	28.00	28.00
Peak rolling 10 s	17.90	17.70	17.70

Window spillover counts: {'early': {'attempted': 0, 'successful': 0, 'http_completed': 0}, 'late': {'attempted': 0, 'successful': 1.0, 'http_completed': 1.0}}. Completed journeys: 1,195. Generator resource errors: 0.

RDS average / maximum: CPU 4.77 / 4.95%; connections 1.60 / 4.00; freeable 10.25 GiB; read/write 0.26 / 19.52 IOPS.

Validity: valid-recorded-evidence. Flags: none recorded. Fixture restored: True.

Go / 100 users

Detailed fixed-load observation | PASS | trial e02ce1700d3a-0009

Attempts / successful	Operation success	Authenticated	Timeout / network
3,925 / 3,925	100.0000%	100 / 100	0 / 0

Mean ms	p50 ms	p95 ms	p99 ms	p99.9 ms
3.08	2.00	7.00	8.00	9.15

VM phase PRIMARY	VM CPU mean/peak %	VM RAM est. mean/peak MiB	n
ramp	6.04 / 7.17	836.97 / 848.03	39
steady	3.33 / 4.64	785.76 / 863.45	53

Container SECONDARY	CPU mean/peak %	Working set mean/peak MiB
ramp	3.42 / 5.64	149.23 / 154.50
steady	0.32 / 0.56	101.93 / 175.00

Steady endpoint	Attempts	Success	Mean ms	p95 ms	p99 ms	p99.9 ms
login	1	1	104.00	104.00	104.00	104.00
comment	1,194	1,194	2.02	3.00	3.00	4.00
comments	217	217	1.36	2.00	2.00	2.78
feed	1,200	1,200	5.53	8.00	9.00	9.00
like	863	863	2.00	3.00	3.00	4.97
unlike	340	340	2.01	3.00	3.00	4.98
post	110	110	2.25	3.00	3.00	11.91

Ramp login: 99 / 99 successful; mean 107.37 ms; p99 133.02 ms; p99.9 133.90 ms.

Rate definition	Attempted	Successful	HTTP completed
Mean / configured 240 s	16.35	16.35	16.35
Peak 1 s inside window	26.00	26.00	26.00
Peak rolling 10 s	18.20	18.20	18.20

Window spillover counts: {'early': {'attempted': 0, 'successful': 0, 'http_completed': 0}, 'late': {'attempted': 0, 'successful': 0, 'http_completed': 0}}. Completed journeys: 1,199. Generator resource errors: 0.

RDS average / maximum: CPU 1.82 / 1.99%; connections 3.20 / 4.00; freeable 10.24 GiB; read/write 0.26 / 14.89 IOPS.

Validity: valid-recorded-evidence. Flags: none recorded. Fixture restored: True.

Rust / 100 users

Detailed fixed-load observation | PASS | trial e02ce1700d3a-0012

Attempts / successful	Operation success	Authenticated	Timeout / network
3,929 / 3,929	100.0000%	100 / 100	0 / 0

Mean ms	p50 ms	p95 ms	p99 ms	p99.9 ms
3.43	3.00	7.00	8.00	9.00

VM phase PRIMARY	VM CPU mean/peak %	VM RAM est. mean/peak MiB	n
ramp	5.77 / 7.46	704.05 / 753.26	40
steady	3.29 / 4.32	688.39 / 700.03	52

Container SECONDARY	CPU mean/peak %	Working set mean/peak MiB
ramp	2.20 / 5.08	6.66 / 68.63
steady	0.30 / 0.47	5.99 / 6.11

Steady endpoint	Attempts	Success	Mean ms	p95 ms	p99 ms	p99.9 ms
login	1	1	95.00	95.00	95.00	95.00
like	851	851	2.67	4.00	4.00	4.00
feed	1,198	1,198	5.34	7.00	8.00	9.00
comment	1,194	1,194	2.59	4.00	4.00	4.81
post	119	119	2.79	4.00	4.00	4.00
unlike	340	340	2.56	3.00	4.00	4.00
comments	226	226	1.81	3.00	3.00	3.78

Ramp login: 99 / 99 successful; mean 97.19 ms; p99 115.18 ms; p99.9 123.12 ms.

Rate definition	Attempted	Successful	HTTP completed
Mean / configured 240 s	16.37	16.37	16.37
Peak 1 s inside window	26.00	26.00	26.00
Peak rolling 10 s	17.80	17.80	17.80

Window spillover counts: {'early': {'attempted': 0, 'successful': 0, 'http_completed': 0}, 'late': {'attempted': 0, 'successful': 0, 'http_completed': 0}}. Completed journeys: 1,197. Generator resource errors: 0.

RDS average / maximum: CPU 1.57 / 1.62%; connections 4.00 / 4.00; freeable 10.24 GiB; read/write 0.26 / 18.51 IOPS.

Validity: valid-recorded-evidence. Flags: none recorded. Fixture restored: True.

Node / 100 users

Detailed fixed-load observation | PASS | trial e02ce1700d3a-0015

Attempts / successful	Operation success	Authenticated	Timeout / network
3,931 / 3,931	100.0000%	100 / 100	0 / 0

Mean ms	p50 ms	p95 ms	p99 ms	p99.9 ms
4.71	4.00	9.00	11.00	13.07

VM phase PRIMARY	VM CPU mean/peak %	VM RAM est. mean/peak MiB	n
ramp	7.14 / 8.65	772.95 / 829.30	39
steady	4.27 / 8.32	791.10 / 815.58	54

Container SECONDARY	CPU mean/peak %	Working set mean/peak MiB
ramp	4.05 / 7.25	83.48 / 148.10
steady	1.14 / 2.02	105.09 / 108.30

Steady endpoint	Attempts	Success	Mean ms	p95 ms	p99 ms	p99.9 ms
like	857	857	3.36	4.00	5.00	6.00
login	1	1	118.00	118.00	118.00	118.00
comment	1,204	1,204	3.60	5.00	5.00	7.00
feed	1,200	1,200	7.51	10.00	12.00	13.80
comments	226	226	2.81	4.00	4.00	15.63
unlike	327	327	3.28	4.00	5.00	9.35
post	116	116	4.00	5.00	5.00	5.89

Ramp login: 99 / 99 successful; mean 121.63 ms; p99 142.14 ms; p99.9 148.31 ms.

Rate definition	Attempted	Successful	HTTP completed
Mean / configured 240 s	16.38	16.38	16.38
Peak 1 s inside window	30.00	30.00	30.00
Peak rolling 10 s	18.10	18.10	18.10

Window spillover counts: {'early': {'attempted': 0, 'successful': 0, 'http_completed': 0}, 'late': {'attempted': 0, 'successful': 0, 'http_completed': 0}}. Completed journeys: 1,199. Generator resource errors: 0.

RDS average / maximum: CPU 1.94 / 2.04%; connections 5.20 / 6.00; freeable 10.24 GiB; read/write 0.26 / 18.74 IOPS.

Validity: valid-recorded-evidence. Flags: none recorded. Fixture restored: True.

Bun / 100 users

Detailed fixed-load observation | PASS | trial e02ce1700d3a-0018

Attempts / successful	Operation success	Authenticated	Timeout / network
3,926 / 3,926	100.0000%	100 / 100	0 / 0

Mean ms	p50 ms	p95 ms	p99 ms	p99.9 ms
4.43	3.00	9.00	11.00	13.07

VM phase PRIMARY	VM CPU mean/peak %	VM RAM est. mean/peak MiB	n
ramp	7.40 / 8.92	758.90 / 810.32	40
steady	4.13 / 5.64	757.91 / 782.99	52

Container SECONDARY	CPU mean/peak %	Working set mean/peak MiB
ramp	4.13 / 7.63	67.44 / 130.20
steady	1.10 / 1.99	73.27 / 75.44

Steady endpoint	Attempts	Success	Mean ms	p95 ms	p99 ms	p99.9 ms
login	1	1	122.00	122.00	122.00	122.00
like	857	857	3.01	4.00	4.00	5.00
comment	1,195	1,195	3.25	4.00	5.00	14.03
feed	1,191	1,191	7.46	10.00	12.00	13.00
unlike	337	337	2.93	4.00	4.00	5.00
post	119	119	3.60	4.00	5.00	5.00
comments	226	226	2.26	3.00	3.00	3.00

Ramp login: 99 / 99 successful; mean 125.20 ms; p99 151.06 ms; p99.9 153.71 ms.

Rate definition	Attempted	Successful	HTTP completed
Mean / configured 240 s	16.36	16.36	16.36
Peak 1 s inside window	29.00	29.00	29.00
Peak rolling 10 s	18.30	18.30	18.30

Window spillover counts: {'early': {'attempted': 0, 'successful': 0, 'http_completed': 0}, 'late': {'attempted': 0, 'successful': 0, 'http_completed': 0}}. Completed journeys: 1,190. Generator resource errors: 0.

RDS average / maximum: CPU 1.95 / 2.03%; connections 5.40 / 6.00; freeable 10.24 GiB; read/write 0.00 / 19.19 IOPS.

Validity: valid-recorded-evidence. Flags: none recorded. Fixture restored: True.

.NET / 100 users

Detailed fixed-load observation | PASS | trial 98b838a32a49-0000

Attempts / successful	Operation success	Authenticated	Timeout / network
3,935 / 3,935	100.0000%	100 / 100	0 / 0

Mean ms	p50 ms	p95 ms	p99 ms	p99.9 ms
3.98	3.00	8.00	9.00	31.26

VM phase PRIMARY	VM CPU mean/peak %	VM RAM est. mean/peak MiB	n
ramp	11.64 / 15.98	903.70 / 924.27	39
steady	4.02 / 10.45	916.07 / 930.86	53

Container SECONDARY	CPU mean/peak %	Working set mean/peak MiB
ramp	8.18 / 30.06	241.81 / 243.90
steady	0.78 / 1.15	257.05 / 259.10

Steady endpoint	Attempts	Success	Mean ms	p95 ms	p99 ms	p99.9 ms
login	1	1	246.00	246.00	246.00	246.00
feed	1,193	1,193	6.23	8.00	9.00	10.81
comment	1,197	1,197	3.10	4.00	4.00	16.65
like	870	870	2.82	3.55	4.00	4.13
post	121	121	3.51	4.00	6.60	31.64
unlike	319	319	2.79	3.00	4.00	22.41
comments	234	234	2.23	3.00	3.00	69.96

Ramp login: 99 / 99 successful; mean 258.43 ms; p99 335.98 ms; p99.9 467.40 ms.

Rate definition	Attempted	Successful	HTTP completed
Mean / configured 240 s	16.40	16.40	16.40
Peak 1 s inside window	27.00	27.00	27.00
Peak rolling 10 s	18.40	18.40	18.40

Window spillover counts: {'early': {'attempted': 0, 'successful': 0, 'http_completed': 0}, 'late': {'attempted': 0, 'successful': 0, 'http_completed': 0}}. Completed journeys: 1,192. Generator resource errors: 0.

RDS average / maximum: CPU 2.08 / 2.20%; connections 3.20 / 4.00; freeable 10.26 GiB; read/write 0.26 / 19.10 IOPS.

Validity: qualified. Flags: none recorded. Fixture restored: True.

Octane / 100 users

Detailed fixed-load observation | PASS | trial 35749f424096-0000

Attempts / successful	Operation success	Authenticated	Timeout / network
3,910 / 3,910	100.0000%	100 / 100	0 / 0

Mean ms	p50 ms	p95 ms	p99 ms	p99.9 ms
6.44	6.00	11.00	12.00	13.09

VM phase PRIMARY	VM CPU mean/peak %	VM RAM est. mean/peak MiB	n
ramp	9.12 / 21.16	1,560.78 / 1,642.90	40
steady	5.65 / 7.00	1,563.03 / 1,580.23	52

Container SECONDARY	CPU mean/peak %	Working set mean/peak MiB
ramp	6.93 / 38.70	842.14 / 915.10
steady	2.58 / 3.96	855.48 / 856.40

Steady endpoint	Attempts	Success	Mean ms	p95 ms	p99 ms	p99.9 ms
login	1	1	125.00	125.00	125.00	125.00
like	879	879	5.22	6.00	7.00	10.24
post	116	116	5.83	7.00	8.00	8.00
feed	1,189	1,189	8.90	11.00	13.00	13.81
comments	215	215	4.47	6.00	7.00	7.00
comment	1,195	1,195	5.55	7.00	8.00	10.00
unlike	315	315	5.15	6.00	7.00	7.00

Ramp login: 99 / 99 successful; mean 143.07 ms; p99 353.40 ms; p99.9 415.14 ms.

Rate definition	Attempted	Successful	HTTP completed
Mean / configured 240 s	16.29	16.29	16.29
Peak 1 s inside window	30.00	30.00	30.00
Peak rolling 10 s	18.00	18.00	18.00

Window spillover counts: {'early': {'attempted': 0, 'successful': 0, 'http_completed': 0}, 'late': {'attempted': 0, 'successful': 0, 'http_completed': 0}}. Completed journeys: 1,188. Generator resource errors: 0.

RDS average / maximum: CPU 2.11 / 2.14%; connections 21.00 / 21.00; freeable 10.21 GiB; read/write 0.26 / 20.17 IOPS.

Validity: qualified. Flags: none recorded. Fixture restored: True.

Python 3.14 / 500 users

Detailed fixed-load observation | PASS | trial e02ce1700d3a-0001

Attempts / successful	Operation success	Authenticated	Timeout / network
19,612 / 19,612	100.0000%	500 / 500	0 / 0

Mean ms	p50 ms	p95 ms	p99 ms	p99.9 ms
6.45	4.00	43.00	44.00	45.00

VM phase PRIMARY	VM CPU mean/peak %	VM RAM est. mean/peak MiB	n
ramp	27.02 / 31.36	861.76 / 895.59	37
steady	12.57 / 28.94	815.65 / 862.82	51

Container SECONDARY	CPU mean/peak %	Working set mean/peak MiB
ramp	20.82 / 26.48	157.53 / 200.60
steady	6.74 / 10.60	137.89 / 141.70

Steady endpoint	Attempts	Success	Mean ms	p95 ms	p99 ms	p99.9 ms
login	1	1	123.00	123.00	123.00	123.00
post	620	620	3.77	5.00	7.00	11.14
feed	5,919	5,919	5.41	7.00	8.00	10.00
unlike	1,424	1,424	3.21	4.00	5.00	16.35
like	4,515	4,515	3.24	4.00	6.00	8.49
comment	5,932	5,932	11.74	44.00	45.00	46.07
comments	1,201	1,201	2.64	4.00	4.00	6.80

Ramp login: 499 / 499 successful; mean 127.37 ms; p99 169.00 ms; p99.9 195.56 ms.

Rate definition	Attempted	Successful	HTTP completed
Mean / configured 240 s	81.72	81.72	81.72
Peak 1 s inside window	110.00	113.00	113.00
Peak rolling 10 s	85.00	85.00	85.00

Window spillover counts: {'early': {'attempted': 0, 'successful': 0, 'http_completed': 0}, 'late': {'attempted': 0, 'successful': 2.0, 'http_completed': 2.0}}. Completed journeys: 5,918. Generator resource errors: 0.

RDS average / maximum: CPU 3.11 / 3.24%; connections 9.00 / 9.00; freeable 10.25 GiB; read/write 0.26 / 58.21 IOPS.

Validity: valid-recorded-evidence. Flags: none recorded. Fixture restored: True.

Java / 500 users

Detailed fixed-load observation | PASS | trial e02ce1700d3a-0004

Attempts / successful	Operation success	Authenticated	Timeout / network
19,571 / 19,571	100.0000%	500 / 500	0 / 0

Mean ms	p50 ms	p95 ms	p99 ms	p99.9 ms
2.90	2.00	6.00	7.00	8.00

VM phase PRIMARY	VM CPU mean/peak %	VM RAM est. mean/peak MiB	n
ramp	30.36 / 38.10	1,499.14 / 1,593.53	38
steady	5.81 / 32.39	1,371.38 / 1,388.31	53

Container SECONDARY	CPU mean/peak %	Working set mean/peak MiB
ramp	25.86 / 47.02	822.76 / 948.30
steady	2.07 / 7.75	688.41 / 689.60

Steady endpoint	Attempts	Success	Mean ms	p95 ms	p99 ms	p99.9 ms
login	1	1	141.00	141.00	141.00	141.00
comment	5,920	5,920	2.28	3.00	4.00	4.08
like	4,537	4,537	2.23	3.00	3.00	4.00
feed	5,909	5,909	4.58	7.00	8.00	9.00
comments	1,198	1,198	1.24	2.00	2.00	2.80
unlike	1,385	1,385	2.19	3.00	3.00	4.00
post	621	621	2.39	3.00	4.00	4.76

Ramp login: 499 / 499 successful; mean 156.90 ms; p99 291.00 ms; p99.9 430.46 ms.

Rate definition	Attempted	Successful	HTTP completed
Mean / configured 240 s	81.55	81.55	81.55
Peak 1 s inside window	102.00	102.00	102.00
Peak rolling 10 s	85.90	85.90	85.90

Window spillover counts: {'early': {'attempted': 0, 'successful': 0, 'http_completed': 0}, 'late': {'attempted': 0, 'successful': 1.0, 'http_completed': 1.0}}. Completed journeys: 5,908. Generator resource errors: 0.

RDS average / maximum: CPU 3.60 / 4.00%; connections 7.00 / 7.00; freeable 10.25 GiB; read/write 0.26 / 59.10 IOPS.

Validity: valid-recorded-evidence. Flags: none recorded. Fixture restored: True.

PHP (FAIL) / 500 users

Detailed fixed-load observation | FAIL | trial e02ce1700d3a-0007

Attempts / successful	Operation success	Authenticated	Timeout / network
10,411 / 0	0.0000%	211 / 500	10,411 / 10,411

Mean ms	p50 ms	p95 ms	p99 ms	p99.9 ms
5,000.66	5,001.00	5,001.00	5,002.00	5,002.00

VM phase PRIMARY	VM CPU mean/peak %	VM RAM est. mean/peak MiB	n
ramp	81.82 / 100.00	829.77 / 892.77	29
steady	99.99 / 100.00	900.57 / 947.02	30

Container SECONDARY	CPU mean/peak %	Working set mean/peak MiB
ramp	80.71 / 97.83	163.00 / 225.40
steady	96.88 / 97.53	203.01 / 234.20

Steady endpoint	Attempts	Success	Mean ms	p95 ms	p99 ms	p99.9 ms
login	6,017	0	5,000.65	5,001.00	5,002.00	5,002.00
feed	4,394	0	5,000.66	5,001.00	5,002.00	5,002.00

Ramp login: 211 / 1,675 successful; mean 4,482.83 ms; p99 5,002.00 ms; p99.9 5,002.00 ms.

Rate definition	Attempted	Successful	HTTP completed
Mean / configured 240 s	43.38	0.00	43.38
Peak 1 s inside window	60.00	0.00	60.00
Peak rolling 10 s	46.30	0.00	46.30

Window spillover counts: {'early': {'attempted': 0, 'successful': 0, 'http_completed': 0}, 'late': {'attempted': 0, 'successful': 0, 'http_completed': 225.0}}. Completed journeys: 0. Generator resource errors: 0.

RDS average / maximum: CPU 3.88 / 4.81%; connections 5.40 / 14.00; freeable 10.24 GiB; read/write 0.26 / 9.82 IOPS.

Validity: valid-recorded-evidence. Flags: none recorded. Fixture restored: True.

Go / 500 users

Detailed fixed-load observation | PASS | trial e02ce1700d3a-0010

Attempts / successful	Operation success	Authenticated	Timeout / network
19,584 / 19,584	100.0000%	500 / 500	0 / 0

Mean ms	p50 ms	p95 ms	p99 ms	p99.9 ms
2.53	2.00	5.00	5.00	7.00

VM phase PRIMARY	VM CPU mean/peak %	VM RAM est. mean/peak MiB	n
ramp	19.34 / 21.45	870.25 / 913.88	39
steady	4.78 / 8.72	811.27 / 952.33	53

Container SECONDARY	CPU mean/peak %	Working set mean/peak MiB
ramp	16.58 / 17.70	180.30 / 219.70
steady	1.27 / 1.61	122.05 / 271.50

Steady endpoint	Attempts	Success	Mean ms	p95 ms	p99 ms	p99.9 ms
login	1	1	107.00	107.00	107.00	107.00
like	4,536	4,536	2.02	3.00	3.00	4.00
comment	5,931	5,931	2.08	3.00	3.00	4.07
comments	1,193	1,193	1.07	2.00	2.00	2.00
feed	5,910	5,910	3.80	5.00	6.00	8.00
post	627	627	2.19	3.00	3.00	4.00
unlike	1,386	1,386	2.00	3.00	3.00	4.00

Ramp login: 499 / 499 successful; mean 107.70 ms; p99 148.00 ms; p99.9 150.00 ms.

Rate definition	Attempted	Successful	HTTP completed
Mean / configured 240 s	81.60	81.60	81.60
Peak 1 s inside window	115.00	114.00	114.00
Peak rolling 10 s	84.90	84.90	84.90

Window spillover counts: {'early': {'attempted': 0, 'successful': 0, 'http_completed': 0}, 'late': {'attempted': 0, 'successful': 1.0, 'http_completed': 1.0}}. Completed journeys: 5,909. Generator resource errors: 0.

RDS average / maximum: CPU 3.07 / 3.63%; connections 5.80 / 6.00; freeable 10.23 GiB; read/write 0.26 / 58.45 IOPS.

Validity: valid-recorded-evidence. Flags: none recorded. Fixture restored: True.

Rust / 500 users

Detailed fixed-load observation | PASS | trial e02ce1700d3a-0013

Attempts / successful	Operation success	Authenticated	Timeout / network
19,611 / 19,611	100.0000%	500 / 500	0 / 0

Mean ms	p50 ms	p95 ms	p99 ms	p99.9 ms
2.91	3.00	5.00	7.00	8.00

VM phase PRIMARY	VM CPU mean/peak %	VM RAM est. mean/peak MiB	n
ramp	17.55 / 18.24	740.97 / 771.13	39
steady	4.80 / 16.70	702.37 / 715.61	53

Container SECONDARY	CPU mean/peak %	Working set mean/peak MiB
ramp	14.36 / 15.37	11.39 / 20.05
steady	1.06 / 1.37	21.07 / 21.51

Steady endpoint	Attempts	Success	Mean ms	p95 ms	p99 ms	p99.9 ms
login	1	1	95.00	95.00	95.00	95.00
like	4,517	4,517	2.36	3.00	4.00	4.00
feed	5,916	5,916	4.45	6.00	7.00	8.00
post	621	621	2.48	3.00	4.00	4.38
comment	5,945	5,945	2.30	3.00	4.00	4.00
unlike	1,413	1,413	2.28	3.00	3.00	4.00
comments	1,198	1,198	1.38	2.00	3.00	3.00

Ramp login: 499 / 499 successful; mean 98.19 ms; p99 129.00 ms; p99.9 131.00 ms.

Rate definition	Attempted	Successful	HTTP completed
Mean / configured 240 s	81.71	81.71	81.71
Peak 1 s inside window	110.00	110.00	110.00
Peak rolling 10 s	85.10	85.10	85.10

Window spillover counts: {'early': {'attempted': 0, 'successful': 0, 'http_completed': 0}, 'late': {'attempted': 0, 'successful': 0, 'http_completed': 0}}. Completed journeys: 5,915. Generator resource errors: 0.

RDS average / maximum: CPU 3.49 / 3.51%; connections 5.00 / 5.00; freeable 10.24 GiB; read/write 0.26 / 60.28 IOPS.

Validity: valid-recorded-evidence. Flags: none recorded. Fixture restored: True.

Node / 500 users

Detailed fixed-load observation | PASS | trial e02ce1700d3a-0016

Attempts / successful	Operation success	Authenticated	Timeout / network
19,583 / 19,583	100.0000%	500 / 500	0 / 0

Mean ms	p50 ms	p95 ms	p99 ms	p99.9 ms
4.00	3.00	8.00	10.00	12.00

VM phase PRIMARY	VM CPU mean/peak %	VM RAM est. mean/peak MiB	n
ramp	22.87 / 25.08	860.65 / 934.43	39
steady	7.50 / 11.85	895.60 / 927.70	53

Container SECONDARY	CPU mean/peak %	Working set mean/peak MiB
ramp	20.25 / 23.20	125.34 / 199.50
steady	3.69 / 5.84	210.23 / 217.00

Steady endpoint	Attempts	Success	Mean ms	p95 ms	p99 ms	p99.9 ms
login	1	1	118.00	118.00	118.00	118.00
like	4,518	4,518	2.91	4.00	5.00	10.00
comment	5,940	5,940	3.09	4.00	5.00	10.06
feed	5,917	5,917	6.43	9.00	10.00	13.00
comments	1,194	1,194	2.14	3.00	4.00	9.00
unlike	1,389	1,389	2.86	4.00	4.00	12.61
post	624	624	3.41	4.00	5.00	8.38

Ramp login: 499 / 499 successful; mean 122.44 ms; p99 154.04 ms; p99.9 161.00 ms.

Rate definition	Attempted	Successful	HTTP completed
Mean / configured 240 s	81.60	81.60	81.60
Peak 1 s inside window	114.00	117.00	117.00
Peak rolling 10 s	86.60	86.60	86.60

Window spillover counts: {'early': {'attempted': 0, 'successful': 0, 'http_completed': 0}, 'late': {'attempted': 0, 'successful': 0, 'http_completed': 0}}. Completed journeys: 5,916. Generator resource errors: 0.

RDS average / maximum: CPU 4.80 / 4.88%; connections 7.00 / 8.00; freeable 10.23 GiB; read/write 0.26 / 61.28 IOPS.

Validity: valid-recorded-evidence. Flags: none recorded. Fixture restored: True.

Bun / 500 users

Detailed fixed-load observation | PASS | trial e02ce1700d3a-0019

Attempts / successful	Operation success	Authenticated	Timeout / network
19,601 / 19,601	100.0000%	500 / 500	0 / 0

Mean ms	p50 ms	p95 ms	p99 ms	p99.9 ms
3.71	3.00	8.00	9.00	11.00

VM phase PRIMARY	VM CPU mean/peak %	VM RAM est. mean/peak MiB	n
ramp	23.13 / 25.51	797.33 / 846.65	39
steady	6.73 / 14.87	779.89 / 793.42	52

Container SECONDARY	CPU mean/peak %	Working set mean/peak MiB
ramp	20.69 / 23.00	72.95 / 87.05
steady	2.92 / 4.91	86.56 / 89.13

Steady endpoint	Attempts	Success	Mean ms	p95 ms	p99 ms	p99.9 ms
comment	5,916	5,916	2.81	4.00	4.00	6.00
login	1	1	123.00	123.00	123.00	123.00
feed	5,932	5,932	6.14	9.00	10.00	12.00
like	4,550	4,550	2.61	3.00	4.00	6.00
unlike	1,392	1,392	2.54	3.00	4.00	8.22
comments	1,191	1,191	1.84	3.00	3.00	5.81
post	619	619	3.14	4.00	4.00	8.53

Ramp login: 499 / 499 successful; mean 125.21 ms; p99 155.02 ms; p99.9 163.02 ms.

Rate definition	Attempted	Successful	HTTP completed
Mean / configured 240 s	81.67	81.67	81.67
Peak 1 s inside window	104.00	104.00	104.00
Peak rolling 10 s	86.00	85.90	85.90

Window spillover counts: {'early': {'attempted': 0, 'successful': 0, 'http_completed': 0}, 'late': {'attempted': 0, 'successful': 0, 'http_completed': 0}}. Completed journeys: 5,931. Generator resource errors: 0.

RDS average / maximum: CPU 4.65 / 4.92%; connections 8.20 / 10.00; freeable 10.23 GiB; read/write 0.26 / 57.35 IOPS.

Validity: valid-recorded-evidence. Flags: none recorded. Fixture restored: True.

.NET / 500 users

Detailed fixed-load observation | PASS | trial 98b838a32a49-0001

Attempts / successful	Operation success	Authenticated	Timeout / network
19,609 / 19,609	100.0000%	500 / 500	0 / 0

Mean ms	p50 ms	p95 ms	p99 ms	p99.9 ms
3.62	3.00	8.00	9.00	11.00

VM phase PRIMARY	VM CPU mean/peak %	VM RAM est. mean/peak MiB	n
ramp	42.41 / 57.05	913.56 / 964.34	38
steady	7.45 / 17.59	922.49 / 935.25	52

Container SECONDARY	CPU mean/peak %	Working set mean/peak MiB
ramp	37.80 / 54.20	253.68 / 308.00
steady	3.38 / 4.34	262.32 / 266.00

Steady endpoint	Attempts	Success	Mean ms	p95 ms	p99 ms	p99.9 ms
login	1	1	247.00	247.00	247.00	247.00
feed	5,913	5,913	6.00	9.00	9.00	12.00
unlike	1,411	1,411	2.42	3.00	4.00	6.00
like	4,530	4,530	2.54	3.00	4.00	6.00
comment	5,941	5,941	2.80	4.00	4.00	8.00
post	617	617	2.99	4.00	4.00	5.38
comments	1,196	1,196	1.58	2.00	3.00	6.00

Ramp login: 499 / 499 successful; mean 257.90 ms; p99 327.04 ms; p99.9 465.84 ms.

Rate definition	Attempted	Successful	HTTP completed
Mean / configured 240 s	81.70	81.70	81.70
Peak 1 s inside window	108.00	108.00	108.00
Peak rolling 10 s	85.20	85.20	85.20

Window spillover counts: {'early': {'attempted': 0, 'successful': 0, 'http_completed': 0}, 'late': {'attempted': 0, 'successful': 0, 'http_completed': 0}}. Completed journeys: 5,912. Generator resource errors: 0.

RDS average / maximum: CPU 5.34 / 5.41%; connections 5.60 / 6.00; freeable 10.25 GiB; read/write 0.26 / 60.87 IOPS.

Validity: qualified. Flags: none recorded. Fixture restored: True.

Octane / 500 users

Detailed fixed-load observation | PASS | trial 35749f424096-0001

Attempts / successful	Operation success	Authenticated	Timeout / network
19,599 / 19,599	100.0000%	500 / 500	0 / 0

Mean ms	p50 ms	p95 ms	p99 ms	p99.9 ms
6.18	5.00	10.00	12.00	14.00

VM phase PRIMARY	VM CPU mean/peak %	VM RAM est. mean/peak MiB	n
ramp	29.81 / 83.56	1,592.24 / 1,709.08	39
steady	15.25 / 21.94	1,583.55 / 1,600.93	52

Container SECONDARY	CPU mean/peak %	Working set mean/peak MiB
ramp	23.88 / 32.66	885.87 / 930.50
steady	10.87 / 13.64	874.84 / 876.40

Steady endpoint	Attempts	Success	Mean ms	p95 ms	p99 ms	p99.9 ms
login	1	1	147.00	147.00	147.00	147.00
feed	5,929	5,929	8.53	11.00	13.00	15.00
like	4,548	4,548	5.08	7.00	9.00	12.00
post	616	616	5.66	8.00	10.00	12.38
comment	5,934	5,934	5.37	7.00	9.00	13.07
unlike	1,376	1,376	5.02	7.00	9.00	12.25
comments	1,195	1,195	4.13	6.00	7.00	11.00

Ramp login: 499 / 499 successful; mean 145.40 ms; p99 596.26 ms; p99.9 949.66 ms.

Rate definition	Attempted	Successful	HTTP completed
Mean / configured 240 s	81.66	81.66	81.66
Peak 1 s inside window	105.00	107.00	107.00
Peak rolling 10 s	86.20	86.10	86.10

Window spillover counts: {'early': {'attempted': 0, 'successful': 0, 'http_completed': 0}, 'late': {'attempted': 0, 'successful': 0, 'http_completed': 0}}. Completed journeys: 5,928. Generator resource errors: 0.

RDS average / maximum: CPU 5.45 / 5.72%; connections 21.00 / 21.00; freeable 10.21 GiB; read/write 0.26 / 58.95 IOPS.

Validity: qualified. Flags: none recorded. Fixture restored: True.

Python 3.14 / 1,000 users

Detailed fixed-load observation | PASS | trial e02ce1700d3a-0002

Attempts / successful	Operation success	Authenticated	Timeout / network
39,275 / 39,275	100.0000%	1,000 / 1,000	0 / 0

Mean ms	p50 ms	p95 ms	p99 ms	p99.9 ms
6.39	4.00	43.00	44.00	46.00

VM phase PRIMARY	VM CPU mean/peak %	VM RAM est. mean/peak MiB	n
ramp	49.18 / 57.79	867.00 / 930.84	36
steady	19.22 / 47.63	833.02 / 879.20	50

Container SECONDARY	CPU mean/peak %	Working set mean/peak MiB
ramp	42.86 / 50.21	182.16 / 209.90
steady	12.38 / 18.79	147.37 / 149.50

Steady endpoint	Attempts	Success	Mean ms	p95 ms	p99 ms	p99.9 ms
login	1	1	128.00	128.00	128.00	128.00
comment	11,921	11,921	11.66	44.00	45.00	47.08
feed	11,904	11,904	5.26	7.00	9.00	12.10
post	1,191	1,191	3.67	5.00	6.10	9.81
unlike	2,699	2,699	3.23	5.00	6.00	10.00
like	9,189	9,189	3.27	5.00	6.00	12.81
comments	2,370	2,370	2.64	4.00	6.00	9.00

Ramp login: 999 / 999 successful; mean 129.84 ms; p99 188.02 ms; p99.9 229.02 ms.

Rate definition	Attempted	Successful	HTTP completed
Mean / configured 240 s	163.65	163.65	163.65
Peak 1 s inside window	195.00	196.00	196.00
Peak rolling 10 s	168.90	169.00	169.00

Window spillover counts: {'early': {'attempted': 0, 'successful': 0, 'http_completed': 0}, 'late': {'attempted': 0, 'successful': 1.0, 'http_completed': 1.0}}. Completed journeys: 11,903. Generator resource errors: 0.

RDS average / maximum: CPU 5.04 / 5.25%; connections 12.00 / 12.00; freeable 10.24 GiB; read/write 0.26 / 101.03 IOPS.

Validity: valid-recorded-evidence. Flags: none recorded. Fixture restored: True.

Java / 1,000 users

Detailed fixed-load observation | PASS | trial e02ce1700d3a-0005

Attempts / successful	Operation success	Authenticated	Timeout / network
39,319 / 39,319	100.0000%	1,000 / 1,000	0 / 0

Mean ms	p50 ms	p95 ms	p99 ms	p99.9 ms
2.93	2.00	6.00	8.00	9.00

VM phase PRIMARY	VM CPU mean/peak %	VM RAM est. mean/peak MiB	n
ramp	59.61 / 99.79	2,139.06 / 2,374.25	38
steady	6.63 / 11.58	1,798.60 / 1,806.21	52

Container SECONDARY	CPU mean/peak %	Working set mean/peak MiB
ramp	53.12 / 98.39	1,465.42 / 1,716.22
steady	2.94 / 7.88	1,113.01 / 1,113.09

Steady endpoint	Attempts	Success	Mean ms	p95 ms	p99 ms	p99.9 ms
login	1	1	201.00	201.00	201.00	201.00
comments	2,383	2,383	1.11	2.00	2.00	3.62
like	9,236	9,236	2.14	3.00	3.00	5.76
comment	11,914	11,914	2.21	3.00	4.00	5.00
feed	11,915	11,915	4.84	7.00	8.00	11.00
post	1,192	1,192	2.37	3.00	4.00	5.00
unlike	2,678	2,678	2.12	3.00	3.00	4.65

Ramp login: 999 / 999 successful; mean 223.58 ms; p99 1,248.14 ms; p99.9 2,679.23 ms.

Rate definition	Attempted	Successful	HTTP completed
Mean / configured 240 s	163.83	163.83	163.83
Peak 1 s inside window	205.00	205.00	205.00
Peak rolling 10 s	169.40	169.30	169.30

Window spillover counts: {'early': {'attempted': 0, 'successful': 0, 'http_completed': 0}, 'late': {'attempted': 0, 'successful': 0, 'http_completed': 0}}. Completed journeys: 11,913. Generator resource errors: 0.

RDS average / maximum: CPU 6.72 / 7.50%; connections 21.00 / 21.00; freeable 10.21 GiB; read/write 0.26 / 106.85 IOPS.

Validity: valid-recorded-evidence. Flags: none recorded. Fixture restored: True.

PHP (FAIL) / 1,000 users

Detailed fixed-load observation | FAIL | trial e02ce1700d3a-0008

Attempts / successful	Operation success	Authenticated	Timeout / network
20,827 / 0	0.0000%	141 / 1,000	20,827 / 20,827

Mean ms	p50 ms	p95 ms	p99 ms	p99.9 ms
5,000.65	5,001.00	5,001.00	5,002.00	5,002.00

VM phase PRIMARY	VM CPU mean/peak %	VM RAM est. mean/peak MiB	n
ramp	96.15 / 100.00	882.52 / 944.19	24
steady	99.99 / 100.00	936.12 / 972.77	31

Container SECONDARY	CPU mean/peak %	Working set mean/peak MiB
ramp	94.75 / 98.12	194.77 / 227.30
steady	96.08 / 97.24	210.05 / 241.20

Steady endpoint	Attempts	Success	Mean ms	p95 ms	p99 ms	p99.9 ms
login	17,883	0	5,000.65	5,001.00	5,002.00	5,002.00
feed	2,944	0	5,000.64	5,001.00	5,002.00	5,002.00

Ramp login: 141 / 6,390 successful; mean 4,917.65 ms; p99 5,002.00 ms; p99.9 5,002.00 ms.

Rate definition	Attempted	Successful	HTTP completed
Mean / configured 240 s	86.78	0.00	86.78
Peak 1 s inside window	113.00	0.00	113.00
Peak rolling 10 s	90.00	0.00	90.00

Window spillover counts: {'early': {'attempted': 0, 'successful': 0, 'http_completed': 0}, 'late': {'attempted': 1.0, 'successful': 0, 'http_completed': 419.0}}. Completed journeys: 0. Generator resource errors: 0.

RDS average / maximum: CPU 2.97 / 3.19%; connections 2.00 / 3.00; freeable 10.25 GiB; read/write 0.26 / 6.53 IOPS.

Validity: valid-recorded-evidence. Flags: none recorded. Fixture restored: True.

Go / 1,000 users

Detailed fixed-load observation | PASS | trial e02ce1700d3a-0011

Attempts / successful	Operation success	Authenticated	Timeout / network
39,313 / 39,313	100.0000%	1,000 / 1,000	0 / 0

Mean ms	p50 ms	p95 ms	p99 ms	p99.9 ms
2.42	2.00	4.00	5.00	6.00

VM phase PRIMARY	VM CPU mean/peak %	VM RAM est. mean/peak MiB	n
ramp	35.53 / 37.69	889.36 / 920.63	38
steady	6.39 / 27.77	761.87 / 919.43	53

Container SECONDARY	CPU mean/peak %	Working set mean/peak MiB
ramp	30.72 / 34.00	207.81 / 238.50
steady	2.18 / 2.68	75.66 / 249.80

Steady endpoint	Attempts	Success	Mean ms	p95 ms	p99 ms	p99.9 ms
login	1	1	107.00	107.00	107.00	107.00
like	9,242	9,242	1.99	3.00	3.00	4.00
feed	11,898	11,898	3.56	5.00	6.00	6.00
comment	11,919	11,919	2.02	3.00	3.00	4.00
comments	2,378	2,378	1.02	2.00	2.00	3.00
post	1,194	1,194	2.18	3.00	3.00	4.00
unlike	2,681	2,681	1.95	3.00	3.00	4.00

Ramp login: 999 / 999 successful; mean 108.39 ms; p99 139.00 ms; p99.9 146.00 ms.

Rate definition	Attempted	Successful	HTTP completed
Mean / configured 240 s	163.80	163.80	163.80
Peak 1 s inside window	198.00	198.00	198.00
Peak rolling 10 s	168.00	168.10	168.10

Window spillover counts: {'early': {'attempted': 0, 'successful': 0, 'http_completed': 0}, 'late': {'attempted': 0, 'successful': 0, 'http_completed': 0}}. Completed journeys: 11,897. Generator resource errors: 0.

RDS average / maximum: CPU 4.86 / 4.90%; connections 7.40 / 8.00; freeable 10.23 GiB; read/write 0.26 / 109.27 IOPS.

Validity: valid-recorded-evidence. Flags: none recorded. Fixture restored: True.

Rust / 1,000 users

Detailed fixed-load observation | PASS | trial e02ce1700d3a-0014

Attempts / successful	Operation success	Authenticated	Timeout / network
39,324 / 39,324	100.0000%	1,000 / 1,000	0 / 0

Mean ms	p50 ms	p95 ms	p99 ms	p99.9 ms
2.71	2.00	5.00	6.00	7.00

VM phase PRIMARY	VM CPU mean/peak %	VM RAM est. mean/peak MiB	n
ramp	31.05 / 32.56	746.87 / 805.78	39
steady	5.91 / 25.49	723.56 / 732.75	53

Container SECONDARY	CPU mean/peak %	Working set mean/peak MiB
ramp	26.64 / 29.41	61.40 / 103.00
steady	1.73 / 1.97	39.85 / 40.11

Steady endpoint	Attempts	Success	Mean ms	p95 ms	p99 ms	p99.9 ms
like	9,311	9,311	2.20	3.00	3.00	4.69
login	1	1	93.00	93.00	93.00	93.00
unlike	2,609	2,609	2.16	3.00	3.00	5.00
comment	11,928	11,928	2.18	3.00	3.00	4.00
feed	11,918	11,918	4.10	6.00	6.00	7.08
comments	2,370	2,370	1.21	2.00	2.00	3.00
post	1,187	1,187	2.34	3.00	4.00	4.00

Ramp login: 999 / 999 successful; mean 96.21 ms; p99 118.02 ms; p99.9 131.13 ms.

Rate definition	Attempted	Successful	HTTP completed
Mean / configured 240 s	163.85	163.85	163.85
Peak 1 s inside window	201.00	201.00	201.00
Peak rolling 10 s	169.40	169.30	169.30

Window spillover counts: {'early': {'attempted': 0, 'successful': 0, 'http_completed': 0}, 'late': {'attempted': 0, 'successful': 0, 'http_completed': 0}}. Completed journeys: 11,917. Generator resource errors: 0.

RDS average / maximum: CPU 5.69 / 5.84%; connections 6.40 / 7.00; freeable 10.23 GiB; read/write 0.26 / 104.55 IOPS.

Validity: valid-recorded-evidence. Flags: none recorded. Fixture restored: True.

Node / 1,000 users

Detailed fixed-load observation | PASS | trial e02ce1700d3a-0017

Attempts / successful	Operation success	Authenticated	Timeout / network
39,300 / 39,298	99.9949%	1,000 / 1,000	0 / 2

Mean ms	p50 ms	p95 ms	p99 ms	p99.9 ms
3.82	3.00	8.00	9.00	13.00

VM phase PRIMARY	VM CPU mean/peak %	VM RAM est. mean/peak MiB	n
ramp	41.50 / 46.29	898.19 / 982.28	38
steady	11.17 / 36.15	928.03 / 979.94	52

Container SECONDARY	CPU mean/peak %	Working set mean/peak MiB
ramp	36.95 / 42.53	187.60 / 282.80
steady	6.48 / 9.62	228.60 / 233.00

Steady endpoint	Attempts	Success	Mean ms	p95 ms	p99 ms	p99.9 ms
login	1	1	132.00	132.00	132.00	132.00
unlike	2,666	2,666	2.69	4.00	5.00	11.34
feed	11,904	11,904	6.18	9.00	10.00	14.10
like	9,252	9,250	2.77	4.00	6.00	13.00
comment	11,908	11,908	2.95	4.00	6.00	12.00
post	1,191	1,191	3.27	4.00	7.00	10.81
comments	2,378	2,378	1.95	3.00	4.00	8.00

Ramp login: 999 / 999 successful; mean 121.01 ms; p99 153.00 ms; p99.9 163.01 ms.

Rate definition	Attempted	Successful	HTTP completed
Mean / configured 240 s	163.75	163.74	163.75
Peak 1 s inside window	196.00	196.00	196.00
Peak rolling 10 s	169.10	169.20	169.20

Window spillover counts: {'early': {'attempted': 0, 'successful': 0, 'http_completed': 0}, 'late': {'attempted': 0, 'successful': 1.0, 'http_completed': 1.0}}. Completed journeys: 11,903. Generator resource errors: 0.

RDS average / maximum: CPU 8.30 / 8.54%; connections 9.60 / 12.00; freeable 10.22 GiB; read/write 0.26 / 107.09 IOPS.

Validity: valid-recorded-evidence. Flags: none recorded. Fixture restored: True.

Bun / 1,000 users

Detailed fixed-load observation | PASS | trial e02ce1700d3a-0020

Attempts / successful	Operation success	Authenticated	Timeout / network
39,331 / 39,330	99.9975%	1,000 / 1,000	0 / 1

Mean ms	p50 ms	p95 ms	p99 ms	p99.9 ms
3.56	3.00	8.00	9.00	11.00

VM phase PRIMARY	VM CPU mean/peak %	VM RAM est. mean/peak MiB	n
ramp	42.31 / 47.02	812.31 / 874.02	38
steady	9.64 / 44.37	790.29 / 814.23	53

Container SECONDARY	CPU mean/peak %	Working set mean/peak MiB
ramp	38.24 / 42.16	116.05 / 157.20
steady	4.70 / 6.34	96.56 / 99.35

Steady endpoint	Attempts	Success	Mean ms	p95 ms	p99 ms	p99.9 ms
login	1	1	130.00	130.00	130.00	130.00
feed	11,911	11,911	6.03	9.00	10.00	12.00
like	9,265	9,264	2.45	3.00	4.00	6.00
comment	11,925	11,925	2.66	4.00	4.00	6.00
post	1,189	1,189	2.93	4.00	4.00	6.00
comments	2,373	2,373	1.69	2.00	3.00	5.63
unlike	2,667	2,667	2.38	3.00	4.00	5.67

Ramp login: 999 / 999 successful; mean 126.10 ms; p99 153.02 ms; p99.9 165.02 ms.

Rate definition	Attempted	Successful	HTTP completed
Mean / configured 240 s	163.88	163.88	163.88
Peak 1 s inside window	203.00	203.00	203.00
Peak rolling 10 s	168.40	168.40	168.40

Window spillover counts: {'early': {'attempted': 0, 'successful': 0, 'http_completed': 0}, 'late': {'attempted': 0, 'successful': 1.0, 'http_completed': 1.0}}. Completed journeys: 11,910. Generator resource errors: 0.

RDS average / maximum: CPU 8.10 / 8.66%; connections 9.80 / 12.00; freeable 10.22 GiB; read/write 0.26 / 99.87 IOPS.

Validity: valid-recorded-evidence. Flags: none recorded. Fixture restored: True.

.NET / 1,000 users

Detailed fixed-load observation | FAIL | trial 98b838a32a49-0002

Attempts / successful	Operation success	Authenticated	Timeout / network
21,858 / 2,176	9.9552%	56 / 1,000	19,682 / 19,682

Mean ms	p50 ms	p95 ms	p99 ms	p99.9 ms
4,505.57	5,001.00	5,001.00	5,002.00	5,033.14

VM phase PRIMARY	VM CPU mean/peak %	VM RAM est. mean/peak MiB	n
ramp	99.32 / 99.79	1,005.14 / 1,140.27	37
steady	99.65 / 99.90	1,059.86 / 1,188.52	48

Container SECONDARY	CPU mean/peak %	Working set mean/peak MiB
ramp	98.14 / 99.83	338.08 / 505.30
steady	98.06 / 99.50	403.22 / 539.10

Steady endpoint	Attempts	Success	Mean ms	p95 ms	p99 ms	p99.9 ms
login	19,682	0	5,000.76	5,001.00	5,002.00	5,044.64
feed	665	665	30.15	218.20	343.56	904.28
comment	663	663	23.18	147.60	504.00	922.31
post	65	65	22.52	75.00	346.08	650.21
like	436	436	28.09	248.25	407.45	638.84
comments	127	127	22.68	209.50	319.80	396.67
unlike	220	220	26.27	175.85	433.59	658.22

Ramp login: 56 / 7,527 successful; mean 4,985.83 ms; p99 5,002.00 ms; p99.9 5,002.00 ms.

Rate definition	Attempted	Successful	HTTP completed
Mean / configured 240 s	91.08	9.07	91.08
Peak 1 s inside window	115.00	19.00	118.00
Peak rolling 10 s	94.90	10.50	95.20

Window spillover counts: {'early': {'attempted': 0, 'successful': 0, 'http_completed': 0}, 'late': {'attempted': 0, 'successful': 0, 'http_completed': 404}}. Completed journeys: 665. Generator resource errors: 0.

RDS average / maximum: CPU 2.49 / 2.60%; connections 21.00 / 21.00; freeable 10.21 GiB; read/write 0.26 / 14.30 IOPS.

Validity: qualified. Flags: none recorded. Fixture restored: True.

Octane / 1,000 users

Detailed fixed-load observation | PASS | trial 35749f424096-0002

Attempts / successful	Operation success	Authenticated	Timeout / network
39,281 / 39,281	100.0000%	1,000 / 1,000	0 / 0

Mean ms	p50 ms	p95 ms	p99 ms	p99.9 ms
6.47	6.00	11.00	13.00	18.00

VM phase PRIMARY	VM CPU mean/peak %	VM RAM est. mean/peak MiB	n
ramp	57.25 / 90.88	1,634.27 / 1,700.63	37
steady	26.50 / 60.17	1,606.25 / 1,631.93	52

Container SECONDARY	CPU mean/peak %	Working set mean/peak MiB
ramp	53.41 / 95.86	914.84 / 957.80
steady	21.03 / 26.65	896.80 / 897.40

Steady endpoint	Attempts	Success	Mean ms	p95 ms	p99 ms	p99.9 ms
login	1	1	152.00	152.00	152.00	152.00
comment	11,910	11,910	5.70	8.00	11.00	18.00
feed	11,900	11,900	8.86	12.00	15.00	18.10
like	9,247	9,247	5.33	8.00	11.00	14.00
post	1,195	1,195	6.02	9.00	11.00	14.81
unlike	2,657	2,657	5.21	8.00	10.00	18.00
comments	2,371	2,371	4.42	7.00	9.00	14.63

Ramp login: 999 / 999 successful; mean 211.68 ms; p99 2,473.80 ms; p99.9 4,105.97 ms.

Rate definition	Attempted	Successful	HTTP completed
Mean / configured 240 s	163.67	163.67	163.67
Peak 1 s inside window	215.00	214.00	214.00
Peak rolling 10 s	169.90	169.90	169.90

Window spillover counts: {'early': {'attempted': 0, 'successful': 0, 'http_completed': 0}, 'late': {'attempted': 0, 'successful': 1, 'http_completed': 1}}. Completed journeys: 11,899. Generator resource errors: 0.

RDS average / maximum: CPU 9.55 / 9.85%; connections 21.00 / 21.00; freeable 10.21 GiB; read/write 0.26 / 110.70 IOPS.

Validity: qualified. Flags: none recorded. Fixture restored: True.

Historical performance / latency

Group 1/12 | cloud-smoke, native-capacity | recorded diagnostic observations, not a valid-ranking table.

ID suffix	Runtime / cell	Users or RPS	Useful ops/s	Measured mean ms	p50 ms	p95 ms	p99 ms	Validity / result
1182-0000	PyB 3.10.22	25 RPS	25.02	12.72	6.00	120.00	152.00	Smoke
1e4d-0000	java-spring-boot	1,309	214.07	n/a	2.00	4.00	5.00	pass
1e4d-0001	java-spring-boot	1,409	230.40	n/a	2.00	4.00	5.00	pass
1e4d-0002	java-spring-boot	2,818	0.01	n/a	5,001.00	5,001.00	5,002.00	fail
1e4d-0003	java-spring-boot	2,113	0.00	n/a	5,001.00	5,001.00	5,002.00	fail
1e4d-0004	java-spring-boot	1,761	288.37	n/a	2.00	4.00	5.00	pass
1e4d-0005	java-spring-boot	1,937	317.12	n/a	2.00	4.00	5.00	pass
1e4d-0006	java-spring-boot	2,025	0.00	n/a	5,001.00	5,001.00	5,002.00	fail
1e4d-0007	java-spring-boot	2,025	0.01	n/a	5,001.00	5,001.00	5,002.00	fail
1e4d-0008	php-laravel	1,937	0.00	n/a	5,001.00	5,001.00	5,002.00	fail
1e4d-0009	php-laravel	968	0.00	n/a	5,001.00	5,001.00	5,002.00	fail
1e4d-0010	php-laravel	484	0.00	n/a	5,001.00	5,001.00	5,002.00	fail
1e4d-0011	php-laravel	242	37.83	n/a	233.00	579.00	690.00	pass
1e4d-0012	php-laravel	363	0.00	n/a	5,001.00	5,001.00	5,002.00	fail
1e4d-0013	php-laravel	302	39.15	n/a	1,349.00	3,006.05	3,402.00	pass
1e4d-0014	php-laravel	363	0.00	n/a	5,001.00	5,001.00	5,002.00	Flagged 2
1e4d-0015	php-laravel	363	0.00	n/a	5,001.00	5,001.00	5,002.00	fail
1e4d-0016	go-net-http	302	49.38	n/a	2.00	6.00	6.00	pass

Capacity latency columns are steady; pilot/smoke columns are their recorded measurement window. Missing old steady means remain n/a, never replaced by all-phase averages. Flagged/FD-invalid/mock/preflight values are diagnostic and cannot define capacity. Full trial IDs and protocols remain in the updated 234-trial JSON; the original 228-row CSV is retained.

Historical full-VM resources

Group 1/12 | cloud-smoke, native-capacity | PRIMARYwhole VM CPU and estimated used RAM.

ID suffix	Runtime / cell	Users or RPS	n	CPU mean %	CPU peak %	CPU p99 %	CPU p99.9 %	VM RAM mean MiB	VM RAM peak MiB	VM RAM p99 MiB	VM RAM p99.9 MiB	Validity
1182-0000	PyB 3.10.22	25 RPS	13	16.6	21.0	21.0	21.0	884.4	925.3	923.1	925.1	Smoke
1e4d-0000	java-spring-boot	1,309	66	7.0	8.9	8.2	8.8	1,203.5	1,218.3	1,217.8	1,218.2	pass
1e4d-0001	java-spring-boot	1,409	66	7.3	9.2	8.7	9.2	1,467.5	1,488.2	1,487.3	1,488.1	pass
1e4d-0002	java-spring-boot	2,818	64	99.9	100.0	100.0	100.0	2,747.7	2,935.1	2,926.9	2,934.3	fail
1e4d-0003	java-spring-boot	2,113	64	99.9	100.0	100.0	100.0	2,871.6	2,952.0	2,949.5	2,951.8	fail
1e4d-0004	java-spring-boot	1,761	66	8.0	10.1	9.2	10.0	2,041.7	2,062.9	2,060.6	2,062.7	pass
1e4d-0005	java-spring-boot	1,937	66	8.3	10.3	10.0	10.3	1,860.7	1,957.4	1,957.1	1,957.4	pass
1e4d-0006	java-spring-boot	2,025	64	99.9	100.0	100.0	100.0	2,929.7	2,992.5	2,992.4	2,992.5	fail
1e4d-0007	java-spring-boot	2,025	64	99.9	100.0	100.0	100.0	2,938.6	2,965.9	2,964.7	2,965.8	fail
1e4d-0008	php-laravel	1,937	37	100.0	100.0	100.0	100.0	895.8	927.1	926.5	927.0	fail
1e4d-0009	php-laravel	968	39	100.0	100.0	100.0	100.0	882.7	923.7	920.4	923.4	fail
1e4d-0010	php-laravel	484	40	100.0	100.0	100.0	100.0	869.2	907.9	904.7	907.6	fail
1e4d-0011	php-laravel	242	63	90.0	100.0	99.6	100.0	743.0	755.8	753.2	755.5	pass
1e4d-0012	php-laravel	363	39	100.0	100.0	100.0	100.0	861.6	894.6	893.3	894.5	fail
1e4d-0013	php-laravel	302	61	100.0	100.0	100.0	100.0	770.1	877.3	876.5	877.2	pass
1e4d-0014	php-laravel	363	37	100.0	100.0	100.0	100.0	866.4	902.0	898.9	901.7	Flagged 2
1e4d-0015	php-laravel	363	38	100.0	100.0	100.0	100.0	856.0	897.8	895.0	897.5	fail
1e4d-0016	go-net-http	302	68	4.1	5.0	5.0	5.0	693.1	700.3	700.2	700.3	pass

PRIMARY CPU spans both VM vCPUs, all host activity/steal; RAM=MemTotal - MemAvailable, not RSS or all cache. Capacity rows are steady; pilot/smoke use their measurement window. Sparsep 99/p99.9 are weak descriptive estimates. FD/invalid flags override capacity inference. No boot counter is represented as trial usage.

Historical container: secondary

Group 1/12 | cloud-smoke, native-capacity | SECONDARY app container CPU and sampled working set.

ID suffix	Runtime / cell	Users or RPS	n	CPU mean %	CPU peak %	CPU p99 %	CPU p99.9 %	Mem mean MiB	Mem peak MiB	Mem p99 MiB	Mem p99.9 MiB	Validity
1182-0000	PyB 3.10.22	25 RPS	13	13.1	32.6	32.0	32.5	172.2	194.5	192.6	194.3	Smoke
1e4d-0000	java-spring-boot	1,309	66	3.1	4.2	4.0	4.2	544.2	558.2	558.2	558.2	pass
1e4d-0001	java-spring-boot	1,409	66	3.4	6.8	5.7	6.7	786.6	804.3	804.2	804.3	pass
1e4d-0002	java-spring-boot	2,818	64	97.8	99.1	99.1	99.1	2,100.5	2,302.0	2,302.0	2,302.0	fail
1e4d-0003	java-spring-boot	2,113	64	97.7	99.0	98.8	99.0	2,237.1	2,317.3	2,317.3	2,317.3	fail
1e4d-0004	java-spring-boot	1,761	66	4.0	6.0	5.7	6.0	1,391.4	1,406.0	1,406.0	1,406.0	pass
1e4d-0005	java-spring-boot	1,937	66	4.3	12.8	8.0	12.3	1,210.2	1,310.7	1,310.7	1,310.7	pass
1e4d-0006	java-spring-boot	2,025	64	97.9	99.2	99.2	99.2	2,280.0	2,320.4	2,319.7	2,320.3	fail
1e4d-0007	java-spring-boot	2,025	64	98.0	99.4	99.3	99.4	2,310.4	2,332.7	2,332.7	2,332.7	fail
1e4d-0008	php-laravel	1,937	37	96.4	108.4	104.4	108.0	256.2	283.0	282.9	283.0	fail
1e4d-0009	php-laravel	968	39	96.9	107.4	104.2	107.0	213.5	244.2	244.0	244.2	fail
1e4d-0010	php-laravel	484	40	96.8	97.6	97.5	97.6	210.4	235.0	234.9	235.0	fail
1e4d-0011	php-laravel	242	63	90.0	98.6	98.4	98.6	95.5	95.9	95.8	95.9	pass
1e4d-0012	php-laravel	363	39	96.9	97.7	97.7	97.7	200.4	230.8	230.8	230.8	fail
1e4d-0013	php-laravel	302	61	97.9	98.7	98.7	98.7	117.8	225.6	225.6	225.6	pass
1e4d-0014	php-laravel	363	37	96.8	99.3	98.8	99.3	204.6	230.7	230.7	230.7	Flagged 2
1e4d-0015	php-laravel	363	38	96.9	98.1	98.0	98.1	206.7	230.4	230.3	230.4	fail
1e4d-0016	go-net-http	302	68	0.9	1.2	1.1	1.2	32.8	35.0	34.1	34.9	pass

ContainerCPU 100% = both allocated vCPUs; memory = sampled Docker working set MiB. Capacity rows are steady; pilot/smoke use their measurement window. Sparsep 99/p99.9 are weak descriptive estimates. FD/invalid flags override capacity inference. No boot counter is represented as trial usage.

Historical performance / latency

Group 2/12 | native-capacity | recorded diagnostic observations, not a valid-ranking table.

ID suffix	Runtime / cell	Users or RPS	Useful ops/s	Measured mean ms	p50 ms	p95 ms	p99 ms	Validity / result
1e4d-0017	go-net-http	363	59.23	n/a	2.00	6.00	6.00	pass
1e4d-0018	go-net-http	726	119.01	n/a	2.00	5.00	6.00	pass
1e4d-0019	go-net-http	1,452	237.65	n/a	2.00	5.00	6.00	pass
1e4d-0020	go-net-http	2,904	475.84	n/a	2.00	4.00	5.00	pass
1e4d-0021	go-net-http	5,808	21.11	n/a	5,001.00	5,001.00	5,002.00	fail
1e4d-0022	go-net-http	4,356	38.88	n/a	5,001.00	5,001.00	5,002.00	fail
1e4d-0023	go-net-http	3,630	106.94	n/a	5,000.00	5,001.00	5,002.00	fail
1e4d-0024	go-net-http	3,267	422.90	n/a	582.00	5,001.00	5,001.00	fail
1e4d-0025	go-net-http	3,085	505.52	n/a	2.00	4.00	5.00	pass
1e4d-0026	go-net-http	3,176	520.56	n/a	2.00	4.00	5.00	pass
1e4d-0027	go-net-http	3,267	421.95	n/a	622.00	5,000.00	5,001.00	fail
1e4d-0028	rust-axum	3,176	165.08	n/a	5,000.00	5,001.00	5,001.00	FD invalid
1e4d-0029	rust-axum	1,588	165.10	n/a	3.00	5,001.00	5,001.00	FD invalid
1e4d-0030	rust-axum	794	130.06	n/a	2.00	5.00	6.00	pass
1e4d-0031	rust-axum	1,191	165.00	n/a	2.00	5,001.00	5,001.00	FD invalid
1e4d-0032	rust-axum	992	162.30	n/a	2.00	5.00	6.00	pass
1e4d-0033	rust-axum	1,091	164.86	n/a	2.00	6.00	5,001.00	FD invalid
1e4d-0034	rust-axum	1,091	164.98	n/a	2.00	6.00	5,001.00	FD invalid

Capacity latency columns are steady; pilot/smoke columns are their recorded measurement window. Missing old steady means remain n/a, never replaced by all-phase averages. Flagged/FD-invalid/mock/preflight values are diagnostic and cannot define capacity. Full trial IDs and protocols remain in the updated 234-trial JSON; the original 228-row CSV is retained.

Historical full-VM resources

Group 2/12 | native-capacity | PRIMARYwhole VM CPU and estimated used RAM.

ID suffix	Runtime / cell	Users or RPS	n	CPU mean %	CPU peak %	CPU p99 %	CPU p99.9 %	VM RAM mean MiB	VM RAM peak MiB	VM RAM p99 MiB	VM RAM p99.9 MiB	Validity
1e4d-0017	go-net-http	363	66	4.3	5.1	5.0	5.1	737.2	760.1	759.2	760.0	pass
1e4d-0018	go-net-http	726	66	5.3	6.0	6.0	6.0	746.2	755.1	754.3	755.0	pass
1e4d-0019	go-net-http	1,452	66	6.7	7.7	7.6	7.7	741.8	750.6	750.4	750.6	pass
1e4d-0020	go-net-http	2,904	66	8.8	9.8	9.8	9.8	903.3	912.6	911.9	912.5	pass
1e4d-0021	go-net-http	5,808	63	99.9	100.0	100.0	100.0	1,078.5	1,123.0	1,122.7	1,123.0	fail
1e4d-0022	go-net-http	4,356	64	99.9	100.0	100.0	100.0	1,044.7	1,074.1	1,071.2	1,073.8	fail
1e4d-0023	go-net-http	3,630	64	99.9	100.0	100.0	100.0	1,047.5	1,097.7	1,095.8	1,097.5	fail
1e4d-0024	go-net-http	3,267	64	99.9	100.0	100.0	100.0	1,062.2	1,088.3	1,084.9	1,087.9	fail
1e4d-0025	go-net-http	3,085	65	8.9	9.7	9.7	9.7	880.6	888.4	888.2	888.4	pass
1e4d-0026	go-net-http	3,176	65	9.1	9.7	9.7	9.7	885.0	893.6	892.8	893.5	pass
1e4d-0027	go-net-http	3,267	64	99.9	100.0	100.0	100.0	1,054.5	1,091.3	1,083.5	1,090.5	fail
1e4d-0028	rust-axum	3,176	67	5.8	6.8	6.6	6.7	713.5	719.3	719.1	719.3	FD invalid
1e4d-0029	rust-axum	1,588	66	5.6	6.3	6.3	6.3	698.2	706.4	704.8	706.2	FD invalid
1e4d-0030	rust-axum	794	66	5.4	12.4	9.5	12.1	689.1	741.2	714.2	738.5	pass
1e4d-0031	rust-axum	1,191	67	5.7	6.6	6.5	6.6	690.3	698.1	697.6	698.1	FD invalid
1e4d-0032	rust-axum	992	68	5.7	6.6	6.6	6.6	691.1	700.5	698.4	700.3	pass
1e4d-0033	rust-axum	1,091	68	5.7	6.6	6.5	6.6	686.6	694.7	693.5	694.6	FD invalid
1e4d-0034	rust-axum	1,091	68	5.6	6.6	6.4	6.5	683.5	691.2	690.0	691.1	FD invalid

PRIMARY CPU spans both VM vCPUs, all host activity/steal; RAM=MemTotal - MemAvailable, not RSS or all cache. Capacity rows are steady; pilot/smoke use their measurement window. Sparsep 99/p99.9 are weak descriptive estimates. FD/invalid flags override capacity inference. No boot counter is represented as trial usage.

Historical container: secondary

Group 2/12 | native-capacity | SECONDARY app container CPU and sampled working set.

ID suffix	Runtime / cell	Users or RPS	n	CPU mean %	CPU peak %	CPU p99 %	CPU p99.9 %	Mem mean MiB	Mem peak MiB	Mem p99 MiB	Mem p99.9 MiB	Validity
1e4d-0017	go-net-http	363	66	1.0	1.3	1.3	1.3	67.7	91.2	91.2	91.2	pass
1e4d-0018	go-net-http	726	66	1.7	2.1	2.1	2.1	86.4	98.1	98.1	98.1	pass
1e4d-0019	go-net-http	1,452	66	2.8	3.6	3.5	3.6	83.8	93.6	93.5	93.6	pass
1e4d-0020	go-net-http	2,904	66	4.3	5.6	5.6	5.6	231.3	232.8	232.7	232.8	pass
1e4d-0021	go-net-http	5,808	63	97.1	98.3	98.2	98.3	435.4	450.8	450.6	450.8	fail
1e4d-0022	go-net-http	4,356	64	97.3	98.4	98.3	98.4	414.2	425.6	424.3	425.5	fail
1e4d-0023	go-net-http	3,630	64	95.9	98.3	98.3	98.3	417.3	468.8	439.4	465.9	fail
1e4d-0024	go-net-http	3,267	64	97.5	98.4	98.3	98.4	433.8	443.7	443.6	443.7	fail
1e4d-0025	go-net-http	3,085	65	4.5	5.8	5.7	5.8	237.0	238.1	238.0	238.1	pass
1e4d-0026	go-net-http	3,176	65	4.6	5.9	5.9	5.9	216.1	216.7	216.7	216.7	pass
1e4d-0027	go-net-http	3,267	64	96.1	99.3	98.7	99.2	428.9	442.2	441.8	442.2	fail
1e4d-0028	rust-axum	3,176	67	1.7	1.9	1.9	1.9	46.4	46.7	46.6	46.7	FD invalid
1e4d-0029	rust-axum	1,588	66	1.7	1.9	1.9	1.9	39.7	39.9	39.9	39.9	FD invalid
1e4d-0030	rust-axum	794	66	1.5	1.7	1.7	1.7	31.3	31.5	31.5	31.5	pass
1e4d-0031	rust-axum	1,191	67	1.7	1.9	1.9	1.9	38.8	39.1	39.1	39.1	FD invalid
1e4d-0032	rust-axum	992	68	1.7	2.7	2.2	2.6	38.2	38.5	38.5	38.5	pass
1e4d-0033	rust-axum	1,091	68	1.7	2.0	2.0	2.0	38.8	39.2	39.2	39.2	FD invalid
1e4d-0034	rust-axum	1,091	68	1.7	2.0	1.9	2.0	39.0	39.3	39.2	39.3	FD invalid

ContainerCPU 100% = both allocated vCPUs; memory = sampled Docker working set MiB. Capacity rows are steady; pilot/smoke use their measurement window. Sparsep 99/p99.9 are weak descriptive estimates. FD/invalid flags override capacity inference. No boot counter is represented as trial usage.

Historical performance / latency

Group 3/12 | native-capacity | recorded diagnostic observations, not a valid-ranking table.

ID suffix	Runtime / cell	Users or RPS	Useful ops/s	Measured mean ms	p50 ms	p95 ms	p99 ms	Validity / result
1e4d-0035	node-express	992	162.17	n/a	3.00	7.00	9.00	pass
1e4d-0036	node-express	1,091	178.38	n/a	3.00	7.00	9.00	pass
1e4d-0037	node-express	2,182	357.43	n/a	3.00	8.00	10.00	pass
1e4d-0038	node-express	4,364	47.66	n/a	2.00	18.00	5,001.00	fail
1e4d-0039	node-express	3,273	190.79	n/a	3.00	5,000.00	5,001.00	fail
1e4d-0040	node-express	2,727	446.90	n/a	3.00	8.00	11.00	pass
1e4d-0041	node-express	3,000	327.72	n/a	4.00	5,000.00	5,001.00	fail
1e4d-0042	node-express	2,863	399.75	n/a	5.00	5,000.00	5,001.00	fail
1e4d-0043	node-express	2,795	458.10	n/a	3.00	8.00	11.00	pass
1e4d-0044	node-express	2,863	376.20	n/a	5.00	5,000.00	5,001.00	fail
1e4d-0045	bun-express	2,795	458.04	n/a	3.00	7.00	9.00	pass
1e4d-0046	bun-express	2,863	469.03	n/a	3.00	7.00	9.00	pass
1e4d-0047	bun-express	5,726	28.44	n/a	1.00	7.00	5,001.00	fail
1e4d-0048	bun-express	4,294	50.62	n/a	1.00	9.00	5,001.00	fail
1e4d-0049	bun-express	3,578	92.51	n/a	1.00	16.00	5,001.00	fail
1e4d-0050	bun-express	3,220	211.78	n/a	2.00	5,000.00	5,001.00	fail
1e4d-0051	bun-express	3,041	312.91	n/a	3.00	5,000.00	5,001.00	fail
1e4d-0052	bun-express	2,952	369.07	n/a	4.00	5,000.00	5,001.00	fail

Capacity latency columns are steady; pilot/smoke columns are their recorded measurement window. Missing old steady means remain n/a, never replaced by all-phase averages. Flagged/FD-invalid/mock/preflight values are diagnostic and cannot define capacity. Full trial IDs and protocols remain in the updated 234-trial JSON; the original 228-row CSV is retained.

Historical full-VM resources

Group 3/12 | native-capacity | PRIMARYwhole VM CPU and estimated used RAM.

ID suffix	Runtime / cell	Users or RPS	n	CPU mean %	CPU peak %	CPU p99 %	CPU p99.9 %	VM RAM mean MiB	VM RAM peak MiB	VM RAM p99 MiB	VM RAM p99.9 MiB	Validity
1e4d-0035	node-express	992	66	10.6	12.1	12.0	12.1	892.7	909.8	908.8	909.7	pass
1e4d-0036	node-express	1,091	66	11.1	12.0	12.0	12.0	891.5	905.5	903.1	905.2	pass
1e4d-0037	node-express	2,182	65	17.7	18.8	18.8	18.8	927.2	940.9	939.4	940.7	pass
1e4d-0038	node-express	4,364	63	100.0	100.0	100.0	100.0	1,080.9	1,158.0	1,138.0	1,156.0	fail
1e4d-0039	node-express	3,273	62	100.0	100.0	100.0	100.0	1,118.7	1,161.7	1,161.2	1,161.6	fail
1e4d-0040	node-express	2,727	65	21.6	23.4	23.1	23.4	941.0	954.0	952.6	953.9	pass
1e4d-0041	node-express	3,000	63	100.0	100.0	100.0	100.0	1,101.1	1,181.1	1,165.2	1,179.5	fail
1e4d-0042	node-express	2,863	62	100.0	100.0	100.0	100.0	1,061.3	1,129.8	1,114.1	1,128.3	fail
1e4d-0043	node-express	2,795	65	21.8	23.5	23.4	23.5	947.9	961.7	961.5	961.7	pass
1e4d-0044	node-express	2,863	64	100.0	100.0	100.0	100.0	1,080.4	1,141.9	1,130.8	1,140.8	fail
1e4d-0045	bun-express	2,795	66	15.4	17.1	16.9	17.1	779.3	796.8	794.5	796.6	pass
1e4d-0046	bun-express	2,863	65	15.4	16.4	16.3	16.4	790.6	807.7	806.8	807.6	pass
1e4d-0047	bun-express	5,726	58	100.0	100.0	100.0	100.0	970.3	1,037.5	1,032.1	1,036.9	fail
1e4d-0048	bun-express	4,294	58	100.0	100.0	100.0	100.0	931.9	991.8	980.5	990.6	fail
1e4d-0049	bun-express	3,578	59	100.0	100.0	100.0	100.0	915.7	976.9	976.0	976.8	fail
1e4d-0050	bun-express	3,220	60	100.0	100.0	100.0	100.0	938.0	1,011.7	999.1	1,010.5	fail
1e4d-0051	bun-express	3,041	59	100.0	100.0	100.0	100.0	896.6	971.6	957.1	970.2	fail
1e4d-0052	bun-express	2,952	59	100.0	100.0	100.0	100.0	894.4	953.6	949.1	953.1	fail

PRIMARY CPU spans both VM vCPUs, all host activity/steal; RAM=MemTotal - MemAvailable, not RSS or all cache. Capacity rows are steady; pilot/smoke use their measurement window. Sparsep 99/p99.9 are weak descriptive estimates. FD/invalid flags override capacity inference. No boot counter is represented as trial usage.

Historical container: secondary

Group 3/12 | native-capacity | SECONDARY app container CPU and sampled working set.

ID suffix	Runtime / cell	Users or RPS	n	CPU mean %	CPU peak %	CPU p99 %	CPU p99.9 %	Mem mean MiB	Mem peak MiB	Mem p99 MiB	Mem p99.9 MiB	Validity
1e4d-0035	node-express	992	66	6.1	7.9	7.9	7.9	286.6	292.7	292.4	292.7	pass
1e4d-0036	node-express	1,091	66	6.9	9.0	8.8	9.0	242.4	244.9	244.8	244.9	pass
1e4d-0037	node-express	2,182	65	12.4	13.9	13.9	13.9	265.9	268.1	267.8	268.1	pass
1e4d-0038	node-express	4,364	63	94.8	97.7	97.4	97.6	392.9	419.3	419.3	419.3	fail
1e4d-0039	node-express	3,273	62	95.9	97.7	97.6	97.7	391.5	420.4	420.2	420.4	fail
1e4d-0040	node-express	2,727	65	15.8	18.3	17.9	18.2	272.6	274.2	274.1	274.2	pass
1e4d-0041	node-express	3,000	63	96.5	98.0	97.8	98.0	383.8	410.2	410.1	410.2	fail
1e4d-0042	node-express	2,863	62	96.5	97.6	97.5	97.6	383.6	408.6	408.4	408.6	fail
1e4d-0043	node-express	2,795	65	15.9	19.0	18.6	19.0	273.6	274.9	274.8	274.9	pass
1e4d-0044	node-express	2,863	64	95.1	97.6	97.5	97.6	387.9	415.3	414.9	415.3	fail
1e4d-0045	bun-express	2,795	66	10.0	14.2	12.8	14.0	168.0	173.8	173.7	173.8	pass
1e4d-0046	bun-express	2,863	65	10.0	11.6	11.5	11.6	121.5	126.1	126.0	126.1	pass
1e4d-0047	bun-express	5,726	58	94.6	96.9	96.8	96.9	260.5	288.2	287.2	288.1	fail
1e4d-0048	bun-express	4,294	58	95.9	97.4	97.4	97.4	239.0	271.0	270.7	271.0	fail
1e4d-0049	bun-express	3,578	59	95.1	97.6	97.6	97.6	235.4	263.2	263.0	263.2	fail
1e4d-0050	bun-express	3,220	60	96.7	98.3	97.9	98.2	240.9	261.1	261.1	261.1	fail
1e4d-0051	bun-express	3,041	59	96.6	98.0	97.8	98.0	240.1	265.5	264.9	265.4	fail
1e4d-0052	bun-express	2,952	59	96.7	97.6	97.6	97.6	243.0	263.9	263.6	263.9	fail

ContainerCPU 100% = both allocated vCPUs; memory = sampled Docker working set MiB. Capacity rows are steady; pilot/smoke use their measurement window. Sparsep 99/p99.9 are weak descriptive estimates. FD/invalid flags override capacity inference. No boot counter is represented as trial usage.

Historical performance / latency

Group 4/12 | native-capacity, native-capacity-preflight-1791217587, pilot | recorded diagnostic observations, not a valid-ranking table.

ID suffix	Runtime / cell	Users or RPS	Useful ops/s	Measured mean ms	p50 ms	p95 ms	p99 ms	Validity / result
1e4d-0053	bun-express	2,952	447.82	n/a	4.00	13.00	5,001.00	fail
e775-0000	java-spring-boot	1,309	n/a	n/a	n/a	n/a	n/a	Flagged 5
5630-0000	PyB 3.15.0rc3	200 RPS	174.97	1,766.04	1,880.00	4,527.70	5,034.00	Pilot
5630-0001	PyB 3.14.8	25 RPS	25.03	12.82	6.00	33.50	159.50	Pilot
5630-0002	PyB 3.15.0rc3	1,600 RPS	173.13	2,316.04	2,280.00	4,864.85	5,084.27	Pilot
5630-0003	PyB 3.14.8	400 RPS	175.73	2,224.36	2,237.50	4,712.35	5,078.00	Pilot
5630-0004	PyB 3.13.16	800 RPS	178.37	2,250.45	2,132.00	4,648.80	5,057.48	Pilot
5630-0005	PyB 3.10.22	25 RPS	25.00	13.15	6.00	35.15	165.53	Pilot
5630-0006	PyB 3.12.13	25 RPS	25.03	12.56	6.00	21.50	166.50	Pilot
5630-0007	PyB 3.11.17	100 RPS	100.03	20.45	6.00	128.00	366.00	Pilot
5630-0008	PyB 3.14.8	1,600 RPS	179.53	2,237.89	2,060.00	4,827.40	5,187.76	Pilot
5630-0009	PyB 3.12.13	1,600 RPS	173.03	2,268.23	2,137.50	5,021.00	5,290.49	Pilot
5630-0010	PyB 3.10.22	800 RPS	170.83	2,329.18	2,233.00	4,618.85	5,134.56	Pilot
5630-0011	PyB 3.11.17	1,600 RPS	176.13	2,309.97	2,345.50	4,495.35	5,207.88	Pilot
5630-0012	PyB 3.13.16	200 RPS	179.30	1,696.35	1,630.50	4,459.05	5,047.00	Pilot
5630-0013	PyB 3.12.13	100 RPS	100.03	20.09	6.00	130.00	328.00	Pilot
5630-0014	PyB 3.14.8	800 RPS	179.07	2,241.03	2,069.00	4,583.50	5,153.14	Pilot
5630-0015	PyB 3.13.16	400 RPS	176.03	2,239.87	2,127.00	4,758.85	5,099.73	Pilot

Capacity latency columns are steady; pilot/smoke columns are their recorded measurement window. Missing old steady means remain n/a, never replaced by all-phase averages. Flagged/FD-invalid/mock/preflight values are diagnostic and cannot define capacity. Full trial IDs and protocols remain in the updated 234-trial JSON; the original 228-row CSV is retained.

Historical full-VM resources

Group 4/12 | native-capacity, native-capacity-preflight-1791217587, pilot | PRIMARYwhole VM CPU and estimated used RAM.

ID suffix	Runtime / cell	Users or RPS	n	CPU mean %	CPU peak %	CPU p99 %	CPU p99.9 %	VM RAM mean MiB	VM RAM peak MiB	VM RAM p99 MiB	VM RAM p99.9 MiB	Validity
1e4d-0053	bun-express	2,952	58	84.6	100.0	100.0	100.0	939.4	1,024.0	1,019.2	1,023.6	fail
e775-0000	java-spring-boot	1,309	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	Flagged 5
5630-0000	PyB 3.15.0rc3	200 RPS	7	92.2	100.0	100.0	100.0	1,175.6	1,203.0	1,202.4	1,202.9	Pilot
5630-0001	PyB 3.14.8	25 RPS	6	16.9	20.8	20.8	20.8	958.5	996.4	994.9	996.3	Pilot
5630-0002	PyB 3.15.0rc3	1,600 RPS	7	89.0	100.0	100.0	100.0	1,189.4	1,217.5	1,217.5	1,217.5	Pilot
5630-0003	PyB 3.14.8	400 RPS	7	92.4	100.0	100.0	100.0	1,154.2	1,199.7	1,199.4	1,199.7	Pilot
5630-0004	PyB 3.13.16	800 RPS	7	92.2	100.0	100.0	100.0	1,168.7	1,202.9	1,202.4	1,202.8	Pilot
5630-0005	PyB 3.10.22	25 RPS	6	17.3	21.5	21.4	21.5	918.9	940.4	940.1	940.4	Pilot
5630-0006	PyB 3.12.13	25 RPS	7	16.0	21.6	21.6	21.6	959.5	970.8	970.8	970.8	Pilot
5630-0007	PyB 3.11.17	100 RPS	6	52.5	73.3	72.9	73.3	999.4	1,098.5	1,093.6	1,098.0	Pilot
5630-0008	PyB 3.14.8	1,600 RPS	7	88.8	100.0	100.0	100.0	1,194.5	1,253.6	1,251.6	1,253.4	Pilot
5630-0009	PyB 3.12.13	1,600 RPS	7	88.9	100.0	100.0	100.0	1,218.3	1,268.9	1,267.2	1,268.7	Pilot
5630-0010	PyB 3.10.22	800 RPS	7	92.0	100.0	100.0	100.0	1,201.9	1,230.1	1,229.9	1,230.1	Pilot
5630-0011	PyB 3.11.17	1,600 RPS	7	91.7	100.0	100.0	100.0	1,188.5	1,262.9	1,261.7	1,262.7	Pilot
5630-0012	PyB 3.13.16	200 RPS	8	87.8	100.0	100.0	100.0	1,160.4	1,254.3	1,252.4	1,254.1	Pilot
5630-0013	PyB 3.12.13	100 RPS	7	52.7	86.0	85.2	85.9	1,021.8	1,051.1	1,050.8	1,051.1	Pilot
5630-0014	PyB 3.14.8	800 RPS	7	89.1	100.0	100.0	100.0	1,217.5	1,244.7	1,244.4	1,244.6	Pilot
5630-0015	PyB 3.13.16	400 RPS	7	89.5	100.0	100.0	100.0	1,190.4	1,224.3	1,223.4	1,224.2	Pilot

PRIMARY CPU spans both VM vCPUs, all host activity/steal; RAM=MemTotal - MemAvailable, not RSS or all cache. Capacity rows are steady; pilot/smoke use their measurement window. Sparsep 99/p99.9 are weak descriptive estimates. FD/invalid flags override capacity inference. No boot counter is represented as trial usage.

Historical container: secondary

Group 4/12 | native-capacity, native-capacity-preflight-1791217587, pilot | SECONDARY app container CPU and sampled working set.

ID suffix	Runtime / cell	Users or RPS	n	CPU mean %	CPU peak %	CPU p99 %	CPU p99.9 %	Mem mean MiB	Mem peak MiB	Mem p99 MiB	Mem p99.9 MiB	Validity
1e4d-0053	bun-express	2,952	58	79.9	98.0	97.7	98.0	208.7	261.0	260.5	261.0	fail
e775-0000	java-spring-boot	1,309	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	Flagged 5
5630-0000	PyB 3.15.0rc3	200 RPS	7	84.0	97.8	97.8	97.8	421.7	490.4	489.3	490.3	Pilot
5630-0001	PyB 3.14.8	25 RPS	6	22.2	37.5	37.1	37.5	208.2	261.8	258.6	261.5	Pilot
5630-0002	PyB 3.15.0rc3	1,600 RPS	7	97.2	98.0	98.0	98.0	460.4	493.5	492.7	493.4	Pilot
5630-0003	PyB 3.14.8	400 RPS	7	83.2	97.7	97.7	97.7	396.3	474.3	473.6	474.2	Pilot
5630-0004	PyB 3.13.16	800 RPS	7	83.8	97.9	97.9	97.9	405.6	462.0	461.3	461.9	Pilot
5630-0005	PyB 3.10.22	25 RPS	6	14.8	33.9	33.0	33.8	143.8	144.4	144.4	144.4	Pilot
5630-0006	PyB 3.12.13	25 RPS	7	13.9	32.5	31.9	32.5	164.4	164.6	164.6	164.6	Pilot
5630-0007	PyB 3.11.17	100 RPS	6	53.2	95.5	95.2	95.5	206.0	302.6	299.4	302.3	Pilot
5630-0008	PyB 3.14.8	1,600 RPS	7	96.9	98.8	98.7	98.8	393.6	456.0	455.7	456.0	Pilot
5630-0009	PyB 3.12.13	1,600 RPS	7	97.4	98.5	98.5	98.5	421.9	453.3	452.9	453.3	Pilot
5630-0010	PyB 3.10.22	800 RPS	7	97.4	99.0	98.9	99.0	379.4	447.2	445.2	447.0	Pilot
5630-0011	PyB 3.11.17	1,600 RPS	7	90.8	98.0	98.0	98.0	339.9	447.6	444.5	447.3	Pilot
5630-0012	PyB 3.13.16	200 RPS	8	85.0	98.0	98.0	98.0	389.8	462.5	462.4	462.5	Pilot
5630-0013	PyB 3.12.13	100 RPS	7	49.2	85.4	85.4	85.4	200.8	294.1	290.2	293.7	Pilot
5630-0014	PyB 3.14.8	800 RPS	7	97.8	98.6	98.6	98.6	444.3	474.1	474.1	474.1	Pilot
5630-0015	PyB 3.13.16	400 RPS	7	96.9	97.7	97.6	97.7	399.4	450.8	450.2	450.7	Pilot

ContainerCPU 100% = both allocated vCPUs; memory = sampled Docker working set MiB. Capacity rows are steady; pilot/smoke use their measurement window. Sparsep 99/p99.9 are weak descriptive estimates. FD/invalid flags override capacity inference. No boot counter is represented as trial usage.

Historical performance / latency

Group 5/12 | pilot | recorded diagnostic observations, not a valid-ranking table.

ID suffix	Runtime / cell	Users or RPS	Useful ops/s	Measured mean ms	p50 ms	p95 ms	p99 ms	Validity / result
5630-0016	PyB 3.15.0rc3	800 RPS	175.63	2,286.52	2,280.00	4,785.60	5,098.56	Pilot
5630-0017	PyB 3.10.22	50 RPS	50.03	14.06	6.00	123.00	179.00	Pilot
5630-0018	PyB 3.13.16	3,200 RPS	178.23	2,256.82	2,215.00	4,946.55	5,160.36	Pilot
5630-0019	PyB 3.13.16	25 RPS	25.00	12.60	6.00	24.90	161.55	Pilot
5630-0020	PyB 3.12.13	800 RPS	174.97	2,302.37	2,126.00	4,923.00	5,336.16	Pilot
5630-0021	PyB 3.14.8	100 RPS	100.03	19.29	6.00	127.00	316.00	Pilot
5630-0022	PyB 3.13.16	100 RPS	100.03	20.30	6.00	129.00	340.00	Pilot
5630-0023	PyB 3.10.22	100 RPS	100.03	25.33	7.00	155.00	418.00	Pilot
5630-0024	PyB 3.10.22	3,200 RPS	172.60	2,328.46	2,253.00	4,730.40	5,118.52	Pilot
5630-0025	PyB 3.12.13	400 RPS	175.87	2,223.26	2,089.00	4,783.70	5,170.38	Pilot
5630-0026	PyB 3.10.22	400 RPS	170.63	2,272.22	2,214.50	4,976.65	5,136.83	Pilot
5630-0027	PyB 3.15.0rc3	100 RPS	100.03	21.05	6.00	129.00	396.00	Pilot
5630-0028	PyB 3.14.8	3,200 RPS	172.33	2,308.73	2,185.00	5,017.00	5,498.48	Pilot
5630-0029	PyB 3.15.0rc3	50 RPS	50.00	14.09	6.00	123.05	184.04	Pilot
5630-0030	PyB 3.15.0rc3	25 RPS	25.03	12.66	6.00	17.00	166.00	Pilot
5630-0031	PyB 3.15.0rc3	3,200 RPS	174.43	2,305.85	2,277.50	4,801.65	5,259.39	Pilot
5630-0032	PyB 3.11.17	3,200 RPS	178.13	2,299.02	2,197.00	4,394.80	5,080.80	Pilot
5630-0033	PyB 3.10.22	1,600 RPS	173.63	2,337.80	2,271.50	4,704.00	5,092.58	Pilot

Capacity latency columns are steady; pilot/smoke columns are their recorded measurement window. Missing old steady means remain n/a, never replaced by all-phase averages. Flagged/FD-invalid/mock/preflight values are diagnostic and cannot define capacity. Full trial IDs and protocols remain in the updated 234-trial JSON; the original 228-row CSV is retained.

Historical full-VM resources

Group 5/12 | pilot | PRIMARYwhole VM CPU and estimated used RAM.

ID suffix	Runtime / cell	Users or RPS	n	CPU mean %	CPU peak %	CPU p99 %	CPU p99.9 %	VM RAM mean MiB	VM RAM peak MiB	VM RAM p99 MiB	VM RAM p99.9 MiB	Validity
5630-0016	PyB 3.15.0rc3	800 RPS	7	92.2	100.0	100.0	100.0	1,241.1	1,265.9	1,265.7	1,265.9	Pilot
5630-0017	PyB 3.10.22	50 RPS	7	28.9	43.4	42.9	43.3	996.0	1,047.2	1,044.4	1,046.9	Pilot
5630-0018	PyB 3.13.16	3,200 RPS	7	91.2	100.0	100.0	100.0	1,273.8	1,322.7	1,322.2	1,322.7	Pilot
5630-0019	PyB 3.13.16	25 RPS	7	16.2	21.4	21.4	21.4	1,049.8	1,062.0	1,062.0	1,062.0	Pilot
5630-0020	PyB 3.12.13	800 RPS	7	89.4	100.0	100.0	100.0	1,252.2	1,290.9	1,290.2	1,290.8	Pilot
5630-0021	PyB 3.14.8	100 RPS	7	52.5	86.4	85.5	86.3	1,121.1	1,202.1	1,200.1	1,201.9	Pilot
5630-0022	PyB 3.13.16	100 RPS	6	53.3	72.5	72.0	72.4	1,116.0	1,172.3	1,169.9	1,172.0	Pilot
5630-0023	PyB 3.10.22	100 RPS	7	56.4	76.2	75.7	76.2	1,075.8	1,162.4	1,157.2	1,161.9	Pilot
5630-0024	PyB 3.10.22	3,200 RPS	7	91.0	100.0	100.0	100.0	1,259.4	1,324.8	1,323.5	1,324.7	Pilot
5630-0025	PyB 3.12.13	400 RPS	7	89.6	100.0	100.0	100.0	1,274.6	1,312.5	1,311.6	1,312.4	Pilot
5630-0026	PyB 3.10.22	400 RPS	7	92.6	100.0	100.0	100.0	1,280.0	1,309.5	1,309.1	1,309.5	Pilot
5630-0027	PyB 3.15.0rc3	100 RPS	6	55.5	72.3	72.0	72.3	1,142.3	1,204.3	1,202.0	1,204.0	Pilot
5630-0028	PyB 3.14.8	3,200 RPS	8	87.5	100.0	100.0	100.0	1,267.6	1,337.6	1,337.0	1,337.5	Pilot
5630-0029	PyB 3.15.0rc3	50 RPS	7	27.7	37.1	36.9	37.1	1,095.3	1,149.3	1,146.9	1,149.0	Pilot
5630-0030	PyB 3.15.0rc3	25 RPS	7	16.1	21.3	21.3	21.3	1,077.8	1,158.1	1,153.7	1,157.6	Pilot
5630-0031	PyB 3.15.0rc3	3,200 RPS	7	88.5	100.0	100.0	100.0	1,285.4	1,335.4	1,335.2	1,335.4	Pilot
5630-0032	PyB 3.11.17	3,200 RPS	7	87.9	100.0	100.0	100.0	1,266.3	1,334.6	1,332.9	1,334.4	Pilot
5630-0033	PyB 3.10.22	1,600 RPS	7	91.8	100.0	100.0	100.0	1,278.1	1,299.3	1,298.7	1,299.3	Pilot

PRIMARY CPU spans both VM vCPUs, all host activity/steal; RAM=MemTotal - MemAvailable, not RSS or all cache. Capacity rows are steady; pilot/smoke use their measurement window. Sparsep 99/p99.9 are weak descriptive estimates. FD/invalid flags override capacity inference. No boot counter is represented as trial usage.

Historical container: secondary

Group 5/12 | pilot | SECONDARY app container CPU and sampled working set.

ID suffix	Runtime / cell	Users or RPS	n	CPU mean %	CPU peak %	CPU p99 %	CPU p99.9 %	Mem mean MiB	Mem peak MiB	Mem p99 MiB	Mem p99.9 MiB	Validity
5630-0016	PyB 3.15.0rc3	800 RPS	7	97.1	97.5	97.4	97.4	463.2	506.4	505.7	506.3	Pilot
5630-0017	PyB 3.10.22	50 RPS	7	18.9	32.4	32.3	32.4	155.2	210.8	207.0	210.4	Pilot
5630-0018	PyB 3.13.16	3,200 RPS	7	97.3	98.0	98.0	98.0	400.9	446.5	446.3	446.5	Pilot
5630-0019	PyB 3.13.16	25 RPS	7	13.4	32.9	32.2	32.8	159.8	159.9	159.9	159.9	Pilot
5630-0020	PyB 3.12.13	800 RPS	7	97.1	97.8	97.7	97.7	402.3	458.7	457.7	458.6	Pilot
5630-0021	PyB 3.14.8	100 RPS	7	43.7	83.7	83.0	83.7	204.4	312.3	304.8	311.6	Pilot
5630-0022	PyB 3.13.16	100 RPS	6	40.7	75.0	73.7	74.9	205.5	342.2	336.6	341.6	Pilot
5630-0023	PyB 3.10.22	100 RPS	7	48.1	97.7	97.7	97.7	198.7	289.1	285.3	288.7	Pilot
5630-0024	PyB 3.10.22	3,200 RPS	7	84.3	98.2	98.2	98.2	356.4	453.6	450.6	453.3	Pilot
5630-0025	PyB 3.12.13	400 RPS	7	83.2	97.4	97.4	97.4	390.1	458.7	458.0	458.6	Pilot
5630-0026	PyB 3.10.22	400 RPS	7	83.5	97.8	97.8	97.8	365.6	446.7	446.4	446.7	Pilot
5630-0027	PyB 3.15.0rc3	100 RPS	6	48.2	81.6	80.0	81.4	228.5	373.3	364.9	372.5	Pilot
5630-0028	PyB 3.14.8	3,200 RPS	8	84.0	97.5	97.5	97.5	409.7	480.8	478.8	480.6	Pilot
5630-0029	PyB 3.15.0rc3	50 RPS	7	17.7	33.8	33.6	33.8	208.2	318.4	310.7	317.6	Pilot
5630-0030	PyB 3.15.0rc3	25 RPS	7	14.3	32.4	32.0	32.4	183.7	183.9	183.9	183.9	Pilot
5630-0031	PyB 3.15.0rc3	3,200 RPS	7	90.2	98.0	98.0	98.0	408.1	485.7	484.7	485.6	Pilot
5630-0032	PyB 3.11.17	3,200 RPS	7	97.6	98.2	98.2	98.2	420.8	446.4	446.3	446.4	Pilot
5630-0033	PyB 3.10.22	1,600 RPS	7	97.5	98.9	98.8	98.9	375.4	442.3	440.4	442.1	Pilot

ContainerCPU 100% = both allocated vCPUs; memory = sampled Docker working set MiB. Capacity rows are steady; pilot/smoke use their measurement window. Sparsep 99/p99.9 are weak descriptive estimates. FD/invalid flags override capacity inference. No boot counter is represented as trial usage.

Historical performance / latency

Group 6/12 | pilot, user-capacity | recorded diagnostic observations, not a valid-ranking table.

ID suffix	Runtime / cell	Users or RPS	Useful ops/s	Measured mean ms	p50 ms	p95 ms	p99 ms	Validity / result
5630-0034	PyB 3.14.8	200 RPS	177.30	1,759.27	1,862.00	4,336.20	5,019.12	Pilot
5630-0035	PyB 3.11.17	800 RPS	180.13	2,243.58	2,050.00	4,172.95	5,074.14	Pilot
5630-0036	PyB 3.12.13	3,200 RPS	175.83	2,277.14	2,134.50	4,759.70	5,278.02	Pilot
5630-0037	PyB 3.15.0rc3	400 RPS	172.87	2,249.63	2,157.00	4,967.00	5,198.60	Pilot
5630-0038	PyB 3.11.17	50 RPS	50.00	13.14	6.00	123.00	177.01	Pilot
5630-0039	PyB 3.13.16	1,600 RPS	181.77	2,233.64	2,258.00	4,676.85	5,052.51	Pilot
5630-0040	PyB 3.11.17	25 RPS	25.03	12.48	6.00	20.50	159.50	Pilot
5630-0041	PyB 3.11.17	400 RPS	177.00	2,239.26	2,152.00	4,419.30	5,087.06	Pilot
5630-0042	PyB 3.14.8	50 RPS	50.00	14.05	6.00	123.00	194.01	Pilot
5630-0043	PyB 3.12.13	50 RPS	50.03	13.29	6.00	122.00	176.00	Pilot
5630-0044	PyB 3.13.16	50 RPS	50.03	13.64	6.00	123.00	179.00	Pilot
5630-0045	PyB 3.12.13	200 RPS	172.90	1,826.54	1,759.00	4,295.80	5,063.28	Pilot
5630-0046	PyB 3.10.22	200 RPS	170.70	1,961.51	2,036.00	4,491.25	5,055.85	Pilot
5630-0047	PyB 3.11.17	200 RPS	176.27	1,760.10	1,912.00	3,937.90	5,032.00	Pilot
638a-0000	PyB 3.10.22	10,000	0.00	n/a	5,001.00	5,001.00	5,002.00	fail
638a-0001	PyB 3.10.22	5,000	0.00	n/a	5,001.00	5,001.00	5,002.00	fail
638a-0002	PyB 3.10.22	2,500	3.15	n/a	5,001.00	5,001.00	5,002.00	fail
638a-0003	PyB 3.10.22	1,250	204.15	n/a	6.00	44.00	48.00	pass

Capacity latency columns are steady; pilot/smoke columns are their recorded measurement window. Missing old steady means remain n/a, never replaced by all-phase averages. Flagged/FD-invalid/mock/preflight values are diagnostic and cannot define capacity. Full trial IDs and protocols remain in the updated 234-trial JSON; the original 228-row CSV is retained.

Historical full-VM resources

Group 6/12 | pilot, user-capacity | PRIMARYwhole VM CPU and estimated used RAM.

ID suffix	Runtime / cell	Users or RPS	n	CPU mean %	CPU peak %	CPU p99 %	CPU p99.9 %	VM RAM mean MiB	VM RAM peak MiB	VM RAM p99 MiB	VM RAM p99.9 MiB	Validity
5630-0034	PyB 3.14.8	200 RPS	7	89.6	100.0	100.0	100.0	1,287.1	1,358.1	1,355.7	1,357.9	Pilot
5630-0035	PyB 3.11.17	800 RPS	7	92.2	100.0	100.0	100.0	1,263.5	1,301.6	1,301.4	1,301.6	Pilot
5630-0036	PyB 3.12.13	3,200 RPS	7	91.2	100.0	100.0	100.0	1,237.9	1,330.2	1,328.2	1,330.0	Pilot
5630-0037	PyB 3.15.0rc3	400 RPS	7	89.6	100.0	100.0	100.0	1,304.6	1,339.3	1,337.7	1,339.1	Pilot
5630-0038	PyB 3.11.17	50 RPS	6	28.5	34.8	34.7	34.8	1,064.5	1,104.7	1,104.2	1,104.6	Pilot
5630-0039	PyB 3.13.16	1,600 RPS	7	89.0	100.0	100.0	100.0	1,292.4	1,331.2	1,330.4	1,331.1	Pilot
5630-0040	PyB 3.11.17	25 RPS	6	16.7	20.9	20.8	20.9	1,048.9	1,099.5	1,097.3	1,099.3	Pilot
5630-0041	PyB 3.11.17	400 RPS	7	89.4	100.0	100.0	100.0	1,259.1	1,293.2	1,293.2	1,293.2	Pilot
5630-0042	PyB 3.14.8	50 RPS	7	27.3	36.7	36.5	36.7	1,080.5	1,111.9	1,111.1	1,111.8	Pilot
5630-0043	PyB 3.12.13	50 RPS	7	28.9	36.1	36.0	36.1	1,071.3	1,106.8	1,105.0	1,106.6	Pilot
5630-0044	PyB 3.13.16	50 RPS	7	26.7	39.4	39.1	39.4	1,077.2	1,124.6	1,122.7	1,124.4	Pilot
5630-0045	PyB 3.12.13	200 RPS	7	89.7	100.0	100.0	100.0	1,279.6	1,364.6	1,360.7	1,364.3	Pilot
5630-0046	PyB 3.10.22	200 RPS	7	89.5	100.0	100.0	100.0	1,258.7	1,304.8	1,304.4	1,304.7	Pilot
5630-0047	PyB 3.11.17	200 RPS	7	89.5	100.0	100.0	100.0	1,264.3	1,297.2	1,296.4	1,297.2	Pilot
638a-0000	PyB 3.10.22	10,000	48	100.0	100.0	100.0	100.0	1,385.9	1,458.2	1,457.5	1,458.1	fail
638a-0001	PyB 3.10.22	5,000	48	100.0	100.0	100.0	100.0	1,376.2	1,522.1	1,495.1	1,519.4	fail
638a-0002	PyB 3.10.22	2,500	56	100.0	100.0	100.0	100.0	1,271.7	1,348.4	1,334.1	1,346.9	fail
638a-0003	PyB 3.10.22	1,250	62	35.2	37.1	37.1	37.1	990.0	1,000.5	1,000.3	1,000.5	pass

PRIMARY CPU spans both VM vCPUs, all host activity/steal; RAM=MemTotal - MemAvailable, not RSS or all cache. Capacity rows are steady; pilot/smoke use their measurement window. Sparsep 99/p99.9 are weak descriptive estimates. FD/invalid flags override capacity inference. No boot counter is represented as trial usage.

Historical container: secondary

Group 6/12 | pilot, user-capacity | SECONDARY app container CPU and sampled working set.

ID suffix	Runtime / cell	Users or RPS	n	CPU mean %	CPU peak %	CPU p99 %	CPU p99.9 %	Mem mean MiB	Mem peak MiB	Mem p99 MiB	Mem p99.9 MiB	Validity
5630-0034	PyB 3.14.8	200 RPS	7	97.4	98.3	98.3	98.3	415.4	468.6	467.8	468.5	Pilot
5630-0035	PyB 3.11.17	800 RPS	7	83.7	97.6	97.6	97.6	378.3	445.9	444.6	445.8	Pilot
5630-0036	PyB 3.12.13	3,200 RPS	7	83.3	97.5	97.5	97.5	374.3	434.7	434.1	434.6	Pilot
5630-0037	PyB 3.15.0rc3	400 RPS	7	97.1	98.0	98.0	98.0	425.8	475.7	475.3	475.7	Pilot
5630-0038	PyB 3.11.17	50 RPS	6	31.5	56.2	55.5	56.2	183.8	286.5	281.5	286.0	Pilot
5630-0039	PyB 3.13.16	1,600 RPS	7	97.2	98.7	98.7	98.7	398.5	451.3	450.5	451.2	Pilot
5630-0040	PyB 3.11.17	25 RPS	6	15.8	33.8	33.2	33.8	166.9	220.4	217.2	220.1	Pilot
5630-0041	PyB 3.11.17	400 RPS	7	83.7	97.9	97.9	97.9	369.5	446.2	445.1	446.1	Pilot
5630-0042	PyB 3.14.8	50 RPS	7	18.5	33.6	33.6	33.6	203.5	310.6	304.2	310.0	Pilot
5630-0043	PyB 3.12.13	50 RPS	7	17.7	37.2	36.5	37.2	159.9	185.8	184.0	185.6	Pilot
5630-0044	PyB 3.13.16	50 RPS	7	19.9	37.0	36.3	36.9	171.2	226.7	222.8	226.3	Pilot
5630-0045	PyB 3.12.13	200 RPS	7	95.9	98.3	98.3	98.3	377.8	458.0	457.4	457.9	Pilot
5630-0046	PyB 3.10.22	200 RPS	7	93.7	98.3	98.3	98.3	371.4	433.1	433.0	433.1	Pilot
5630-0047	PyB 3.11.17	200 RPS	7	97.4	98.3	98.3	98.3	376.6	451.5	451.1	451.5	Pilot
638a-0000	PyB 3.10.22	10,000	48	81.2	91.8	90.6	91.7	569.9	675.3	673.7	675.1	fail
638a-0001	PyB 3.10.22	5,000	48	82.2	91.2	90.9	91.2	574.6	739.6	726.8	738.3	fail
638a-0002	PyB 3.10.22	2,500	56	94.0	96.2	96.2	96.2	446.6	519.4	519.2	519.4	fail
638a-0003	PyB 3.10.22	1,250	62	28.5	34.6	34.4	34.6	173.9	174.2	174.2	174.2	pass

ContainerCPU 100% = both allocated vCPUs; memory = sampled Docker working set MiB. Capacity rows are steady; pilot/smoke use their measurement window. Sparsep 99/p99.9 are weak descriptive estimates. FD/invalid flags override capacity inference. No boot counter is represented as trial usage.

Historical performance / latency

Group 7/12 | user-capacity | recorded diagnostic observations, not a valid-ranking table.

ID suffix	Runtime / cell	Users or RPS	Useful ops/s	Measured mean ms	p50 ms	p95 ms	p99 ms	Validity / result
638a-0004	PyB 3.10.22	1,875	306.54	n/a	7.00	46.00	59.00	pass
638a-0005	PyB 3.10.22	2,187	22.20	n/a	5,001.00	5,001.00	5,002.00	fail
638a-0006	PyB 3.10.22	2,031	331.82	n/a	7.00	47.00	70.00	pass
638a-0007	PyB 3.10.22	2,109	82.60	n/a	5,000.00	5,001.00	5,002.00	fail
638a-0008	PyB 3.10.22	2,031	332.02	n/a	7.00	47.00	80.00	pass
638a-0009	PyB 3.10.22	2,031	331.79	n/a	7.00	46.00	64.00	pass
638a-0010	PyB 3.10.22	2,031	331.83	n/a	7.00	46.00	60.00	pass
638a-0011	PyB 3.10.22	2,031	331.93	n/a	7.00	47.00	68.00	pass
638a-0012	PyB 3.10.22	2,031	331.94	n/a	7.00	46.00	68.00	pass
638a-0013	PyB 3.10.22	2,109	234.06	n/a	10.00	5,001.00	5,001.00	fail
638a-0014	PyB 3.10.22	2,109	192.70	n/a	25.00	5,001.00	5,001.00	fail
638a-0015	PyB 3.10.22	2,109	36.57	n/a	5,000.00	5,001.00	5,002.00	fail
638a-0016	PyB 3.10.22	2,109	146.26	n/a	4,542.00	5,001.00	5,001.00	fail
638a-0017	PyB 3.10.22	2,109	136.49	n/a	4,730.00	5,001.00	5,001.00	fail
638a-0018	PyB 3.11.17	2,031	332.03	n/a	6.00	45.00	51.00	pass
638a-0019	PyB 3.11.17	12,031	0.00	n/a	5,001.00	5,001.00	5,002.00	fail
638a-0020	PyB 3.11.17	7,031	0.00	n/a	5,001.00	5,001.00	5,002.00	fail
638a-0021	PyB 3.11.17	4,531	0.00	n/a	5,001.00	5,001.00	5,002.00	fail

Capacity latency columns are steady; pilot/smoke columns are their recorded measurement window. Missing old steady means remain n/a, never replaced by all-phase averages. Flagged/FD-invalid/mock/preflight values are diagnostic and cannot define capacity. Full trial IDs and protocols remain in the updated 234-trial JSON; the original 228-row CSV is retained.

Historical full-VM resources

Group 7/12 | user-capacity | PRIMARYwhole VM CPU and estimated used RAM.

ID suffix	Runtime / cell	Users or RPS	n	CPU mean %	CPU peak %	CPU p99 %	CPU p99.9 %	VM RAM mean MiB	VM RAM peak MiB	VM RAM p99 MiB	VM RAM p99.9 MiB	Validity
638a-0004	PyB 3.10.22	1,875	61	52.1	55.0	55.0	55.0	1,046.3	1,060.0	1,059.8	1,059.9	pass
638a-0005	PyB 3.10.22	2,187	57	100.0	100.0	100.0	100.0	1,295.1	1,387.9	1,368.6	1,386.0	fail
638a-0006	PyB 3.10.22	2,031	61	56.5	59.0	58.9	59.0	1,095.1	1,105.5	1,105.4	1,105.5	pass
638a-0007	PyB 3.10.22	2,109	57	96.9	100.0	100.0	100.0	1,274.6	1,350.6	1,350.2	1,350.6	fail
638a-0008	PyB 3.10.22	2,031	61	56.1	60.3	59.4	60.2	1,080.7	1,092.4	1,091.6	1,092.3	pass
638a-0009	PyB 3.10.22	2,031	61	55.9	58.1	58.1	58.1	1,075.4	1,085.5	1,085.2	1,085.4	pass
638a-0010	PyB 3.10.22	2,031	60	55.9	57.6	57.5	57.6	1,098.5	1,112.2	1,112.2	1,112.2	pass
638a-0011	PyB 3.10.22	2,031	60	56.3	58.2	57.9	58.2	1,072.1	1,086.6	1,085.0	1,086.5	pass
638a-0012	PyB 3.10.22	2,031	60	56.5	58.8	58.8	58.8	1,081.1	1,090.5	1,090.1	1,090.4	pass
638a-0013	PyB 3.10.22	2,109	59	76.0	100.0	100.0	100.0	1,181.0	1,353.3	1,344.1	1,352.3	fail
638a-0014	PyB 3.10.22	2,109	60	82.3	100.0	100.0	100.0	1,215.3	1,361.4	1,359.0	1,361.2	fail
638a-0015	PyB 3.10.22	2,109	57	100.0	100.0	100.0	100.0	1,306.6	1,380.4	1,366.9	1,379.0	fail
638a-0016	PyB 3.10.22	2,109	58	89.2	100.0	100.0	100.0	1,237.7	1,356.5	1,351.2	1,356.0	fail
638a-0017	PyB 3.10.22	2,109	57	89.8	100.0	100.0	100.0	1,250.9	1,343.5	1,338.9	1,343.1	fail
638a-0018	PyB 3.11.17	2,031	60	48.8	50.6	50.5	50.6	1,059.2	1,073.1	1,072.6	1,073.0	pass
638a-0019	PyB 3.11.17	12,031	44	100.0	100.0	100.0	100.0	1,304.1	1,374.0	1,373.8	1,374.0	fail
638a-0020	PyB 3.11.17	7,031	45	100.0	100.0	100.0	100.0	1,316.6	1,391.0	1,389.2	1,390.8	fail
638a-0021	PyB 3.11.17	4,531	43	100.0	100.0	100.0	100.0	1,322.5	1,408.8	1,405.2	1,408.5	fail

PRIMARY CPU spans both VM vCPUs, all host activity/steal; RAM=MemTotal - MemAvailable, not RSS or all cache. Capacity rows are steady; pilot/smoke use their measurement window. Sparsep 99/p99.9 are weak descriptive estimates. FD/invalid flags override capacity inference. No boot counter is represented as trial usage.

Historical container: secondary

Group 7/12 | user-capacity | SECONDARY app container CPU and sampled working set.

ID suffix	Runtime / cell	Users or RPS	n	CPU mean %	CPU peak %	CPU p99 %	CPU p99.9 %	Mem mean MiB	Mem peak MiB	Mem p99 MiB	Mem p99.9 MiB	Validity
638a-0004	PyB 3.10.22	1,875	61	43.7	49.7	49.7	49.7	213.7	222.5	221.8	222.4	pass
638a-0005	PyB 3.10.22	2,187	57	95.0	97.3	96.8	97.3	453.6	511.0	510.6	511.0	fail
638a-0006	PyB 3.10.22	2,031	61	47.8	56.7	55.7	56.6	243.5	249.4	246.1	249.1	pass
638a-0007	PyB 3.10.22	2,109	57	91.5	97.1	97.0	97.1	436.5	514.6	514.3	514.6	fail
638a-0008	PyB 3.10.22	2,031	61	49.8	60.9	60.7	60.9	235.5	236.7	236.6	236.7	pass
638a-0009	PyB 3.10.22	2,031	61	47.7	58.1	57.6	58.0	232.8	241.2	239.8	241.1	pass
638a-0010	PyB 3.10.22	2,031	60	47.8	56.8	56.6	56.8	250.2	257.6	257.3	257.6	pass
638a-0011	PyB 3.10.22	2,031	60	47.5	60.3	58.5	60.1	231.4	239.7	235.2	239.2	pass
638a-0012	PyB 3.10.22	2,031	60	47.6	56.9	56.5	56.9	235.0	235.4	235.3	235.4	pass
638a-0013	PyB 3.10.22	2,109	59	68.2	97.1	96.9	97.1	327.3	512.2	507.3	511.7	fail
638a-0014	PyB 3.10.22	2,109	60	76.0	97.8	97.4	97.7	359.0	512.9	512.2	512.8	fail
638a-0015	PyB 3.10.22	2,109	57	95.3	97.0	97.0	97.0	458.0	514.4	510.4	514.0	fail
638a-0016	PyB 3.10.22	2,109	58	82.7	97.4	97.4	97.4	398.9	511.6	511.0	511.5	fail
638a-0017	PyB 3.10.22	2,109	57	83.8	96.7	96.6	96.7	397.1	513.9	509.4	513.4	fail
638a-0018	PyB 3.11.17	2,031	60	40.2	49.7	49.4	49.7	240.0	240.5	240.5	240.5	pass
638a-0019	PyB 3.11.17	12,031	44	82.9	89.8	89.5	89.8	463.1	644.8	616.4	642.0	fail
638a-0020	PyB 3.11.17	7,031	45	84.5	90.7	90.6	90.7	500.8	725.1	695.3	722.1	fail
638a-0021	PyB 3.11.17	4,531	43	84.2	92.3	92.3	92.3	492.5	613.4	609.5	613.0	fail

ContainerCPU 100% = both allocated vCPUs; memory = sampled Docker working set MiB. Capacity rows are steady; pilot/smoke use their measurement window. Sparsep 99/p99.9 are weak descriptive estimates. FD/invalid flags override capacity inference. No boot counter is represented as trial usage.

Historical performance / latency

Group 8/12 | user-capacity | recorded diagnostic observations, not a valid-ranking table.

ID suffix	Runtime / cell	Users or RPS	Useful ops/s	Measured mean ms	p50 ms	p95 ms	p99 ms	Validity / result
638a-0022	PyB 3.11.17	3,281	1.68	n/a	5,001.00	5,001.00	5,002.00	fail
638a-0023	PyB 3.11.17	2,656	6.15	n/a	5,001.00	5,001.00	5,002.00	fail
638a-0024	PyB 3.11.17	2,343	81.00	n/a	5,000.00	5,001.00	5,002.00	fail
638a-0025	PyB 3.11.17	2,187	357.89	n/a	6.00	45.00	55.00	pass
638a-0026	PyB 3.11.17	2,265	155.58	n/a	2,306.00	5,001.00	5,001.00	fail
638a-0027	PyB 3.11.17	2,187	357.79	n/a	6.00	45.00	57.00	pass
638a-0028	PyB 3.11.17	2,187	357.49	n/a	6.00	45.00	57.00	pass
638a-0029	PyB 3.11.17	2,187	340.54	n/a	6.00	1,368.00	5,000.00	fail
638a-0030	PyB 3.11.17	2,187	337.47	n/a	7.00	1,566.90	5,001.00	fail
638a-0031	PyB 3.11.17	2,187	317.38	n/a	7.00	4,816.30	5,001.00	fail
638a-0032	PyB 3.11.17	2,265	252.34	n/a	8.00	5,001.00	5,001.00	fail
638a-0033	PyB 3.11.17	2,265	155.06	n/a	2,149.00	5,001.00	5,001.00	fail
638a-0034	PyB 3.11.17	2,265	284.89	n/a	8.00	5,001.00	5,001.00	fail
638a-0035	PyB 3.11.17	2,265	223.59	n/a	11.00	5,001.00	5,001.00	fail
638a-0036	PyB 3.11.17	2,265	195.72	n/a	16.00	5,001.00	5,001.00	fail
638a-0037	PyB 3.12.13	2,031	332.03	n/a	6.00	45.00	56.00	pass
638a-0038	PyB 3.12.13	2,265	214.90	n/a	14.00	5,001.00	5,001.00	fail
638a-0039	PyB 3.12.13	2,148	351.87	n/a	6.00	46.00	66.00	pass

Capacity latency columns are steady; pilot/smoke columns are their recorded measurement window. Missing old steady means remain n/a, never replaced by all-phase averages. Flagged/FD-invalid/mock/preflight values are diagnostic and cannot define capacity. Full trial IDs and protocols remain in the updated 234-trial JSON; the original 228-row CSV is retained.

Historical full-VM resources

Group 8/12 | user-capacity | PRIMARYwhole VM CPU and estimated used RAM.

ID suffix	Runtime / cell	Users or RPS	n	CPU mean %	CPU peak %	CPU p99 %	CPU p99.9 %	VM RAM mean MiB	VM RAM peak MiB	VM RAM p99 MiB	VM RAM p99.9 MiB	Validity
638a-0022	PyB 3.11.17	3,281	53	100.0	100.0	100.0	100.0	1,247.1	1,367.0	1,360.9	1,366.4	fail
638a-0023	PyB 3.11.17	2,656	54	100.0	100.0	100.0	100.0	1,264.2	1,375.1	1,359.5	1,373.5	fail
638a-0024	PyB 3.11.17	2,343	55	95.7	100.0	100.0	100.0	1,230.2	1,343.5	1,332.6	1,342.4	fail
638a-0025	PyB 3.11.17	2,187	61	53.1	55.3	55.2	55.3	1,058.9	1,109.2	1,105.4	1,108.8	pass
638a-0026	PyB 3.11.17	2,265	58	85.1	100.0	100.0	100.0	1,190.6	1,337.0	1,322.3	1,335.6	fail
638a-0027	PyB 3.11.17	2,187	60	52.7	55.8	55.3	55.7	1,065.5	1,078.7	1,078.6	1,078.7	pass
638a-0028	PyB 3.11.17	2,187	60	52.9	54.7	54.7	54.7	1,065.7	1,088.9	1,082.9	1,088.3	pass
638a-0029	PyB 3.11.17	2,187	61	57.7	100.0	100.0	100.0	1,086.0	1,305.1	1,303.4	1,305.0	fail
638a-0030	PyB 3.11.17	2,187	59	57.6	100.0	100.0	100.0	1,086.6	1,309.0	1,293.2	1,307.4	fail
638a-0031	PyB 3.11.17	2,187	59	61.1	100.0	100.0	100.0	1,095.6	1,284.1	1,281.9	1,283.9	fail
638a-0032	PyB 3.11.17	2,265	58	72.5	100.0	100.0	100.0	1,127.5	1,355.4	1,326.3	1,352.5	fail
638a-0033	PyB 3.11.17	2,265	57	85.9	100.0	100.0	100.0	1,174.8	1,321.9	1,314.9	1,321.2	fail
638a-0034	PyB 3.11.17	2,265	60	68.9	100.0	100.0	100.0	1,127.4	1,307.3	1,306.3	1,307.2	fail
638a-0035	PyB 3.11.17	2,265	58	76.9	100.0	100.0	100.0	1,160.1	1,327.6	1,323.9	1,327.3	fail
638a-0036	PyB 3.11.17	2,265	57	79.4	100.0	100.0	100.0	1,166.2	1,313.5	1,309.8	1,313.1	fail
638a-0037	PyB 3.12.13	2,031	60	51.1	52.9	52.8	52.9	1,043.5	1,054.7	1,053.6	1,054.6	pass
638a-0038	PyB 3.12.13	2,265	57	78.9	100.0	100.0	100.0	1,163.6	1,320.9	1,315.5	1,320.4	fail
638a-0039	PyB 3.12.13	2,148	60	54.4	57.2	56.8	57.2	1,074.9	1,089.7	1,089.0	1,089.6	pass

PRIMARY CPU spans both VM vCPUs, all host activity/steal; RAM=MemTotal - MemAvailable, not RSS or all cache. Capacity rows are steady; pilot/smoke use their measurement window. Sparsep 99/p99.9 are weak descriptive estimates. FD/invalid flags override capacity inference. No boot counter is represented as trial usage.

Historical container: secondary

Group 8/12 | user-capacity | SECONDARY app container CPU and sampled working set.

ID suffix	Runtime / cell	Users or RPS	n	CPU mean %	CPU peak %	CPU p99 %	CPU p99.9 %	Mem mean MiB	Mem peak MiB	Mem p99 MiB	Mem p99.9 MiB	Validity
638a-0022	PyB 3.11.17	3,281	53	90.1	95.0	94.9	94.9	429.4	580.1	569.4	579.0	fail
638a-0023	PyB 3.11.17	2,656	54	92.6	96.4	96.2	96.4	444.5	567.5	567.5	567.5	fail
638a-0024	PyB 3.11.17	2,343	55	88.1	96.1	96.1	96.1	409.6	538.0	533.0	537.5	fail
638a-0025	PyB 3.11.17	2,187	61	44.4	57.1	53.0	56.7	246.3	255.4	250.1	254.9	pass
638a-0026	PyB 3.11.17	2,265	58	77.1	97.0	96.4	96.9	375.7	523.4	518.0	522.9	fail
638a-0027	PyB 3.11.17	2,187	60	44.6	53.0	52.9	53.0	247.6	250.6	250.4	250.6	pass
638a-0028	PyB 3.11.17	2,187	60	44.2	54.5	52.8	54.3	252.6	261.2	260.4	261.1	pass
638a-0029	PyB 3.11.17	2,187	61	48.2	96.4	96.0	96.3	266.7	488.2	485.3	487.9	fail
638a-0030	PyB 3.11.17	2,187	59	49.0	96.2	95.9	96.1	267.5	485.8	463.4	483.6	fail
638a-0031	PyB 3.11.17	2,187	59	53.3	95.8	95.8	95.8	276.9	495.4	491.8	495.0	fail
638a-0032	PyB 3.11.17	2,265	58	65.1	96.5	96.2	96.4	313.4	483.4	478.8	482.9	fail
638a-0033	PyB 3.11.17	2,265	57	78.4	96.5	96.3	96.4	372.0	524.0	523.3	523.9	fail
638a-0034	PyB 3.11.17	2,265	60	60.5	96.0	95.5	96.0	302.4	526.6	499.0	523.8	fail
638a-0035	PyB 3.11.17	2,265	58	68.8	96.2	96.2	96.2	327.4	529.1	506.4	526.8	fail
638a-0036	PyB 3.11.17	2,265	57	71.5	96.6	96.1	96.6	343.1	517.1	515.0	516.9	fail
638a-0037	PyB 3.12.13	2,031	60	42.8	52.1	51.4	52.1	253.1	259.9	259.7	259.9	pass
638a-0038	PyB 3.12.13	2,265	57	70.6	96.7	96.3	96.7	350.3	543.9	543.7	543.9	fail
638a-0039	PyB 3.12.13	2,148	60	45.9	55.2	55.0	55.2	261.6	271.0	266.6	270.6	pass

ContainerCPU 100% = both allocated vCPUs; memory = sampled Docker working set MiB. Capacity rows are steady; pilot/smoke use their measurement window. Sparsep 99/p99.9 are weak descriptive estimates. FD/invalid flags override capacity inference. No boot counter is represented as trial usage.

Historical performance / latency

Group 9/12 | user-capacity | recorded diagnostic observations, not a valid-ranking table.

ID suffix	Runtime / cell	Users or RPS	Useful ops/s	Measured mean ms	p50 ms	p95 ms	p99 ms	Validity / result
638a-0040	PyB 3.12.13	2,206	321.34	n/a	7.00	4,477.25	5,001.00	fail
638a-0041	PyB 3.12.13	2,148	351.40	n/a	6.00	46.00	66.00	pass
638a-0042	PyB 3.12.13	2,148	335.28	n/a	7.00	1,025.00	5,000.00	fail
638a-0043	PyB 3.12.13	2,148	346.97	n/a	7.00	48.00	1,963.87	pass
638a-0044	PyB 3.12.13	2,148	351.36	n/a	7.00	46.00	65.00	pass
638a-0045	PyB 3.12.13	2,148	351.05	n/a	6.00	46.00	66.00	pass
638a-0046	PyB 3.12.13	2,206	303.21	n/a	7.00	5,000.00	5,001.00	fail
638a-0047	PyB 3.12.13	2,206	230.91	n/a	10.00	5,001.00	5,001.00	fail
638a-0048	PyB 3.12.13	2,206	216.82	n/a	11.00	5,001.00	5,001.00	fail
638a-0049	PyB 3.12.13	2,206	349.88	n/a	7.00	87.00	4,190.21	FD invalid
638a-0050	PyB 3.12.13	2,206	328.68	n/a	7.00	3,521.60	5,001.00	fail
638a-0051	PyB 3.13.16	2,031	332.31	n/a	6.00	45.00	54.00	pass
638a-0052	PyB 3.13.16	2,206	270.56	n/a	7.00	5,001.00	5,001.00	fail
638a-0053	PyB 3.13.16	2,118	346.45	n/a	6.00	45.00	53.00	pass
638a-0054	PyB 3.13.16	2,118	346.22	n/a	6.00	45.00	53.00	FD invalid
638a-0055	PyB 3.13.16	2,118	346.31	n/a	6.00	45.00	53.00	FD invalid
638a-0056	PyB 3.13.16	2,118	346.41	n/a	6.00	45.00	56.00	pass
638a-0057	PyB 3.13.16	2,118	346.43	n/a	6.00	45.00	59.00	pass

Capacity latency columns are steady; pilot/smoke columns are their recorded measurement window. Missing old steady means remain n/a, never replaced by all-phase averages. Flagged/FD-invalid/mock/preflight values are diagnostic and cannot define capacity. Full trial IDs and protocols remain in the updated 234-trial JSON; the original 228-row CSV is retained.

Historical full-VM resources

Group 9/12 | user-capacity | PRIMARYwhole VM CPU and estimated used RAM.

ID suffix	Runtime / cell	Users or RPS	n	CPU mean %	CPU peak %	CPU p99 %	CPU p99.9 %	VM RAM mean MiB	VM RAM peak MiB	VM RAM p99 MiB	VM RAM p99.9 MiB	Validity
638a-0040	PyB 3.12.13	2,206	61	63.1	100.0	100.0	100.0	1,102.4	1,312.4	1,298.9	1,311.0	fail
638a-0041	PyB 3.12.13	2,148	60	53.7	56.5	56.3	56.5	1,062.9	1,082.0	1,077.1	1,081.5	pass
638a-0042	PyB 3.12.13	2,148	61	58.7	100.0	100.0	100.0	1,096.7	1,345.9	1,321.0	1,343.4	fail
638a-0043	PyB 3.12.13	2,148	61	55.6	100.0	100.0	100.0	1,085.8	1,241.6	1,214.7	1,238.9	pass
638a-0044	PyB 3.12.13	2,148	60	55.0	57.8	57.6	57.7	1,074.4	1,090.5	1,087.7	1,090.2	pass
638a-0045	PyB 3.12.13	2,148	61	54.4	57.3	56.5	57.2	1,076.6	1,089.0	1,088.5	1,088.9	pass
638a-0046	PyB 3.12.13	2,206	59	65.0	100.0	100.0	100.0	1,122.5	1,338.4	1,324.1	1,336.9	fail
638a-0047	PyB 3.12.13	2,206	59	75.1	100.0	100.0	100.0	1,148.8	1,328.6	1,320.5	1,327.8	fail
638a-0048	PyB 3.12.13	2,206	57	76.4	100.0	100.0	100.0	1,151.1	1,360.5	1,346.9	1,359.2	fail
638a-0049	PyB 3.12.13	2,206	60	57.8	100.0	100.0	100.0	1,095.9	1,304.5	1,298.4	1,303.9	FD invalid
638a-0050	PyB 3.12.13	2,206	60	62.1	100.0	100.0	100.0	1,103.6	1,296.1	1,255.7	1,292.1	fail
638a-0051	PyB 3.13.16	2,031	61	48.7	50.5	50.4	50.4	1,043.4	1,062.5	1,057.2	1,061.9	pass
638a-0052	PyB 3.13.16	2,206	55	65.8	100.0	100.0	100.0	1,148.5	1,345.0	1,333.9	1,343.9	fail
638a-0053	PyB 3.13.16	2,118	60	50.9	54.8	53.4	54.6	1,062.3	1,073.8	1,073.7	1,073.8	pass
638a-0054	PyB 3.13.16	2,118	61	50.6	54.5	53.4	54.4	1,086.5	1,098.6	1,098.4	1,098.6	FD invalid
638a-0055	PyB 3.13.16	2,118	61	51.2	54.6	53.6	54.5	1,076.0	1,091.2	1,090.3	1,091.1	FD invalid
638a-0056	PyB 3.13.16	2,118	61	51.4	53.5	53.5	53.5	1,082.1	1,094.9	1,094.6	1,094.9	pass
638a-0057	PyB 3.13.16	2,118	60	51.6	53.6	53.4	53.6	1,085.4	1,099.6	1,097.6	1,099.4	pass

PRIMARY CPU spans both VM vCPUs, all host activity/steal; RAM=MemTotal - MemAvailable, not RSS or all cache. Capacity rows are steady; pilot/smoke use their measurement window. Sparsep 99/p99.9 are weak descriptive estimates. FD/invalid flags override capacity inference. No boot counter is represented as trial usage.

Historical container: secondary

Group 9/12 | user-capacity | SECONDARY app container CPU and sampled working set.

ID suffix	Runtime / cell	Users or RPS	n	CPU mean %	CPU peak %	CPU p99 %	CPU p99.9 %	Mem mean MiB	Mem peak MiB	Mem p99 MiB	Mem p99.9 MiB	Validity
638a-0040	PyB 3.12.13	2,206	61	53.7	96.5	95.6	96.4	290.1	490.1	480.4	489.1	fail
638a-0041	PyB 3.12.13	2,148	60	46.2	56.7	55.0	56.5	248.4	256.0	255.0	255.9	pass
638a-0042	PyB 3.12.13	2,148	61	49.3	95.8	95.4	95.7	277.7	523.9	484.2	519.9	fail
638a-0043	PyB 3.12.13	2,148	61	46.8	96.4	77.8	94.5	268.1	522.5	369.0	507.2	pass
638a-0044	PyB 3.12.13	2,148	60	47.4	64.2	62.5	64.0	260.4	267.3	266.8	267.3	pass
638a-0045	PyB 3.12.13	2,148	61	45.9	58.7	57.0	58.5	263.0	267.6	267.1	267.6	pass
638a-0046	PyB 3.12.13	2,206	59	55.8	96.6	96.1	96.6	290.7	481.8	470.1	480.6	fail
638a-0047	PyB 3.12.13	2,206	59	66.7	96.1	96.0	96.1	328.5	528.9	521.2	528.1	fail
638a-0048	PyB 3.12.13	2,206	57	67.9	96.4	96.2	96.3	352.4	537.3	536.6	537.2	fail
638a-0049	PyB 3.12.13	2,206	60	49.2	95.7	95.1	95.6	274.1	408.5	401.9	407.8	FD invalid
638a-0050	PyB 3.12.13	2,206	60	53.8	96.0	95.9	96.0	287.2	514.3	486.6	511.5	fail
638a-0051	PyB 3.13.16	2,031	61	40.6	52.2	50.0	52.0	237.9	239.0	238.5	239.0	pass
638a-0052	PyB 3.13.16	2,206	55	57.3	97.1	96.4	97.1	331.8	538.7	517.2	536.6	fail
638a-0053	PyB 3.13.16	2,118	60	43.0	52.7	52.0	52.6	246.3	254.1	252.3	253.9	pass
638a-0054	PyB 3.13.16	2,118	61	42.3	54.0	52.3	53.8	272.0	282.3	276.9	281.8	FD invalid
638a-0055	PyB 3.13.16	2,118	61	42.4	51.9	51.5	51.9	262.4	272.7	272.4	272.7	FD invalid
638a-0056	PyB 3.13.16	2,118	61	42.6	53.2	52.7	53.2	267.7	268.2	268.1	268.2	pass
638a-0057	PyB 3.13.16	2,118	60	43.3	55.2	52.8	54.9	269.7	278.9	278.3	278.8	pass

ContainerCPU 100% = both allocated vCPUs; memory = sampled Docker working set MiB. Capacity rows are steady; pilot/smoke use their measurement window. Sparsep 99/p99.9 are weak descriptive estimates. FD/invalid flags override capacity inference. No boot counter is represented as trial usage.

Historical performance / latency

Group 10/12 | user-capacity | recorded diagnostic observations, not a valid-ranking table.

ID suffix	Runtime / cell	Users or RPS	Useful ops/s	Measured mean ms	p50 ms	p95 ms	p99 ms	Validity / result
638a-0058	PyB 3.13.16	2,118	n/a	n/a	n/a	n/a	n/a	Flagged 3
638a-0059	PyB 3.13.16	2,206	360.65	n/a	6.00	45.00	58.00	FD invalid
638a-0060	PyB 3.14.8	2,031	332.38	n/a	6.00	45.00	55.00	pass
638a-0061	PyB 3.14.8	2,206	326.66	n/a	7.00	3,491.00	5,001.00	FD invalid
638a-0062	PyB 3.14.8	2,118	346.53	n/a	6.00	45.00	59.00	FD invalid
638a-0063	PyB 3.14.8	2,206	262.69	n/a	8.00	5,001.00	5,001.00	fail
638a-0064	PyB 3.15.0rc3	2,118	346.12	n/a	6.00	45.00	59.00	pass
638a-0065	PyB 3.15.0rc3	2,206	23.82	n/a	5,001.00	5,001.00	5,002.00	fail
638a-0066	PyB 3.15.0rc3	2,206	17.48	n/a	5,001.00	5,001.00	5,002.00	fail
638a-0067	PyM Std J0 G1/A	2,118	346.83	n/a	4.00	43.00	45.00	pass
638a-0068	PyM Std J0 G1/A	2,206	361.18	n/a	4.00	43.00	45.00	pass
638a-0069	PyM Std J0 G1/A	2,406	0.00	n/a	37.00	186.00	271.00	fail
638a-0070	PyM Std J0 G1/A	2,306	0.00	n/a	20.00	120.00	196.00	fail
638a-0071	PyM Std J0 G1/A	2,306	0.00	n/a	20.00	131.00	211.00	fail
638a-0072	PyM jit J0 G1/A	2,206	361.21	n/a	4.00	43.00	45.00	pass
638a-0073	PyM jit J0 G1/A	2,306	0.00	n/a	20.00	123.00	201.00	fail
638a-0074	PyM jit J0 G1/A	2,306	0.00	n/a	18.00	112.00	192.00	fail
638a-0075	PyM jit J1 G1/A	2,206	361.17	n/a	5.00	44.00	46.00	pass

Capacity latency columns are steady; pilot/smoke columns are their recorded measurement window. Missing old steady means remain n/a, never replaced by all-phase averages. Flagged/FD-invalid/mock/preflight values are diagnostic and cannot define capacity. Full trial IDs and protocols remain in the updated 234-trial JSON; the original 228-row CSV is retained.

Historical full-VM resources

Group 10/12 | user-capacity | PRIMARYwhole VM CPU and estimated used RAM.

ID suffix	Runtime / cell	Users or RPS	n	CPU mean %	CPU peak %	CPU p99 %	CPU p99.9 %	VM RAM mean MiB	VM RAM peak MiB	VM RAM p99 MiB	VM RAM p99.9 MiB	Validity
638a-0058	PyB 3.13.16	2,118	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	Flagged 3
638a-0059	PyB 3.13.16	2,206	60	53.1	55.1	54.8	55.1	1,099.4	1,119.2	1,118.7	1,119.2	FD invalid
638a-0060	PyB 3.14.8	2,031	60	49.8	52.3	52.1	52.2	1,067.3	1,084.0	1,082.5	1,083.9	pass
638a-0061	PyB 3.14.8	2,206	59	59.9	100.0	100.0	100.0	1,125.5	1,278.3	1,233.6	1,273.9	FD invalid
638a-0062	PyB 3.14.8	2,118	61	52.6	55.9	55.6	55.9	1,113.5	1,127.7	1,127.0	1,127.6	FD invalid
638a-0063	PyB 3.14.8	2,206	57	68.4	100.0	100.0	100.0	1,177.2	1,388.6	1,370.3	1,386.8	fail
638a-0064	PyB 3.15.0rc3	2,118	61	52.5	56.4	55.8	56.3	1,107.4	1,118.6	1,117.2	1,118.5	pass
638a-0065	PyB 3.15.0rc3	2,206	50	100.0	100.0	100.0	100.0	1,242.5	1,400.7	1,385.0	1,399.1	fail
638a-0066	PyB 3.15.0rc3	2,206	51	100.0	100.0	100.0	100.0	1,234.8	1,378.5	1,341.0	1,374.8	fail
638a-0067	PyM Std J0 G1/A	2,118	61	34.1	36.4	36.0	36.4	983.0	999.6	997.1	999.3	pass
638a-0068	PyM Std J0 G1/A	2,206	61	35.3	37.0	36.6	36.9	991.8	1,006.5	1,006.2	1,006.5	pass
638a-0069	PyM Std J0 G1/A	2,406	44	96.4	99.3	99.2	99.3	1,030.1	1,048.3	1,047.7	1,048.2	fail
638a-0070	PyM Std J0 G1/A	2,306	47	93.3	95.2	95.2	95.2	1,006.6	1,022.8	1,022.2	1,022.7	fail
638a-0071	PyM Std J0 G1/A	2,306	49	93.3	95.2	95.2	95.2	1,016.9	1,029.5	1,029.3	1,029.5	fail
638a-0072	PyM jit J0 G1/A	2,206	61	35.4	36.9	36.8	36.9	1,020.8	1,030.2	1,030.1	1,030.2	pass
638a-0073	PyM jit J0 G1/A	2,306	48	93.7	96.2	96.1	96.2	1,019.8	1,035.6	1,034.2	1,035.4	fail
638a-0074	PyM jit J0 G1/A	2,306	48	93.2	97.1	96.2	97.0	1,016.7	1,043.4	1,040.2	1,043.1	fail
638a-0075	PyM jit J1 G1/A	2,206	62	39.9	42.1	42.0	42.1	1,028.9	1,042.9	1,040.0	1,042.6	pass

PRIMARY CPU spans both VM vCPUs, all host activity/steal; RAM=MemTotal - MemAvailable, not RSS or all cache. Capacity rows are steady; pilot/smoke use their measurement window. Sparsep 99/p99.9 are weak descriptive estimates. FD/invalid flags override capacity inference. No boot counter is represented as trial usage.

Historical container: secondary

Group 10/12 | user-capacity | SECONDARY app container CPU and sampled working set.

ID suffix	Runtime / cell	Users or RPS	n	CPU mean %	CPU peak %	CPU p99 %	CPU p99.9 %	Mem mean MiB	Mem peak MiB	Mem p99 MiB	Mem p99.9 MiB	Validity
638a-0058	PyB 3.13.16	2,118	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	Flagged 3
638a-0059	PyB 3.13.16	2,206	60	46.2	55.6	55.0	55.6	284.4	291.2	287.5	290.8	FD invalid
638a-0060	PyB 3.14.8	2,031	60	41.8	50.9	50.7	50.9	277.0	283.8	282.2	283.6	pass
638a-0061	PyB 3.14.8	2,206	59	51.8	95.4	94.8	95.3	315.7	444.8	441.2	444.4	FD invalid
638a-0062	PyB 3.14.8	2,118	61	44.5	54.4	53.6	54.4	295.1	303.7	303.6	303.7	FD invalid
638a-0063	PyB 3.14.8	2,206	57	60.0	95.5	95.5	95.5	354.7	470.3	470.0	470.3	fail
638a-0064	PyB 3.15.0rc3	2,118	61	43.2	55.9	54.5	55.8	318.9	322.4	320.6	322.2	pass
638a-0065	PyB 3.15.0rc3	2,206	50	93.0	97.2	96.9	97.2	435.5	544.6	540.6	544.2	fail
638a-0066	PyB 3.15.0rc3	2,206	51	92.4	97.2	96.6	97.2	437.5	617.1	576.4	613.0	fail
638a-0067	PyM Std J0 G1/A	2,118	61	26.9	34.8	34.4	34.8	175.6	184.7	182.4	184.5	pass
638a-0068	PyM Std J0 G1/A	2,206	61	27.9	35.2	35.1	35.2	177.7	184.0	183.7	184.0	pass
638a-0069	PyM Std J0 G1/A	2,406	44	82.9	90.1	90.1	90.1	224.9	237.5	236.3	237.4	fail
638a-0070	PyM Std J0 G1/A	2,306	47	80.4	90.0	89.5	90.0	202.4	213.1	213.1	213.1	fail
638a-0071	PyM Std J0 G1/A	2,306	49	80.6	90.0	89.9	90.0	214.1	226.7	225.7	226.6	fail
638a-0072	PyM jit J0 G1/A	2,206	61	27.9	37.2	35.4	37.1	182.7	190.8	190.1	190.7	pass
638a-0073	PyM jit J0 G1/A	2,306	48	80.2	90.2	89.4	90.1	216.1	228.7	228.6	228.7	fail
638a-0074	PyM jit J0 G1/A	2,306	48	80.1	90.1	90.0	90.1	205.5	216.3	213.2	216.0	fail
638a-0075	PyM jit J1 G1/A	2,206	62	31.9	39.9	39.8	39.8	215.8	217.5	217.4	217.5	pass

ContainerCPU 100% = both allocated vCPUs; memory = sampled Docker working set MiB. Capacity rows are steady; pilot/smoke use their measurement window. Sparsep 99/p99.9 are weak descriptive estimates. FD/invalid flags override capacity inference. No boot counter is represented as trial usage.

Historical performance / latency

Group 11/12 | user-capacity | recorded diagnostic observations, not a valid-ranking table.

ID suffix	Runtime / cell	Users or RPS	Useful ops/s	Measured mean ms	p50 ms	p95 ms	p99 ms	Validity / result
638a-0076	PyM jit J1 G1/A	2,306	0.00	n/a	24.00	134.00	211.00	fail
638a-0077	PyM jit J1 G1/A	2,306	0.00	n/a	23.00	135.00	214.00	fail
638a-0078	PyM FT J0 G1/A	2,206	0.00	n/a	29.00	166.00	258.00	fail
638a-0079	PyM FT J0 G1/A	1,103	180.20	n/a	4.00	43.00	45.00	pass
638a-0080	PyM FT J0 G1/A	1,654	270.41	n/a	4.00	43.00	45.00	pass
638a-0081	PyM FT J0 G1/A	1,930	315.70	n/a	4.00	43.00	45.00	pass
638a-0082	PyM FT J0 G1/A	2,068	338.50	n/a	4.00	43.00	45.00	pass
638a-0083	PyM FT J0 G1/A	2,137	349.89	n/a	4.00	43.00	46.00	pass
638a-0084	PyM FT J0 G1/A	2,206	0.00	n/a	30.00	186.00	292.00	fail
638a-0085	PyM FT G0/A	2,137	349.90	n/a	4.00	43.00	46.00	pass
638a-0086	PyM FT G0/A	2,206	361.06	n/a	4.00	43.00	46.00	pass
638a-0087	PyM FT G0/A	2,406	6.09	n/a	382.00	5,001.00	5,001.00	fail
638a-0088	PyM FT G0/A	2,306	0.00	n/a	62.00	231.00	324.00	fail
638a-0089	PyM FT G0/A	2,306	0.00	n/a	65.00	236.00	338.00	fail
638a-0090	PyM FT G1/T	2,206	0.00	n/a	5,000.00	5,001.00	5,002.00	FD invalid
638a-0091	PyM FT G1/T	1,103	180.33	n/a	5.00	9.00	12.98	pass
638a-0092	PyM FT G1/T	1,654	0.00	n/a	5,001.00	5,001.00	5,002.00	fail
638a-0093	PyM FT G1/T	1,378	0.00	n/a	5,001.00	5,001.00	5,002.00	fail

Capacity latency columns are steady; pilot/smoke columns are their recorded measurement window. Missing old steady means remain n/a, never replaced by all-phase averages. Flagged/FD-invalid/mock/preflight values are diagnostic and cannot define capacity. Full trial IDs and protocols remain in the updated 234-trial JSON; the original 228-row CSV is retained.

Historical full-VM resources

Group 11/12 | user-capacity | PRIMARYwhole VM CPU and estimated used RAM.

ID suffix	Runtime / cell	Users or RPS	n	CPU mean %	CPU peak %	CPU p99 %	CPU p99.9 %	VM RAM mean MiB	VM RAM peak MiB	VM RAM p99 MiB	VM RAM p99.9 MiB	Validity
638a-0076	PyM jit J1 G1/A	2,306	48	94.8	96.7	96.6	96.7	1,057.9	1,078.0	1,076.3	1,077.8	fail
638a-0077	PyM jit J1 G1/A	2,306	48	94.2	96.5	96.4	96.5	1,058.1	1,074.7	1,072.6	1,074.5	fail
638a-0078	PyM FT J0 G1/A	2,206	46	95.1	97.4	97.4	97.4	1,225.5	1,238.7	1,238.4	1,238.7	fail
638a-0079	PyM FT J0 G1/A	1,103	63	21.4	23.8	23.3	23.7	1,140.8	1,158.5	1,157.3	1,158.3	pass
638a-0080	PyM FT J0 G1/A	1,654	63	29.3	30.5	30.5	30.5	1,174.3	1,191.3	1,189.7	1,191.1	pass
638a-0081	PyM FT J0 G1/A	1,930	62	33.8	37.1	36.3	37.0	1,190.0	1,208.9	1,205.7	1,208.6	pass
638a-0082	PyM FT J0 G1/A	2,068	62	36.1	37.7	37.7	37.7	1,202.9	1,216.0	1,215.3	1,215.9	pass
638a-0083	PyM FT J0 G1/A	2,137	61	37.3	39.7	39.5	39.7	1,235.8	1,251.1	1,250.8	1,251.1	pass
638a-0084	PyM FT J0 G1/A	2,206	47	95.0	96.8	96.7	96.8	1,333.9	1,356.1	1,356.0	1,356.1	fail
638a-0085	PyM FT G0/A	2,137	61	36.9	39.1	38.8	39.1	1,229.4	1,244.8	1,244.5	1,244.7	pass
638a-0086	PyM FT G0/A	2,206	62	37.8	39.8	39.7	39.8	1,273.9	1,291.5	1,291.2	1,291.5	pass
638a-0087	PyM FT G0/A	2,406	42	100.0	100.0	100.0	100.0	1,454.1	1,654.1	1,641.5	1,652.8	fail
638a-0088	PyM FT G0/A	2,306	39	98.4	99.9	99.9	99.9	1,353.6	1,377.6	1,373.9	1,377.2	fail
638a-0089	PyM FT G0/A	2,306	38	98.4	100.0	100.0	100.0	1,354.0	1,371.9	1,371.1	1,371.8	fail
638a-0090	PyM FT G1/T	2,206	63	15.1	69.2	57.5	68.0	2,845.6	3,520.1	3,507.3	3,518.8	FD invalid
638a-0091	PyM FT G1/T	1,103	62	24.7	27.3	26.7	27.2	799.4	817.5	816.5	817.4	pass
638a-0092	PyM FT G1/T	1,654	63	11.1	18.6	18.2	18.5	3,062.7	3,467.6	3,464.6	3,467.3	fail
638a-0093	PyM FT G1/T	1,378	64	10.5	16.1	15.5	16.1	2,578.2	3,322.0	3,310.0	3,320.8	fail

PRIMARY CPU spans both VM vCPUs, all host activity/steal; RAM=MemTotal - MemAvailable, not RSS or all cache. Capacity rows are steady; pilot/smoke use their measurement window. Sparsep 99/p99.9 are weak descriptive estimates. FD/invalid flags override capacity inference. No boot counter is represented as trial usage.

Historical container: secondary

Group 11/12 | user-capacity | SECONDARY app container CPU and sampled working set.

ID suffix	Runtime / cell	Users or RPS	n	CPU mean %	CPU peak %	CPU p99 %	CPU p99.9 %	Mem mean MiB	Mem peak MiB	Mem p99 MiB	Mem p99.9 MiB	Validity
638a-0076	PyM jit J1 G1/A	2,306	48	82.8	90.1	90.1	90.1	245.1	260.7	260.2	260.6	fail
638a-0077	PyM jit J1 G1/A	2,306	48	81.5	89.9	89.8	89.9	244.8	258.0	257.5	258.0	fail
638a-0078	PyM FT J0 G1/A	2,206	46	84.7	90.5	90.4	90.5	400.4	428.4	428.3	428.4	fail
638a-0079	PyM FT J0 G1/A	1,103	63	14.9	21.1	20.7	21.1	317.9	323.9	320.9	323.6	pass
638a-0080	PyM FT J0 G1/A	1,654	63	21.8	28.9	28.7	28.9	340.5	365.5	360.5	365.0	pass
638a-0081	PyM FT J0 G1/A	1,930	62	26.0	33.7	33.4	33.7	355.8	362.4	358.5	362.0	pass
638a-0082	PyM FT J0 G1/A	2,068	62	27.9	36.2	34.9	36.1	365.6	372.4	368.5	372.0	pass
638a-0083	PyM FT J0 G1/A	2,137	61	29.3	37.8	37.0	37.7	397.5	415.0	414.9	415.0	pass
638a-0084	PyM FT J0 G1/A	2,206	47	84.1	90.7	90.6	90.7	500.7	526.8	521.3	526.3	fail
638a-0085	PyM FT G0/A	2,137	61	29.0	36.1	35.3	36.0	407.6	425.2	414.7	424.1	pass
638a-0086	PyM FT G0/A	2,206	62	29.6	36.3	36.2	36.3	447.2	454.0	452.8	453.9	pass
638a-0087	PyM FT G0/A	2,406	42	89.6	97.2	96.8	97.1	642.2	834.5	832.6	834.3	fail
638a-0088	PyM FT G0/A	2,306	39	86.6	90.7	90.6	90.7	524.4	531.2	528.9	531.0	fail
638a-0089	PyM FT G0/A	2,306	38	86.8	90.7	90.6	90.7	529.4	552.4	552.4	552.4	fail
638a-0090	PyM FT G1/T	2,206	52	10.6	50.9	49.9	50.8	2,618.5	3,055.6	3,042.0	3,054.3	FD invalid
638a-0091	PyM FT G1/T	1,103	62	18.1	26.4	25.4	26.3	236.8	253.5	253.1	253.5	pass
638a-0092	PyM FT G1/T	1,654	63	5.5	13.3	11.7	13.2	2,516.1	2,938.9	2,928.1	2,937.8	fail
638a-0093	PyM FT G1/T	1,378	64	6.2	31.4	25.4	30.8	2,034.4	2,782.2	2,780.9	2,782.1	fail

ContainerCPU 100% = both allocated vCPUs; memory = sampled Docker working set MiB. Capacity rows are steady; pilot/smoke use their measurement window. Sparsep 99/p99.9 are weak descriptive estimates. FD/invalid flags override capacity inference. No boot counter is represented as trial usage.

Historical performance / latency

Group 12/12 | user-capacity, user-capacity-invalid-20261004 | recorded diagnostic observations, not a valid-ranking table.

ID suffix	Runtime / cell	Users or RPS	Useful ops/s	Measured mean ms	p50 ms	p95 ms	p99 ms	Validity / result
638a-0094	PyM FT G1/T	1,240	202.65	n/a	5.00	10.00	14.00	pass
638a-0095	PyM FT G1/T	1,309	0.00	n/a	5,001.00	5,001.00	5,002.00	fail
638a-0096	PyM FT G1/T	1,309	0.00	n/a	5,001.00	5,001.00	5,002.00	fail
638a-0097	PyM FT G0/T	1,240	202.62	n/a	5.00	11.00	16.00	pass
638a-0098	PyM FT G0/T	1,309	213.93	n/a	5.00	11.00	18.00	pass
638a-0099	PyM FT G0/T	1,509	0.00	n/a	5,001.00	5,001.00	5,002.00	fail
638a-0100	PyM FT G0/T	1,409	0.00	n/a	5,001.00	5,001.00	5,002.00	fail
638a-0101	PyM FT G0/T	1,409	0.00	n/a	5,001.00	5,001.00	5,002.00	fail
0457-0000	PyB 3.10.22	10,000	0.00	n/a	39.00	5,001.00	5,005.00	Flagged 3

Capacity latency columns are steady; pilot/smoke columns are their recorded measurement window. Missing old steady means remain n/a, never replaced by all-phase averages. Flagged/FD-invalid/mock/preflight values are diagnostic and cannot define capacity. Full trial IDs and protocols remain in the updated 234-trial JSON; the original 228-row CSV is retained.

Historical full-VM resources

Group 12/12 | user-capacity, user-capacity-invalid-20261004 | PRIMARYwhole VM CPU and estimated used RAM.

ID suffix	Runtime / cell	Users or RPS	n	CPU mean %	CPU peak %	CPU p99 %	CPU p99.9 %	VM RAM mean MiB	VM RAM peak MiB	VM RAM p99 MiB	VM RAM p99.9 MiB	Validity
638a-0094	PyM FT G1/T	1,240	63	27.2	29.0	29.0	29.0	827.0	842.3	841.2	842.2	pass
638a-0095	PyM FT G1/T	1,309	64	10.0	16.2	15.5	16.1	2,622.1	3,416.7	3,405.0	3,415.5	fail
638a-0096	PyM FT G1/T	1,309	64	10.3	24.6	18.7	24.0	2,310.3	3,203.6	3,200.6	3,203.3	fail
638a-0097	PyM FT G0/T	1,240	63	27.3	29.8	29.1	29.7	825.0	882.1	879.7	881.8	pass
638a-0098	PyM FT G0/T	1,309	62	28.3	30.1	30.0	30.1	802.1	823.1	822.4	823.0	pass
638a-0099	PyM FT G0/T	1,509	64	10.9	17.6	16.9	17.5	2,860.6	3,462.3	3,460.6	3,462.2	fail
638a-0100	PyM FT G0/T	1,409	64	10.6	18.5	16.2	18.2	2,771.9	3,401.6	3,394.9	3,400.9	fail
638a-0101	PyM FT G0/T	1,409	64	10.8	17.0	16.3	16.9	2,787.8	3,434.4	3,413.3	3,432.3	fail
0457-0000	PyB 3.10.22	10,000	49	100.0	100.0	100.0	100.0	1,381.3	1,469.8	1,468.4	1,469.7	Flagged 3

PRIMARY CPU spans both VM vCPUs, all host activity/steal; RAM=MemTotal - MemAvailable, not RSS or all cache. Capacity rows are steady; pilot/smoke use their measurement window. Sparsep 99/p99.9 are weak descriptive estimates. FD/invalid flags override capacity inference. No boot counter is represented as trial usage.

Historical container: secondary

Group 12/12 | user-capacity, user-capacity-invalid-20261004 | SECONDARY app container CPU and sampled working set.

ID suffix	Runtime / cell	Users or RPS	n	CPU mean %	CPU peak %	CPU p99 %	CPU p99.9 %	Mem mean MiB	Mem peak MiB	Mem p99 MiB	Mem p99.9 MiB	Validity
638a-0094	PyM FT G1/T	1,240	63	20.6	27.4	27.3	27.4	245.2	266.3	264.2	266.1	pass
638a-0095	PyM FT G1/T	1,309	64	4.9	31.5	19.1	30.3	2,068.0	2,871.3	2,859.7	2,870.1	fail
638a-0096	PyM FT G1/T	1,309	64	4.2	14.6	13.7	14.5	1,768.8	2,647.0	2,639.3	2,646.3	fail
638a-0097	PyM FT G0/T	1,240	63	20.6	27.2	27.1	27.2	247.1	310.9	310.9	310.9	pass
638a-0098	PyM FT G0/T	1,309	62	22.2	30.4	29.9	30.3	221.1	257.9	250.0	257.1	pass
638a-0099	PyM FT G0/T	1,509	64	5.0	18.4	14.8	18.1	2,317.8	2,948.1	2,944.2	2,947.7	fail
638a-0100	PyM FT G0/T	1,409	64	4.8	12.6	10.7	12.4	2,228.0	2,867.2	2,855.6	2,866.0	fail
638a-0101	PyM FT G0/T	1,409	64	5.1	32.9	20.8	31.6	2,235.8	2,887.7	2,876.7	2,886.6	fail
0457-0000	PyB 3.10.22	10,000	49	82.1	95.9	93.5	95.6	577.4	772.9	743.1	769.9	Flagged 3

ContainerCPU 100% = both allocated vCPUs; memory = sampled Docker working set MiB. Capacity rows are steady; pilot/smoke use their measurement window. Sparsep 99/p99.9 are weak descriptive estimates. FD/invalid flags override capacity inference. No boot counter is represented as trial usage.

Sources and metric provenance

Reproducible files and interpretation boundaries, without private secrets.

Source	Purpose
all-experiments-2026-10-07.json	Updated 234-trial whitelisted dataset; saved metrics, resource/RDS summaries and rate reductions
all-trials.csv / all-trial-metrics.csv	Original 228-trial inventory and numeric archive; retained unchanged
fixed-load-complete-metrics-2026-10-07.csv	Updated 27-row wide scalar export, including phase and endpoint detail
metric-fields.json	Field definitions, availability and interpretation limits
review-data-final.json	Original historical snapshots; unavailable latency fields remain unavailable
Original fixed-load-results.json	Recorded seven-stack fixed-load outcomes and metrics
New .NET / Octane fixed-load records	Six additional complete trials; canonical results/telemetry/rates reduced from AWS
Runtime and dependency locks	Versions, topology and immutable image provenance
user-capacity-methodology.md	Workload, profiles, gates, validity and threshold-replay limits

Dataset/build sources: artifacts/report-data. The updated wide CSV is in output/pdf. Added raw campaigns: dotnet-fixed-20261007 and octane-fixed-20261007. Original artifacts and the 06-Oct PDF remain unchanged.

WholeVM CPU/RAM is primary; Docker container metrics are separately labeled secondary. No database credentials, JWT secrets, private environment files or SSH keys are included. Raw historical metrics remain on AWS; bounded old summaries explain unavailable fields. Fixed raw points are reduced into second-by-second scheduled-window rates and spillover rather than downloaded as a large archival scan.

Unified dataset SHA256: 574a3b06ac0c1e8503a099242f44221f9985df40fdb1e688919f1908bf78daa5

Cumulative boot network and PostgreSQL counter snapshots/quantiles in exports are not trial bandwidth/loss/usage. Only valid interval deltas or rates can be interpreted that way. Source JSON and dedicated resource/second-bucket CSVs retain the time series.

Official metric and hardware references

AWS: Amazon RDS db.m7i hardware specifications

AWS: Amazon RDS CloudWatch metrics and FreeableMemory

Statistical caution

Repeated data points within one run are not independent experimental repetitions. Resource quantiles from sparse/rounded samples and DB quantiles from independently windowed minute buckets do not establish confidence intervals or rare-event service guarantees. The report gives descriptive single-run observations and recorded quality flags.

Scope

This document reviews completed evidence. Smoke/calibration/preflight work and invalid archives are separate from capacity evidence. Fixed-load passes at 1,000 users demonstrate the tested population under the recorded gates; they do not establish a maximum user count.